

1 Physical coordinates and exact finite viscous realization

We use the orientation

$$x = (x_1, x_2) \in \mathbb{R}^2, \quad e_2 = (0, 1), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \nabla^\perp = J\nabla, \quad \text{curl } u = \partial_1 u_2 - \partial_2 u_1.$$

The physical equations throughout this note are

$$\theta_t + u \cdot \nabla \theta = \kappa \Delta \theta + f_\theta, \tag{1}$$

$$u_t + (u \cdot \nabla)u + \nabla p = \varepsilon \Delta u + \theta e_2 + f_u, \quad \text{div } u = 0. \tag{2}$$

Here $\kappa, \varepsilon \geq 0$ are respectively thermal diffusivity and physical viscosity. In particular ε is not the source's time-scaling parameter ν and is not its schedule error tolerance. The datum is retained exactly:

$$A_0 \geq 1, \quad \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0} \sin(\lambda_0 x_2) \chi_0(x), \quad u_{\text{in}}(x) = 0.$$

The original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, $R_0 > 2$, equals one on B_2 and lies in $[0, 1]$. Define

$$\mu = \lambda_0, \quad \nu = \sqrt{A_0}, \quad (X, T) = (\mu x, \nu t).$$

These are explicit maps between the source coordinates and the physical coordinates, rather than replacements of A_0 or λ_0 by one. For source fields decorated with a tilde their pullback is

$$\begin{aligned} \theta(x, t) &= \frac{\nu^2}{\mu} \tilde{\theta}(\mu x, \nu t), & u(x, t) &= \frac{\nu}{\mu} \tilde{u}(\mu x, \nu t), \\ p(x, t) &= \frac{\nu^2}{\mu^2} \tilde{p}(\mu x, \nu t), & \psi(x, t) &= \frac{\nu}{\mu^2} \tilde{\psi}(\mu x, \nu t). \end{aligned} \tag{3}$$

Their forces pull back with factors ν^3/μ and ν^2/μ , respectively. Their source diffusion coefficients must be

$$\tilde{\kappa} = \frac{\mu^2}{\nu} \kappa, \quad \tilde{\varepsilon} = \frac{\mu^2}{\nu} \varepsilon. \tag{4}$$

Indeed each term on the left of the scalar equation acquires ν^3/μ , whereas $\kappa \Delta_x \theta$ acquires $\kappa \nu^2 \mu$. Each momentum and buoyancy term acquires ν^2/μ , whereas $\varepsilon \Delta_x u$ acquires $\varepsilon \nu \mu$. Division gives exactly (4). Also $\nabla_x^\perp \psi = u$, $\Delta_x \psi = \nu \tilde{\omega}$, and $\text{div}_x u = \nu \text{div}_X \tilde{u}$. The inverse map substitutes $(x, t) = (X/\mu, T/\nu)$ and reciprocates the displayed factors, proving a bijection for corresponding forces, coefficients, and initial values. The cutoff is $\tilde{\chi}_0(X) = \chi_0(X/\mu)$, so the pulled-back initial temperature is exactly the displayed θ_{in} . These computations extend source Lemma 1.2, equations (1.6)–(1.7), pp. 6–7, by retaining both diffusion coefficients.

1.1 The released profile, the initial interpolation, and the base

Fix $0 < s_0 < 1$ and an even smooth cutoff χ supported in $(-\pi, \pi)$, equal to one on $[-s_0, s_0]$. On $[-\pi, \pi]$ put

$$F(s) = \sin s + \chi(s)(s - \sin s)$$

and extend periodically. Let

$$P(s) = \int_0^s F(r) dr - \frac{1}{2\pi} \int_{-\pi}^\pi \int_0^y F(r) dr dy.$$

Oddness and zero period integral of F show that P is even, periodic, has zero mean, and $P' = F$. Near zero, $F(s) = s$ and $P(s) = P(0) + s^2/2$. No sinusoidal identity is substituted for this profile.

Take a smooth nondecreasing step η that is zero for $t \leq 0$ and one for $t \geq 1$. Every derivative of positive order vanishes at its endpoints: smoothness and its constant extension prove this.

Choose $R_H > R_0 + 1$ and radial $H \in C_c^\infty(B_{R_H})$ equal to $|x|^2/2$ on B_{R_0+1} . Let $\alpha = 0$ on $[0, 1/\nu]$ and $e_0 = R_\alpha e_2$. Define

$$\begin{aligned}\theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x) \chi_0(x), & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H, \quad u_b = \dot{\alpha} J \nabla H.\end{aligned}\tag{5}$$

For $t \leq 1/\nu$, $u_b = 0$ and $\partial_t \theta_b = \nu \eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}})$. For later times, $u_b = \dot{\alpha} J x$ on the support of θ_0 . Since $\dot{e}_0 = \dot{\alpha} J e_0$,

$$(\partial_t + \dot{\alpha} J x \cdot \nabla)(\mu e_0 \cdot x) = \mu \dot{\alpha} (J e_0 \cdot x + e_0 \cdot J x) = 0.$$

Radiality gives $J x \cdot \nabla \chi_0 = 0$. Therefore the exact zero-pressure base forces are

$$\begin{aligned}f_{\theta,b}^0 &= \nu \eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}), \\ f_{u,b}^0 &= \ddot{\alpha} J \nabla H + \dot{\alpha}^2 (J \nabla H \cdot \nabla)(J \nabla H) - \theta_b e_2.\end{aligned}\tag{6}$$

On $|\mu e_0 \cdot x| < s_0$, $|x| < 2$, and $t \geq 1/\nu$, the base has the exact values

$$\theta_b = -A_0 e_0 \cdot x, \quad u_b = \dot{\alpha} J x, \quad G_b = -A_0 e_0, \quad D_b = \dot{\alpha} J.$$

This is source (4.9)–(4.10), pp. 38–39, in the original coordinates.

1.2 A compact realization of the complete first controlled stage

Use the first-stage coefficients constructed and proved below, with physical frequency $K_1 = \mu \lambda_1$, phase $\zeta = R_\alpha e(s_*)$, $|\zeta| = 1$, and signed amplitudes Θ, Ω satisfying

$$\dot{\Theta} = -\frac{J \zeta \cdot G_b}{K_1} \Omega, \quad \dot{\Omega} = K_1 \zeta_1 \Theta.\tag{7}$$

Here $e(\phi) = (\sin \phi, \cos \phi)$ and $J \zeta \cdot G_b = -A_0 \sin s_*$, a rotationally invariant scalar. Choose $C_\ell \geq 4$, set $\ell = s_0/(C_\ell \mu)$, and take radial $f \in C_c^\infty(B_1)$ equal to one on $B_{1/2}$. Put

$$g(x) = f(x/\ell), \quad s = K_1 \zeta \cdot x, \quad w = \eta(\Gamma(t - 1/\nu)), \quad \Gamma = \nu \sin s_*, \quad T = w \Theta, \quad Q = \frac{w \Omega}{K_1^2}.$$

Although g is written in laboratory coordinates, it is exactly the paper's material envelope: $M = R_\alpha$, so $f(M^{-1}x/\ell) = f(x/\ell)$ by radiality. Set

$$\begin{aligned}V &= T F(s) g, & \Psi &= Q P(s) g, \\ v &= Q \{K_1 J \zeta F(s) g + P(s) J \nabla g\}, & \theta &= \theta_b + V, \quad u = u_b + v, \quad p = 0.\end{aligned}\tag{8}$$

The new fields are extended by zero before $t = 1/\nu$. They are smooth across that time because w is flat there. Their support is in $B_\ell \subset B_{s_0/\mu} \cap B_2$, where the base is affine for every angle. The phase and envelope satisfy

$$(\partial_t + D_b x \cdot \nabla)s = 0, \quad (\partial_t + D_b x \cdot \nabla)g = 0.$$

For the phase this follows from $\dot{\zeta} = \dot{\alpha} J \zeta = -D_b^T \zeta$; for the envelope it follows from radiality.

Write $\partial_\perp g = J \zeta \cdot \nabla g$. Direct expansion gives the following complete added forces at zero diffusion:

$$E_\theta^0 = \dot{w} \Theta F g + Q P (J \nabla g \cdot G_b) + K_1 Q T (F^2 - P F') g \partial_\perp g,\tag{9}$$

$$E_u^0 = 2 D_b v + \dot{Q} \{K_1 J \zeta F g + P J \nabla g\} + (v \cdot \nabla) v - T F g e_2.\tag{10}$$

Every function of s in these equations is evaluated at the phase in (8). The vector derivative in the last formula can, equivalently, be evaluated by the explicitly specified matrix

$$\nabla v = QJ [K_1^2 \zeta \otimes \zeta F'g + K_1 F(\zeta \otimes \nabla g + \nabla g \otimes \zeta) + P \nabla^2 g], \quad (v \cdot \nabla)v = (\nabla v)v.$$

Thus (10) is a direct component formula, without solving a pressure equation or an inverse-curl problem.

Here is a proof of both residual identities. Since $J\zeta \cdot \zeta = 0$,

$$v \cdot \nabla V = K_1 QT(F^2 - PF')g(J\zeta \cdot \nabla g).$$

Also $v \cdot G_b = QK_1(J\zeta \cdot G_b)Fg + QP(J\nabla g \cdot G_b)$ and the material derivative of V is $\dot{T}Fg$. Equation (7) implies $\dot{T} + QK_1 J\zeta \cdot G_b = \dot{w}\Theta$, proving (9). For any trace-free matrix D , direct multiplication proves $DJ + JD^T = 0$. Consequently

$$(\partial_t + Dx \cdot \nabla)J\nabla\Psi = DJ\nabla\Psi + J\nabla[(\partial_t + Dx \cdot \nabla)\Psi].$$

The material derivative of Ψ is $\dot{Q}Pg$, and $(v \cdot \nabla)u_b = D_b v$ on the new support. Adding these two terms, the self-advection, and the negative buoyancy increment proves (10). Away from the new support every increment vanishes, so this verification is global.

For every prescribed $\kappa, \varepsilon \geq 0$ the *same* controlled fields now solve (1)–(2) with the explicit forces

$$f_\theta^{\kappa, \varepsilon} = f_{\theta, b}^0 + E_\theta^0 - \kappa \Delta \theta_b - \kappa \Delta V, \quad (11)$$

$$f_u^{\kappa, \varepsilon} = f_{u, b}^0 + E_u^0 - \varepsilon \Delta u_b - \varepsilon \Delta v. \quad (12)$$

Indeed substituting these right-hand sides cancels the displayed diffusion terms exactly and leaves the two residual identities already proved. This proves a concrete finite controlled viscous stage, for arbitrary physical coefficients, retaining the paper's actual profile and motion. Its costs are calculated next; no claim that these are small or uniform over infinitely many stages is part of this construction.

All fields and forces are supported in the fixed spatial ball B_{R_H} . They are smooth on the complete closed finite interval, including interpolation, activation, growth, transition, steering, and any specified finite holding duration. The pressure $p = 0$ satisfies $p(0, t) = 0$. Since $\psi_b + \Psi$ is compactly supported, $u = J\nabla(\psi_b + \Psi)$ is divergence free and has finite spatial energy. It also has exactly the Biot–Savart relation $u = K\omega$: if $N = (2\pi)^{-1} \log|x|$, integration by parts against the compact streamfunction gives $N * \Delta(\psi_b + \Psi) = (\Delta N) * (\psi_b + \Psi) = \psi_b + \Psi$; applying $J\nabla$ proves the assertion in distributions and hence for these smooth fields. Odd F , even P , and radial cutoffs give odd θ, u, f_θ, f_u and even ω . The finite-slab forces can also be specified with compact temporal support. Extend the state by its stationary initial values to the left and its stationary held values to the right; flatness of the interpolation and the selected pulse proves smoothness of these extensions. For any fixed $\tau_{\text{ext}} > 0$, multiply the resulting force formulas by

$$\eta(1 + t/\tau_{\text{ext}})\eta(1 + (T_f - t)/\tau_{\text{ext}}).$$

Here T_f denotes the finite final time of the prescribed hold. This multiplier is one on $[0, T_f]$ and zero outside $(-\tau_{\text{ext}}, T_f + \tau_{\text{ext}})$. Thus the same solution on its specified slab has forces in $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$, with explicitly specified spatial and temporal support.

On the smaller open set

$$|x| < \ell/2, \quad |K_1 \zeta \cdot x| < s_0,$$

the exact fields are

$$\theta = (-A_0 e_0 + w K_1 \Theta \zeta) \cdot x, \quad u = (\dot{\alpha} J + w \Omega J \zeta \otimes \zeta)x. \quad (13)$$

Their Laplacians vanish there, and so do the additional diffusive forces *and all their spatial derivatives* there. Therefore the evolving parent background for a next layer inside this open set is exactly the coupled gradient and shear in (13), with the actual time-dependent $\Theta, \Omega, \alpha, w$. It has not been replaced by prescribed unchanging D, G . The finite-stage force supplies the compensating diffusion outside this affine region.

1.3 Every spatial diffusion cost, including the cutoff terms

Put $k = K_1\zeta$, so $|k| = K_1$. Direct differentiation gives

$$\Delta V = T\{K_1^2 F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (14)$$

$$\begin{aligned} \Delta v = QJ\{K_1^2 k F''g + K_1^2 F' \nabla g + 2F'k(k \cdot \nabla g) \\ + 2F \nabla(k \cdot \nabla g) + Fk \Delta g + P \nabla \Delta g\}. \end{aligned} \quad (15)$$

For the second formula first differentiate $\Delta \Psi = Q\{K_1^2 F'g + 2Fk \cdot \nabla g + P\Delta g\}$, and apply J . These are exact differential expressions on all of $\mathbb{R}^2$ and retain the phase, mixed, and envelope terms. For example, their zeroth-order norms satisfy

$$\begin{aligned} \|\Delta V\|_\infty &\leq |T|\{K_1^2 \|F''\|_\infty \|g\|_\infty + 2K_1 \|F'\|_\infty \|\nabla g\|_\infty + \|F\|_\infty \|\Delta g\|_\infty\}, \\ \|\Delta v\|_\infty &\leq |Q|\{K_1^3 \|F''\|_\infty \|g\|_\infty + 3K_1^2 \|F'\|_\infty \|\nabla g\|_\infty \\ &\quad + 2K_1 \|F\|_\infty \|\nabla^2 g\|_\infty + K_1 \|F\|_\infty \|\Delta g\|_\infty + \|P\|_\infty \|\nabla \Delta g\|_\infty\}. \end{aligned} \quad (16)$$

The Hessian norm is its operator norm, and vector norms are Euclidean. Thus no cutoff diffusion term has been absorbed into a frequency-only estimate. Here $\|\partial^\beta g\|_\infty = \ell^{-|\beta|} \|\partial^\beta f\|_\infty$, including every original s_0, C_ℓ, μ through $\ell = s_0/(C_\ell \mu)$.

For an arbitrary nonnegative integer m put

$$E_m = \max_{|\beta|=m} \|\partial^\beta g\|_\infty, \quad B_m(W) = \sum_{j=0}^m \binom{m}{j} K_1^j \|W^{(j)}\|_\infty E_{m-j}.$$

Leibniz' rule, $\partial^\beta W(k \cdot x) = k^\beta W^{(|\beta|)}(k \cdot x)$, and $\sum_{|\beta|=j, \beta \leq \gamma} \binom{\gamma}{\beta} = \binom{m}{j}$ for $|\gamma| = m$ prove

$$\begin{aligned} \max_{|\gamma|=m} \|\partial^\gamma \Delta V\|_\infty &\leq 2|T|B_{m+2}(F), \\ \max_{|\gamma|=m} \|\partial^\gamma \Delta v\|_\infty &\leq 2\sqrt{2}|Q|B_{m+3}(P). \end{aligned} \quad (17)$$

The two in each line counts the two coordinate Laplacian terms; $\sqrt{2}$ counts the two components of $J\nabla$.

Let $S = |\Theta^{\text{seed}}|$, let L be the prescribed growth gain, let Λ_1 be the paper's pulse compression, and let $H_t \geq 0$ be the specified physical holding duration. The proved selected stage below has durations $L/\Gamma, 1/\Gamma, (1 + \Lambda_1^{-1})/\Gamma$, and H_t , respectively. On its growth and transition $|\Theta| = S \exp(\Gamma(t - t_{\text{in}}))$ and $|\Omega| = (K_1/\nu)|\Theta|$. On its selected steering,

$$|\Theta| \leq S e^{L+2+\Lambda_1^{-1}}, \quad |\Omega| \leq \frac{K_1}{\nu} |\Theta|.$$

On its hold $\Omega = 0$ and Θ is constant. Consequently the explicit costs

$$\begin{aligned} I_\Theta &= S \frac{e^{L+1} - 1}{\Gamma} + S e^{L+2+\Lambda_1^{-1}} \left(\frac{1 + \Lambda_1^{-1}}{\Gamma} + H_t \right), \\ I_\Omega &= \frac{K_1}{\nu} S \left\{ \frac{e^{L+1} - 1}{\Gamma} + e^{L+2+\Lambda_1^{-1}} \frac{1 + \Lambda_1^{-1}}{\Gamma} \right\} \end{aligned} \quad (18)$$

satisfy $\int |T| dt \leq I_\Theta$ and $\int |w\Omega| dt \leq I_\Omega$ over the complete physical stage. This follows by integrating the exponential exactly, using $0 \leq w \leq 1$, and then using the stated bounds on the remaining intervals. Combining with (17) gives

$$\int \max_{|\gamma|=m} \|\partial^\gamma (-\kappa \Delta V)\|_\infty dt \leq 2\kappa B_{m+2}(F) I_\Theta,$$

$$\int \max_{|\gamma|=m} \|\partial^\gamma(-\varepsilon \Delta v)\|_\infty dt \leq \frac{2\sqrt{2}\varepsilon}{K_1^2} B_{m+3}(P) I_\Omega. \quad (19)$$

In particular the holding scalar cost is proportional to H_t ; there is no assertion that a forced hold has zero scalar diffusion cost.

The base costs are equally explicit. Write

$$C_m = \max_{|\beta|=m} \|\partial^\beta \chi_0\|_\infty, \quad \mathcal{B}_m(W) = \sum_{j=0}^m \binom{m}{j} \mu^j \|W^{(j)}\|_\infty C_{m-j}.$$

On the initial interpolation $\theta_b = (1 - \eta(\nu t))\theta_{\text{in}} + \eta(\nu t)\theta_{\text{ref}}$; afterwards it is a rotated F -profile with the same radial cutoff. Thus, writing $T_f = 1/\nu + (L + 2 + \Lambda_1^{-1})/\Gamma + H_t$,

$$\begin{aligned} \int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\kappa \Delta \theta_b)\|_\infty dt &\leq \frac{2\kappa A_0}{\mu} \left\{ \frac{\max(\mathcal{B}_{m+2}(\sin), \mathcal{B}_{m+2}(F))}{\nu} + (T_f - 1/\nu) \mathcal{B}_{m+2}(F) \right\}, \\ \int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\varepsilon \Delta u_b)\|_\infty dt &\leq \varepsilon \max_{|\gamma|=m} \|\partial^\gamma J \nabla \Delta H\|_\infty \int_0^{T_f} |\dot{\alpha}| dt. \end{aligned} \quad (20)$$

The last angle variation is exactly the finite integral of the explicit control and has the following fixed upper bound. For $z(\tau)$ in the stage proof and its selected pulse multiplier m_* , $|Z| \leq 1/4$, and

$$\int |\dot{\alpha}| dt \leq \frac{4}{\sqrt{15}} \int |\dot{Z}| dt \leq \frac{4 \sin s_*}{\sqrt{15}} \left(1 + m_* \Lambda_1 \int_0^1 |\eta'_2(y)| dy \right).$$

This uses $\alpha = s_* - \arcsin Z$, $\int_0^1 \eta' = 1$, and the change of variables $y = \Lambda_1(\tau - 1)$ on the pulse.

For completeness all mixed derivatives also have an exact finite replay formula; spatial bounds alone are not being called bounds for time derivatives. For $A = T, W = F$, or $A = Q, W = P$, and any multi-index γ , Leibniz gives

$$\partial_t^b \partial_x^\gamma [AW(k \cdot x)g] = \sum_{\beta \leq \gamma} \binom{\gamma}{\beta} (\partial^\beta g) \sum_{a+c+d=b} \frac{b!}{a!c!d!} A^{(a)}(k^{\gamma-\beta})^{(c)} \partial_t^d W^{(|\gamma-\beta|)}(k \cdot x). \quad (21)$$

For $d > 0$ the remaining derivative is the finite sum

$$\partial_t^d W^{(n)}(k \cdot x) = \sum_{\substack{m_1, \dots, m_d \geq 0 \\ \sum j m_j = d}} \frac{d!}{\prod_{j=1}^d m_j! (j!)^{m_j}} W^{(n+\sum m_j)}(k \cdot x) \prod_{j=1}^d (k^{(j)} \cdot x)^{m_j}.$$

For $d = 0$ it is $W^{(n)}(k \cdot x)$. This identity follows by induction: differentiation either increases the derivative order on W and creates $k' \cdot x$, or differentiates one of the displayed factors; grouping the resulting multiplicities gives exactly the factorial coefficients at order $d + 1$. On the support use $|k^{(j)} \cdot x| \leq \ell |k^{(j)}|$ and $|\partial^\beta g| \leq E_{|\beta|}$ to obtain an explicit upper bound from each term. Sum the two coordinate second derivatives in (21) for ΔV and then differentiate once more spatially for Δv . The same formula with $k = \mu e_0$, $g = \chi_0$ handles θ_0 ; the initial interpolation additionally uses the exact derivatives $\partial_t^a \eta(\nu t) = \nu^a \eta^{(a)}(\nu t)$. For u_b , mixed diffusion derivatives are exactly $\varepsilon \alpha^{(b+1)} \partial^\gamma J \nabla \Delta H$. All coefficient derivatives are defined by the explicit smooth stage equations on a compact interval and the finitely many fixed profile derivatives, so these formulas quantify every specified mixed order without an asymptotic omission.

2 The actual first controlled stage in the original units

The source for this section is Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, <https://cims.nyu.edu/~tristanb/boussinesq.pdf>, released PDF, 76 pages. The source locations used here are Lemma 1.2 and (1.6), pp. 6–7; (3.1)–(3.12), pp. 8–10; the complete proofs of Lemmas 3.7–3.8 and (3.24)–(3.34), pp. 15–18; (3.58) and the following second-stage calculation, p. 27; and the first holding integral on p. 36. The proofs below are given in full. The source's unit-scale convention is not imposed on the fields in this section.

2.1 Parameters, coordinates, and the original coupled equations

Let $x = (x_1, x_2) \in \mathbb{R}^2$, $e_2 = (0, 1)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, \quad e(s) = (\sin s, \cos s).$$

In particular $R_\alpha e(s) = e(s - \alpha)$ and $\text{curl } u = \partial_1 u_2 - \partial_2 u_1$. Retain $A_0 > 0$, $\lambda_0 \geq 1$, and the source's two scaling parameters

$$\mu = \lambda_0, \quad \nu = \sqrt{A_0}, \quad K_1 = \mu \lambda_1.$$

Here ν is the source's *time-scaling parameter*. Physical momentum viscosity below is denoted $\nu_{\text{phys}} \geq 0$; thermal diffusivity is $\kappa_{\text{phys}} \geq 0$. The first pulse compression is the source's Λ_1 , which is different from the physical spatial frequency K_1 .

Fix a smooth nondecreasing step $\eta : \mathbb{R} \rightarrow [0, 1]$ equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $h \in C_c^\infty((0, 1))$ with $h \geq 0$ and $\int_0^1 h = 1$. Write $H = \|h\|_\infty$. The source's choice is $h = \eta_2$, $c_p = 3$; the following calculation retains any

$$c_p > 1, \quad \Lambda_1 \geq 2c_p H, \quad 0 < s = s_* \leq \frac{1}{8}, \quad c_p \Lambda_1 H \sin s \leq \frac{1}{4}. \quad (22)$$

Since $\int h = 1$, $H \geq 1$ and $\Lambda_1 > 2$. Let $k_1 \geq 0$ be an integer, $\lambda_1 > 1$, and set

$$S = \frac{A_0}{\mu} \lambda_1^{-k_1-6}, \quad L = (k_1 + 5 + \frac{1}{8}) \log \lambda_1 \geq 1, \quad P_* = S e^L = \frac{A_0}{\mu} \lambda_1^{-7/8}, \quad \Gamma = \nu \sin s, \quad t_0 = \nu^{-1}. \quad (23)$$

The condition $L \geq 1$ is ensured explicitly by $\lambda_1 \geq \exp((k_1 + 41/8)^{-1})$.

The first layer's complete preceding affine background, phase transport, and signed amplitude pair are

$$D_{<1} = \dot{\alpha} J, \quad G_{<1} = -A_0 R_\alpha e_2, \quad \dot{M}_1 = D_{<1} M_1, \quad M_1(t_0) = I, \quad (24)$$

$$\zeta_1 = M_1^{-T} e(s), \quad \dot{\Theta}_1 = a_1 \Omega_1, \quad \dot{\Omega}_1 = b_1 \Theta_1, \quad a_1 = -\frac{J \zeta_1 \cdot G_{<1}}{K_1 |\zeta_1|^2}, \quad b_1 = K_1 \zeta_{1,1}. \quad (25)$$

The entry data are

$$\alpha(t_0) = 0, \quad \Theta_1(t_0) = -S, \quad \Omega_1(t_0) = -\frac{K_1}{\nu} S, \quad w_1(t) = \eta(\Gamma(t - t_0)).$$

The activation factor multiplies the physical increment, not the homogeneous amplitude equation. It is flat at t_0 and equals one by $t_0 + \Gamma^{-1}$.

For this actual background, $M_1 = R_\alpha$: differentiation gives $\dot{R}_\alpha = \dot{\alpha} J R_\alpha$ and the initial values agree. Uniqueness follows, for example, by differentiating $R_\alpha^{-1} M_1$. Consequently

$$\zeta_1 = R_\alpha e(s), \quad |\zeta_1| = 1, \quad J \zeta_1 \cdot G_{<1} = -A_0 \sin s, \quad a_1 = \frac{A_0 \sin s}{K_1}, \quad b_1 = K_1 \sin(s - \alpha). \quad (26)$$

The determinant identity used here is explicit: $J e(s) = (-\cos s, \sin s)$ has scalar product $\sin s$ with e_2 , and simultaneous rotation preserves that scalar product. In particular a_1 is constant because both the parent gradient and the new phase rotate; it has not been frozen independently of the evolving parent.

2.2 Exact growth and transition

For the first stage the feedback sums F_0 and F_1 in source (3.4) are empty and equal zero. Thus both the source's growth control $\dot{\alpha} = F_0$ and its transition control $(1 - \eta_b)F_0 + \eta_b F_1$ give $\alpha = 0$. Set

$$t_a = t_0 + \frac{L}{\Gamma}, \quad t_1 = t_a + \Gamma^{-1}, \quad t_b = t_1 + \Gamma^{-1}(1 + \Lambda_1^{-1}).$$

On $[t_0, t_1]$ the exact solution of the original pair is

$$P_1(t) := -\Theta_1(t) = S e^{\Gamma(t-t_0)}, \quad \Omega_1(t) = -\frac{K_1}{\nu} P_1(t). \quad (27)$$

Indeed $a_1 K_1 / \nu = \Gamma$ and $\Gamma K_1 / \nu = K_1 \sin s = b_1$. This proves both differential equations and both initial data. Uniqueness follows from the linear integral equation: on a bounded interval with coefficient norm at most B , the n -fold iterated integral of the difference of two solutions is bounded by its supremum times $(B\ell)^n / n!$, which tends to zero. The same convergent integral series supplies existence on every finite interval. Since P_1 is strictly increasing, t_a is exactly the first crossing of P_* , and the transition contributes exactly one to $\log P_1$. The steering entry amplitude is therefore $P_{\text{ent}} = e P_*$.

2.3 The trial steering family and proof of its unique return

Use $m \in [0, c_p]$ for the pulse trial parameter, to distinguish it from the unchanged spatial parameter $\mu = \lambda_0$. With $\tau = \Gamma(t - t_1)$ and $\tau_b = 1 + \Lambda_1^{-1}$ set

$$z(\tau; m) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -m \Lambda_1 h(\Lambda_1(\tau - 1)), & 1 \leq \tau \leq \tau_b, \end{cases} \quad Z = \sin s z, \quad \alpha = s - \arcsin Z. \quad (28)$$

Extend $Z = \sin s$ before t_1 and $Z = 0$ after t_b . The two smooth pieces and extensions agree to every order, since the step is flat and h is supported in the interior. The source's selected pulse height in (3.12) is precisely $h_1 = m \Lambda_1 \sin s$; its allowed interval is $[0, c_p \Lambda_1 \sin s]$. The design gives

$$|Z| \leq \frac{1}{4}, \quad \zeta_{1,1} = Z, \quad \zeta_{1,2} = \sqrt{1 - Z^2} \geq \frac{\sqrt{15}}{4}, \quad \dot{\alpha} = -\frac{\dot{Z}}{\zeta_{1,2}}, \quad 0 \leq \alpha < \frac{1}{2}. \quad (29)$$

For the last bound, $Z \leq \sin s$ implies $\alpha \geq 0$, and $\arcsin(1/4) \leq (1/4)/\sqrt{1 - 1/16} < 1/3$ gives $\alpha \leq 1/8 + 1/3 < 1/2$. These are the source's actual feedback and the original laboratory phase component; gravity remains e_2 .

Define $\mathcal{P}$ first as the solution of the linear equation

$$\mathcal{P}_{\tau\tau} = z\mathcal{P}, \quad \mathcal{P}(0) = \mathcal{P}_\tau(0) = 1. \quad (30)$$

The same linear integral series proves existence, uniqueness, and smoothness for every trial on the full finite interval. Moreover the exact inverse map to the original amplitude pair is

$$\Theta_1(t) = -P_{\text{ent}} \mathcal{P}(\tau), \quad \Omega_1(t) = -\frac{K_1}{\nu} P_{\text{ent}} \mathcal{P}_\tau(\tau). \quad (31)$$

To check it, its component derivatives are exactly

$$-\Gamma P_{\text{ent}} \mathcal{P}_\tau = a_1 \Omega_1, \quad -\frac{K_1}{\nu} \Gamma P_{\text{ent}} z \mathcal{P} = K_1 \sin s z \Theta_1 = b_1 \Theta_1.$$

The data at t_1 match (27). Conversely the first amplitude equation with constant a_1 and $a_1 b_1 = \Gamma^2 z$ gives (30) for $-\Theta_1 / P_{\text{ent}}$, with the stated initial derivative. Thus the auxiliary equation and original pair are explicitly inverse descriptions of the same dynamics, not a substituted oscillator.

On $0 \leq \tau \leq 1$, $z \geq 0$. Before any first zero of either $\mathcal{P}$ or $\mathcal{P}_\tau$, one has $\mathcal{P}_{\tau\tau} \geq 0$ and therefore $\mathcal{P}_\tau \geq 1$, $\mathcal{P} \geq 1 + \tau$. Continuity excludes that first zero and proves these inequalities through $\tau = 1$. Only now define $v = \mathcal{P}_\tau / \mathcal{P}$. It satisfies

$$v_\tau = 1 - \eta - v^2, \quad v(0) = 1.$$

Since $(v - 1)_\tau = -(v + 1)(v - 1) - \eta$, the integrating-factor formula gives $v \leq 1$. Since $v > 0$ and $v_\tau \geq -v^2$, $(1/v)_\tau \leq 1$, so

$$\frac{1}{1 + \tau} \leq v(\tau) \leq 1, \quad \log 2 \leq \int_0^1 v d\tau \leq 1, \quad v_d := v(1) \in [1/2, 1], \quad 2 \leq \mathcal{P}(1) \leq e. \quad (32)$$

On the negative pulse set $y = \Lambda_1(\tau - 1)$ and $\chi(y) = \mathcal{P}(1 + y/\Lambda_1)/\mathcal{P}(1)$. There is no division by an unknown evolving amplitude in this definition. Its exact equation and data are

$$\chi_{yy} + \frac{mh(y)}{\Lambda_1}\chi = 0, \quad \chi(0) = 1, \quad \chi_y(0) = v_d/\Lambda_1. \quad (33)$$

Before a possible first zero, χ is concave, so $\chi \leq 1 + y/\Lambda_1 \leq 2$. Since $mh/\Lambda_1 \leq c_p H/\Lambda_1 \leq 1/2$, one has $\chi_{yy} \geq -1$, hence

$$\chi(y) \geq 1 + \frac{v_d}{\Lambda_1}y - \frac{y^2}{2} \geq \frac{1}{2} \quad (0 \leq y \leq 1).$$

The strict positive lower bound excludes a first zero. Thus every trial satisfies on the entire pulse

$$\frac{1}{2} \leq \chi(y) \leq 1 + y/\Lambda_1, \quad \chi_y = \frac{v_d}{\Lambda_1} - \frac{m}{\Lambda_1} \int_0^y h(r)\chi(r) dr \geq -\frac{2c_p}{\Lambda_1}. \quad (34)$$

Consequently the original temperature is strictly negative for every trial, and the quotient is well defined. Writing $V(y; m) = v(1 + y/\Lambda_1; m) = \Lambda_1 \chi_y / \chi$ gives

$$V_y = -mh(y) - \frac{V^2}{\Lambda_1}, \quad V(0; m) = v_d, \quad -4c_p \leq V(y; m) \leq v_d \leq 1. \quad (35)$$

Here is the parameter dependence needed to select the return. In the first-order integral system for (χ, χ_y) the coefficient matrix depends affinely on m . On the compact parameter interval, its iterated integrals converge uniformly with factorial denominators. Differentiating r times in m introduces at most a degree- r polynomial in the iteration index, still dominated by that factorial. Thus this series and each parameter derivative converge uniformly. The lower bound $\chi \geq 1/2$ then gives smooth parameter dependence of V . Differentiating its exact equation and using an integrating factor gives

$$\partial_m V(1; m) = - \int_0^1 h(y) \exp\left(-\frac{2}{\Lambda_1} \int_y^1 V(r; m) dr\right) dy < 0. \quad (36)$$

The strict sign follows because $h \geq 0$ has mass one and the exponential is positive. Indeed the explicit bounds also give $e^{-2/\Lambda_1} \leq -\partial_m V(1; m) \leq e^{8c_p/\Lambda_1}$. At $m = 0$, direct solution of $V_y = -V^2/\Lambda_1$ gives $V(1; 0) = v_d/(1 + v_d/\Lambda_1) > 0$. At $m = c_p$, integration gives $V(1; c_p) \leq v_d - c_p < 0$. Continuity and strict monotonicity therefore give exactly one $m_* \in (0, c_p)$ with $V(1; m_*) = 0$. This is exactly the source's selected return: by (31) and positivity, it is equivalent to $\Omega_1(t_b) = 0$.

For this selected trial V is nonincreasing and ends at zero, so $0 \leq V \leq v_d$. Integration of (35) now gives the exact identity and bounds

$$m_* = v_d - \frac{1}{\Lambda_1} \int_0^1 V(y; m_*)^2 dy, \quad \frac{1}{2} - \frac{1}{4\Lambda_1} \leq m_* \leq 1. \quad (37)$$

For the lower bound, $v_d - v_d^2/\Lambda_1$ is increasing for $v_d \in [1/2, 1]$, because its derivative is $1 - 2v_d/\Lambda_1 > 0$.

For every trial and every steering time,

$$0 \leq \log \mathcal{P}(\tau) \leq 1 + \Lambda_1^{-1}.$$

On the ramp this follows from (32); on the pulse it follows from $\mathcal{P}(1) \geq 2$, $\chi \geq 1/2$, and $\log \chi \leq \log(1 + \Lambda_1^{-1}) \leq \Lambda_1^{-1}$. For the selected trial the pulse logarithmic contribution is nonnegative, so the endpoint obeys

$$\log 2 \leq \log \mathcal{P}(\tau_b) \leq 1 + \Lambda_1^{-1}, \quad 2eP_* \leq P_b := P_1(t_b) \leq e^{2+\Lambda_1^{-1}} P_*. \quad (38)$$

The total duration from entry to return is exactly

$$t_b - t_0 = \frac{L + 2 + \Lambda_1^{-1}}{\nu \sin s}. \quad (39)$$

Throughout the selected stage P_1 is nondecreasing, $S \leq P_1 \leq P_b$, and $-(K_1/\nu)P_1 \leq \Omega_1 \leq 0$. For every unselected trial the valid bound instead is $|\Omega_1| \leq 4c_p(K_1/\nu)e^{2+\Lambda_1^{-1}}P_*$; its sign need not stay negative.

Near t_b , $h = 0$, so $Z = 0$ and $\alpha = s$. The equation $\mathcal{P}_{\tau\tau} = 0$, together with $\mathcal{P}_\tau(\tau_b) = 0$, implies that $\mathcal{P}$ is constant on that whole neighborhood. Thus α , Θ_1 , and Ω_1 match their stationary continuations to all orders. The endpoint state is

$$\alpha = s, \quad M_1 = R_s, \quad \zeta_1 = e_2, \quad w_1 = 1, \quad \Theta_1 = -P_b, \quad \Omega_1 = 0. \quad (40)$$

2.4 Control bounds and exact whole-stage integrals

For $j = 1, 2$ define the explicit finite profile quantities

$$B_j = \Gamma^j \sin s (\|\eta^{(j)}\|_\infty + c_p \Lambda_1^{j+1} \|h^{(j)}\|_\infty).$$

From (28), $|\partial_t^j Z| \leq B_j$. Differentiating the original control, including its denominator, gives

$$\dot{\alpha} = -\dot{Z}(1 - Z^2)^{-1/2}, \quad (41)$$

$$\ddot{\alpha} = -\ddot{Z}(1 - Z^2)^{-1/2} - Z\dot{Z}^2(1 - Z^2)^{-3/2},$$

$$|\dot{\alpha}| \leq \frac{4}{\sqrt{15}}B_1, \quad |\ddot{\alpha}| \leq \frac{4}{\sqrt{15}}B_2 + \frac{16}{15\sqrt{15}}B_1^2. \quad (42)$$

Higher derivatives exist because the displayed smooth composition has argument in $[-1/4, 1/4]$; no endpoint differentiability is omitted.

For exact coefficient costs use the invertible fixed map $Y = (\Theta_1, \nu\Omega_1/K_1)^T$. Its inverse is $(Y_1, K_1 Y_2/\nu)^T$, and direct differentiation gives

$$\dot{Y} = \hat{B}Y, \quad \hat{B} = \begin{pmatrix} 0 & \Gamma \\ \nu Z & 0 \end{pmatrix}.$$

Let $J_\eta = \int_0^1 (1 - \eta(r)) dr \in [0, 1]$. On a hold of physical duration $T_h \geq 0$, the following are exact:

interval	duration	$\int \hat{B}_{12} dt$	$\int \hat{B}_{21} dt$
growth	L/Γ	L	L
transition	$1/\Gamma$	1	1
steering ramp	$1/\Gamma$	1	J_η
negative pulse	$1/(\Lambda_1 \Gamma)$	$1/\Lambda_1$	m_*
holding	T_h	ΓT_h	0

For example, on the pulse substitution $y = \Lambda_1 \Gamma(t - t_1 - \Gamma^{-1})$ gives $\int \nu|Z| dt = m_* \int_0^1 h = m_*$. Thus the two transition-and-steering costs are exactly $2 + \Lambda_1^{-1}$ and $1 + J_\eta + m_*$; their sum is at most $4 + c_p + \Lambda_1^{-1}$.

The activation and amplitude integrals needed by force estimates can also be retained without an asymptotic replacement. Set

$$J_a = \int_0^1 \eta'(r)e^r dr \in [1, e], \quad J_g = \int_0^1 \eta(r)e^r dr = e - J_a.$$

For the selected stage including the hold, direct change of variable and integration by parts give

$$\begin{aligned} \int_{t_0}^{t_b} |\dot{w}_1 \Theta_1| dt &= S J_a, & \int_{t_0}^{t_b} |\dot{w}_1 \Omega_1| dt &= \frac{K_1}{\nu} S J_a, \\ I_P := \int_{t_0}^{t_b+T_h} w_1 P_1 dt &= \frac{S}{\Gamma} (J_g + e^{L+1} - e) + P_b T_h \end{aligned} \quad (43)$$

$$+ \frac{eP_*}{\Gamma} \int_0^{1+\Lambda_1^{-1}} \mathcal{P}(\tau; m_*) d\tau, \quad (44)$$

$$I_\Omega := \int_{t_0}^{t_b+T_h} w_1 |\Omega_1| dt = \frac{P_b - SJ_a}{a_1}. \quad (45)$$

For the last identity, $P'_1 = -a_1 \Omega_1 = a_1 |\Omega_1|$, $w_1(t_0) = 0$, and $w_1(t_b + T_h) = 1$, hence $\int w_1 P'_1 = P_b - \int \dot{w}_1 P_1 = P_b - SJ_a$. In particular

$$I_P \leq \frac{S}{\Gamma} (J_g + e^{L+1} - e) + \frac{eP_*}{\Gamma} (1 + \Lambda_1^{-1}) e^{1+\Lambda_1^{-1}} + P_b T_h.$$

These are finite specified integrals of the prescribed profiles and the uniquely selected exact linear solution.

2.5 The evolving background inherited by layer two

On the affine part of the first wave, the complete background for the next layer is

$$G_{<2} = -A_0 R_\alpha e_2 + w_1 K_1 \Theta_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + w_1 \Omega_1 J \zeta_1 \otimes \zeta_1. \quad (46)$$

These are source (3.3) with $A_1 = -K_1 \Theta_1 |\zeta_1|$ and the physical units retained. Their complete derivatives are

$$\dot{G}_{<2} = \dot{\alpha} J G_{<2} + (K_1 \dot{w}_1 \Theta_1 + w_1 A_0 \sin s \Omega_1) \zeta_1, \quad (47)$$

$$\begin{aligned} \dot{D}_{<2} &= \ddot{\alpha} J + (\dot{w}_1 \Omega_1 + w_1 K_1 \zeta_{1,1} \Theta_1) J \zeta_1 \otimes \zeta_1 \\ &\quad + w_1 \Omega_1 \dot{\alpha} (-\zeta_1 \otimes \zeta_1 + J \zeta_1 \otimes J \zeta_1). \end{aligned} \quad (48)$$

Indeed $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$, $J \dot{\zeta}_1 = -\dot{\alpha} \zeta_1$, and the product rule gives each displayed term. In particular, using $J \zeta_1 \cdot G_{<2} = -A_0 \sin s$ gives the exact identity

$$\dot{G}_{<2} + D_{<2}^T G_{<2} = K_1 \dot{w}_1 \Theta_1 \zeta_1. \quad (49)$$

Thus the inherited gradient satisfies its true affine transport equation after activation, even while the first stage rotates it. The affine vorticity is $2\dot{\alpha} + w_1 \Omega_1$ and its scalar source is $G_{<2,1} = A_0 \sin \alpha + w_1 K_1 \Theta_1 \zeta_{1,1}$; therefore its exact vorticity residual is

$$(2\dot{\alpha} + w_1 \Omega_1)' - G_{<2,1} = 2\ddot{\alpha} - A_0 \sin \alpha + \dot{w}_1 \Omega_1. \quad (50)$$

At the selected endpoint define $A_1 = K_1 P_b > 0$. Then

$$G_{<2}(t_b) = (A_0 \sin s, -A_0 \cos s - A_1), \quad D_{<2}(t_b) = 0, \quad \sigma_1^2 = |G_{<2}(t_b)| = \sqrt{A_0^2 + A_1^2 + 2A_0 A_1 \cos s}. \quad (51)$$

Consequently

$$2eA_0 \lambda_1^{1/8} \leq A_1 \leq e^{2+\Lambda_1^{-1}} A_0 \lambda_1^{1/8}, \quad A_1 + A_0 \cos s \leq \sigma_1^2 \leq A_1 + A_0.$$

The lower norm bound is its positive vertical component in absolute value; the upper bound is the triangle inequality, which also follows by squaring the displayed expression. No horizontal component of the parent gradient has been discarded.

On the first hold the actual feedback is $F_1 = 0$. The endpoint state (40) is stationary under the original equations: $b_1 = K_1 \zeta_{1,1} = 0$ gives $\dot{\Omega}_1 = 0$, then $\dot{\Theta}_1 = a_1 \Omega_1 = 0$, and $\dot{\alpha} = 0$ gives $\dot{\zeta}_1 = 0$. Thus (51) remains the actual inherited background throughout this hold. For an arbitrary perturbation of the weighted amplitude pair its propagator over length T_h is exactly

$$\exp\left(T_h \begin{pmatrix} 0 & \Gamma \\ 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & \Gamma T_h \\ 0 & 1 \end{pmatrix}. \quad (52)$$

This follows by squaring the matrix and obtaining zero, and retains the nonzero first holding coefficient cost.

For clarity one can construct the next growth interval itself rather than prescribing an unspecified holding time. Retain $\lambda_2 = \lambda_1^{Q_2}$ with $Q_2 \geq 202$, an integer $k_2 \geq 0$, $L_2 = (k_2 + 41/8) \log \lambda_2$, and the source's

$$s_2 = L_2 \nu / \sigma_1, \quad K_2 = \mu \lambda_2, \quad \Gamma_2 = \sigma_1 \sin s_2.$$

An explicit sufficient finite choice ensuring $L_2 \geq 2$ and $0 < s_2 \leq 1/8$ is $\log \lambda_1 \geq 4096(k_2 + 41/8)Q_2$. To prove this, put $B = (k_2 + 41/8)Q_2$ and $x = \log \lambda_1$. The gain implies $\sigma_1 \geq \sqrt{2e\nu}e^{x/16}$; the exponential series gives $e^{x/16} \geq x^2/512$. Hence $s_2 \leq 512B/(\sqrt{2e}x) \leq 1/8$. Also $L_2 = Bx \geq 4096B^2 > 2$.

During the second growth, $\alpha = s$, $\Omega_1 = 0$, and $D_{<2} = 0$, so $M_2 = I$, $\zeta_2 = e(s_2)$, and the original coefficient formulas yield exactly

$$a_2 = \frac{A_1 \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \gamma_2 = \sqrt{[A_1 \sin s_2 + A_0 \sin(s + s_2)] \sin s_2} > 0. \quad (53)$$

Both signs are positive because $0 < s + s_2 \leq 1/4$. For its actual seed $S_2 = (A_0/\mu)\lambda_2^{-k_2-6}$ and entry vorticity $-\sqrt{b_2/a_2}S_2$, the exact solution is $P_2 = S_2 e^{\gamma_2(t-t_b)}$, $\Omega_2 = -\sqrt{b_2/a_2}P_2$. Substitution verifies both equations; the first threshold time is $t_{a,2} = t_b + L_2/\gamma_2$. Its activation is the source's $\eta(\Gamma_2(t - t_b))$. To see that activation completes before threshold, note

$$R := \frac{\gamma_2^2}{\Gamma_2^2} = \frac{A_1 + A_0 \cos s + A_0 \sin s \cot s_2}{\sigma_1^2}, \quad \cos s \leq R \leq 1 + \frac{\nu \sin s}{L_2 \sigma_1}. \quad (54)$$

For the lower bound, discard the positive cotangent term, use $\sigma_1^2 \leq A_1 + A_0$, and then $(A_1 + A_0 \cos s)/(A_1 + A_0) \geq \cos s$. For the upper bound use $A_1 + A_0 \cos s \leq \sigma_1^2$ and $\cot s_2 \leq 1/s_2$, the latter following from $(\sin r - r \cos r)' = r \sin r \geq 0$. Since $\sigma_1 \geq \nu$ and $L_2 \geq 2$, $R \leq 1 + \sin s/2 < 2$, and $\Gamma_2 L_2/\gamma_2 = L_2/\sqrt{R} > 1$. The exact first holding integral therefore is

$$H_1 = \Gamma \frac{L_2}{\gamma_2} = \sin s \frac{s_2}{\sin s_2} \frac{1}{\sqrt{R}}, \quad \frac{\sin s}{\sqrt{1 + \nu \sin s/(L_2 \sigma_1)}} \leq H_1 \leq \frac{\sin s}{\sqrt{\cos s(1 - s_2^2/6)}}. \quad (55)$$

Here $\sin r \leq r$ follows by integrating $\cos r \leq 1$; $\sin r \geq r - r^3/6$ follows by integrating $\cos r \geq 1 - r^2/2$, itself obtained from $1 - \cos r = \int_0^r \sin u \, du \leq r^2/2$. Thus no small-angle replacement is used in the bounds. The second transition also has $F_2 = 0$, because its only shear coefficient is $\Omega_1 = 0$; it prolongs the first pair's stationarity by exactly Γ_2^{-1} . The source counts that transition in the first layer's tail, not in H_1 .

2.6 Exact uncut forced fields and their retained diffusion cost

This subsection gives an explicit PDE realization of the just-proved finite stage on $\mathbb{R}^2$. Its fields have the spatial domain $\mathbb{R}^2$ and need not decay; compact spatial localization is a separate construction. Let F be any smooth odd 2π -periodic function. Define $Q_0(r) = \int_0^r F(v) \, dv$ and $\mathcal{Q}(r) = Q_0(r) - (2\pi)^{-1} \int_{-\pi}^{\pi} Q_0(v) \, dv$. Oddness gives zero period integral for F , so Q_0 is periodic; substitution $v = -u$ gives $Q_0(-r) = Q_0(r)$. Thus $\mathcal{Q}$ is even and mean-zero, and $\mathcal{Q}' = F$. Put $q(x, t) = K_1 \zeta_1(t) \cdot x$ and

$$\begin{aligned} \theta &= G_{<1} \cdot x + w_1 \Theta_1 F(q), \\ u &= \dot{\alpha} Jx + \frac{w_1 \Omega_1}{K_1} J \zeta_1 F(q), \end{aligned} \quad (56)$$

$$p = \frac{\dot{\alpha}^2}{2} |x|^2 + \frac{1}{2} x_2 (G_{<1} \cdot x) + \left(\frac{2w_1 \Omega_1 \dot{\alpha}}{K_1^2} + \frac{w_1 \Theta_1 \zeta_{1,2}}{K_1} \right) \mathcal{Q}(q). \quad (57)$$

An additive function of time can be subtracted if $p(0, t) = 0$ is desired. The exact inviscid forces are

$$f_\theta^0 = \dot{w}_1 \Theta_1 F(q), \quad f_u^0 = \left(\ddot{\alpha} - \frac{A_0}{2} \sin \alpha \right) Jx + \frac{\dot{w}_1 \Omega_1}{K_1} J\zeta_1 F(q). \quad (58)$$

We verify every component. The background derivative satisfies $\dot{G}_{<1} = \dot{\alpha} J G_{<1}$, hence its scalar transport residual vanishes. The phase satisfies $(\partial_t + \dot{\alpha} Jx \cdot \nabla)q = 0$. The wave velocity has scalar product zero with ∇q , so it advects neither its own temperature nor its own velocity profile. Its advection of the background temperature equals $(w_1 \Omega_1 / K_1)(J\zeta_1 \cdot G_{<1})F(q)$, which cancels $w_1 \dot{\Theta}_1 F(q)$ by (25). The remaining scalar residual is exactly $\dot{w}_1 \Theta_1 F(q)$.

For momentum, write $C = w_1 \Omega_1 / K_1$ and $v = C J\zeta_1 F(q)$. Differentiating v and adding its transport of $\dot{\alpha} Jx$ gives, after subtracting the wave buoyancy,

$$\left[\frac{\dot{w}_1 \Omega_1}{K_1} J\zeta_1 + \frac{w_1 \dot{\Omega}_1}{K_1} J\zeta_1 - 2C\dot{\alpha}\zeta_1 - w_1 \Theta_1 e_2 \right] F(q).$$

The last three vector terms have zero scalar product with $J\zeta_1$: this is $w_1 \dot{\Omega}_1 / K_1 - w_1 \Theta_1 \zeta_{1,1} = 0$. Their scalar product with ζ_1 is $-2C\dot{\alpha} - w_1 \Theta_1 \zeta_{1,2}$. Since $(\zeta_1, J\zeta_1)$ is an orthonormal basis, the wave pressure in (57) cancels precisely these three terms, leaving the displayed activation force. The base momentum residual before pressure is $(\ddot{\alpha} J - \dot{\alpha}^2 I - e_2 \otimes G_{<1})x$. The gradient of the first two pressure terms removes its symmetric part. Its antisymmetric part is $(\ddot{\alpha} - G_{<1,1}/2)Jx = (\ddot{\alpha} - A_0 \sin \alpha / 2)Jx$, as claimed. Finally $\text{div } u = 0$ since $\text{tr } J = 0$ and $J\zeta_1 \cdot \zeta_1 = 0$.

For the retained-viscosity equations

$$\partial_t \theta + u \cdot \nabla \theta = \kappa_{\text{phys}} \Delta \theta + f_\theta, \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu_{\text{phys}} \Delta u + \theta e_2 + f_u, \quad \nabla \cdot u = 0,$$

the very same state and controls solve them with the explicitly specified compensating forces

$$\begin{aligned} f_\theta &= f_\theta^0 - \kappa_{\text{phys}} w_1 \Theta_1 K_1^2 F''(q), \\ f_u &= f_u^0 - \nu_{\text{phys}} w_1 \Omega_1 K_1 J\zeta_1 F''(q). \end{aligned} \quad (59)$$

Indeed the affine Laplacians vanish, while two spatial derivatives of the profile multiply by $K_1^2 |\zeta_1|^2 = K_1^2$. This proves the asserted equations by substitution, without changing either ν or ν_{phys} and without assuming damping leaves the parent coefficients unchanged. The exact cumulative L_x^∞ norms of the two added forces through the entire stage and hold are

$$\kappa_{\text{phys}} K_1^2 \|F''\|_\infty I_P, \quad \nu_{\text{phys}} K_1 \|F''\|_\infty I_\Omega. \quad (60)$$

Equality holds because $x \mapsto K_1 \zeta_1 \cdot x$ is surjective onto $\mathbb{R}$ and $|J\zeta_1| = 1$; this verifies the norm statement as well as an upper bound. The scalar holding contribution is exactly $\kappa_{\text{phys}} K_1^2 \|F''\|_\infty P_b T_h$; the velocity holding contribution is zero. These finite costs are not asserted to be summable over the source's infinite frequency sequence.

The construction proves a finite controlled stage, its actual evolving affine background, its return, and its next growth holding interval. It does not establish an infinite viscous cascade or a smooth terminal-time viscous force. In particular, simply replacing the first temperature by an exponentially damped amplitude during holding would replace A_1 in (51) by a time-dependent quantity and would change both coefficients and the rate in (53). The compensating forces in (59) give an exact specified relation that retains the original coupled finite trajectory; their costs have been retained explicitly rather than claimed negligible.

3 Solving the next growth interval with a diffusing parent

This section continues an uncompensated equal-diffusion variant of the first stage through the next growth interval. It keeps the laboratory direction and the original coefficients. The exact amplitude system is realized below both on a forced affine parent and on an actual two-wave state with an explicitly computed interaction force. The latter force specifies the difference between the affine-gradient calculation and the full spatial equations.

3.1 The inherited gradient and the original second pair

Retain the preceding section's $A_0, \mu, \nu, K_1, s, \lambda_1$, and let $\nu_{\text{phys}} = \kappa_{\text{phys}} = d > 0$. For a sine first wave on its rotating affine base, the exact amplitude equations with retained diffusion are

$$\dot{\Theta}_{1,d} = a_1 \Omega_{1,d} - dK_1^2 \Theta_{1,d}, \quad \dot{\Omega}_{1,d} = b_1 \Theta_{1,d} - dK_1^2 \Omega_{1,d}.$$

If (Θ_1, Ω_1) denotes the first inviscid pair of the preceding section on its specified control and times, direct differentiation proves

$$(\Theta_{1,d}, \Omega_{1,d})(t) = e^{-dK_1^2(t-t_0)}(\Theta_1, \Omega_1)(t).$$

Both the forward map and its inverse multiplication by the positive exponential satisfy the indicated systems, so they are a bijection of solution spaces. In particular the selected vorticity still returns to zero at the specified t_b . This uses the inviscid times and does not assert the damped temperature crosses the same growth threshold. Its exact log-gain from the original seed is

$$\log(P_b/S) - \frac{dK_1^2}{\nu \sin s}(L + 2 + \Lambda_1^{-1}).$$

Set $P_{d,b} = e^{-dK_1^2(t_b-t_0)}P_b > 0$, $A_{1,d} = K_1 P_{d,b}$, and $r = t - t_b \geq 0$. During the fixed first holding control $\alpha = s$ the exact first pair is

$$\Theta_{1,d}(r) = -P_{d,b}e^{-qr}, \quad \Omega_{1,d} = 0, \quad q = dK_1^2, \quad G_d(r) = (A_0 \sin s, -A_0 \cos s - A_{1,d}e^{-qr}).$$

The gradient therefore changes at the rate $G'_d(r) = (0, qA_{1,d}e^{-qr})$. Let the new laboratory phase be $\zeta_2 = e(s_2)$ with $0 < s_2 \leq 1/8$, and let $K_2 = \mu\lambda_2 > 0$. The actual first hold has zero velocity gradient, so $\zeta'_2 = 0$. The second affine-wave coefficients are exactly

$$a_{2,d}(r) = -\frac{J\zeta_2 \cdot G_d(r)}{K_2} = \frac{A_{1,d}e^{-qr} \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \delta_2 = dK_2^2. \quad (61)$$

Their signs are positive because $0 < s + s_2 \leq 1/4$. The exact second sine amplitude system is

$$\Theta'_2 = a_{2,d}(r)\Omega_2 - \delta_2\Theta_2, \quad \Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2. \quad (62)$$

In particular the changing inherited gradient has not been replaced by its endpoint value. An affine realization alone would require the scalar force $G'_d(r) \cdot x$: the affine Laplacian is zero, so that force cannot be inferred to vanish from the old sine-wave diffusion.

3.2 Exact operator change and convergent fundamental solutions

Define

$$c_0 = A_0 \sin(s + s_2) \sin s_2 > 0, \quad c_1 = A_{1,d} \sin^2 s_2 > 0, \quad W(r) = e^{\delta_2 r} \Omega_2(r).$$

Because b_2 is constant, differentiating the second original equation gives the exact system equivalence

$$W'' = (c_0 + c_1 e^{-qr})W, \quad \Omega_2 = e^{-\delta_2 r} W, \quad \Theta_2 = \frac{e^{-\delta_2 r}}{b_2} W'. \quad (63)$$

Conversely these last two formulas give $\Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2$ and $\Theta'_2 = e^{-\delta_2 r} W''/b_2 - \delta_2\Theta_2 = a_{2,d}\Omega_2 - \delta_2\Theta_2$, since $b_2 a_{2,d} = c_0 + c_1 e^{-qr}$. Thus the initial-data map is the invertible map $(\Theta_2(0), \Omega_2(0)) \mapsto (W(0), W'(0)) = (\Omega_2(0), b_2\Theta_2(0))$.

Put

$$x(r) = x_0 e^{-qr/2}, \quad x_0 = \frac{2\sqrt{c_1}}{q}, \quad \vartheta = \frac{2\sqrt{c_0}}{q} > 0, \quad W(r) = \mathcal{W}(x(r)).$$

Here $x(r)$ is a scalar change of independent variable and is unrelated to either physical spatial coordinate x_1, x_2 , which remain fixed in the PDE below. Since $x' = -qx/2$ and $x'' = q^2x/4$, one has

$$W'' - (c_0 + c_1 e^{-qr})W = \frac{q^2}{4} [x^2 \mathcal{W}_{xx} + x \mathcal{W}_x - (x^2 + \vartheta^2) \mathcal{W}].$$

This proves the operator transformation including its nonzero factor. Rather than invoking a special-function name, define on $x > 0$

$$f(x) = x^\vartheta \sum_{n=0}^{\infty} a_n x^{2n}, \quad a_0 = 1, \quad a_n = \frac{a_{n-1}}{4n(n+\vartheta)}, \quad g(x) = f(x) \int_{x_0}^x \frac{d\xi}{\xi f(\xi)^2}. \quad (64)$$

For every compact interval in $x > 0$, the consecutive-term absolute ratio $x^2/[4n(n+\vartheta)]$ tends uniformly to zero. Every fixed derivative multiplies these terms by a polynomial in n and a bounded power of x^{-1} on that compact interval; its series therefore converges uniformly as well. Termwise substitution is valid. The coefficient of $x^{\vartheta+2n}$ in $x^2 f'' + x f' - (x^2 + \vartheta^2) f$ is $4n(n+\vartheta)a_n - a_{n-1} = 0$ for $n \geq 1$; the $n = 0$ coefficient is also zero. All coefficients are positive, so $f > 0$. The defining integral for g consequently exists on every compact subinterval of $x > 0$. If $I' = 1/(xf^2)$, differentiation yields

$$f(fI)' - f'(fI) = f^2 I' = 1/x.$$

Also $I''/I' = -1/x - 2f'/f$; substituting $g = fI$ into $g'' + g'/x - (1 + \vartheta^2/x^2)g$ leaves $fI'' + (2f' + f/x)I' = 0$. Thus g solves the same equation and its Wronskian with f is the nonvanishing function $1/x$. For completeness this proves completeness of the two solutions: at any $x > 0$ their value-and-derivative matrix is invertible, so one linear combination matches arbitrary Cauchy data; linear uniqueness then identifies that combination with any solution on overlapping compact subintervals of $(0, \infty)$.

For the original data let $W_0 = \Omega_2(0)$ and $W_1 = b_2 \Theta_2(0)$. The exact solution is

$$W(r) = c_f f(x(r)) + c_g g(x(r)), \quad c_f = \frac{W_0}{f(x_0)}, \quad c_g = -\frac{2f(x_0)}{q} \left(W_1 + \frac{qx_0}{2} c_f f'(x_0) \right). \quad (65)$$

Indeed $g(x_0) = 0$, $g'(x_0) = 1/[x_0 f(x_0)]$, and $x'(0) = -qx_0/2$ verify both initial conditions directly. For every finite r , $x(r) > 0$, so the displayed functions and their derivatives are finite there. If $d = 0$, the preceding substitutions dividing by q are not used: the original equation is $W'' = (c_0 + c_1)W$, with the exact solution $W = W_0 \cosh(\sqrt{c_0 + c_1} r) + W_1 \sinh(\sqrt{c_0 + c_1} r)/\sqrt{c_0 + c_1}$.

3.3 Signs and quantitative physical gain with this feedback

Let $\gamma_0 = \sqrt{c_0}$ and $\gamma_i = \sqrt{c_0 + c_1}$. Use the original instantaneous growing eigenline at entry: $\Theta_2(0) < 0$ and $\Omega_2(0) = \sqrt{b_2/a_{2,d}(0)} \Theta_2(0) < 0$. Then $Y = -W$ has $Y(0) = Y_0 > 0$ and $Y'(0) = \gamma_i Y_0 > 0$. As long as $Y > 0$, its equation gives $Y'' > 0$ and $Y' \geq Y'(0) > 0$; this excludes a first zero by continuity. Define

$$Y_- = Y_0 \left(\cosh(\gamma_0 r) + \frac{\gamma_i}{\gamma_0} \sinh(\gamma_0 r) \right), \quad Y_+ = Y_0 e^{\gamma_i r}.$$

The differences have zero initial value and derivative and satisfy

$$\begin{aligned} (Y - Y_-)'' - \gamma_0^2 (Y - Y_-) &= c_1 e^{-qr} Y, \\ (Y_+ - Y)'' - \gamma_i^2 (Y_+ - Y) &= c_1 (1 - e^{-qr}) Y. \end{aligned}$$

For $a > 0$, the formula $Z(r) = \int_0^r \sinh(a(r-u)) S(u)/a \, du$ has $Z(0) = Z'(0) = 0$ and $Z'' - a^2 Z = S$, by two differentiations. Uniqueness proves it is the solution of that initial-value problem. Both sources above are nonnegative; both this kernel and its first r derivative are nonnegative for $0 \leq$

$u \leq r$. Therefore $Y_- \leq Y \leq Y_+$ and $Y'_- \leq Y' \leq Y'_+$ for every finite $r \geq 0$. The inverse physical map now gives both amplitudes negative and the precise temperature-gain bounds

$$e^{-\delta_2 r} \left(\cosh(\gamma_0 r) + \frac{\gamma_0}{\gamma_i} \sinh(\gamma_0 r) \right) \leq \frac{-\Theta_2(r)}{-\Theta_2(0)} \leq e^{(\gamma_i - \delta_2)r}. \quad (66)$$

In particular, if the actual numerical coefficients satisfy $\delta_2 \geq \gamma_i$, this particular growth control never raises the second temperature above its entry amplitude. This is a proved obstruction to that specified uncompensated control, not a statement about other controls or the existence of an infinite viscous sequence.

3.4 The exact map to spatial fields and its interaction force

The affine parent $\theta_b = G_d(r) \cdot x$, $u_b = 0$ is a solution with pressure $p_b = x_2 G_d(r) \cdot x/2$ and forces $f_{\theta,b} = G'_d(r) \cdot x$ and $f_{u,b} = -(A_0 \sin s/2)Jx$. The scalar statement follows from $\Delta\theta_b = 0$. For momentum, direct differentiation gives $\nabla p_b - \theta_b e_2 = (G_d x_2 - e_2(G_d \cdot x))/2 = -G_{d,1}Jx/2$. An added sine wave whose amplitudes solve (62) has zero additional PDE residual except for its activation, by the same explicit phase, transverse-vector, and pressure calculations below. Thus the solved system has an exact forced affine-background realization.

There is also a full two-wave realization specifying the original spatial discrepancy. Let F_1, F_2 be smooth odd periodic profiles, each with mean-zero primitive $Q'_j = F_j$, and assume $F'_1(0) = 1$. This includes both sine and the released profile which is linear near zero. Set

$$G_0 = (A_0 \sin s, -A_0 \cos s), \quad T(r) = -P_{d,b}e^{-qr}, \quad q_1 = K_1 x_2, \quad q_2 = K_2 \zeta_2 \cdot x,$$

and let $w_2(r)$ be any specified smooth activation, for example the source's $\eta(\Gamma_2 r)$. Define on $\mathbb{R}^2$

$$\begin{aligned} \theta &= G_0 \cdot x + T F_1(q_1) + w_2 \Theta_2 F_2(q_2), \\ u &= \frac{w_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2), \\ p &= \frac{1}{2} x_2 (G_0 \cdot x) + \frac{T}{K_1} Q_1(q_1) + \frac{w_2 \Theta_2 \zeta_{2,2}}{K_2} Q_2(q_2). \end{aligned} \quad (67)$$

Their complete retained-diffusion forces are

$$\begin{aligned} f_\theta &= -dK_1^2 T [F_1(q_1) + F_1''(q_1)] + w'_2 \Theta_2 F_2(q_2) \\ &\quad - dK_2^2 w_2 \Theta_2 [F_2(q_2) + F_2''(q_2)] + f_{\text{cross}}, \end{aligned} \quad (68)$$

$$f_{\text{cross}} = \frac{w_2 \Omega_2}{K_2} \sin s_2 K_1 T [F'_1(q_1) - 1] F_2(q_2), \quad (69)$$

$$f_u = -\frac{A_0 \sin s}{2} Jx + \frac{w'_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2) - dK_2 w_2 \Omega_2 J \zeta_2 [F_2(q_2) + F_2''(q_2)]. \quad (70)$$

Here is a direct verification. The old wave has derivative $T' = -dK_1^2 T$ and Laplacian $K_1^2 T F_1''$, giving the first scalar force. The old temperature gradient differs from the coefficient gradient $G_d = G_0 + K_1 T e_2$ by precisely $K_1 T [F'_1(q_1) - 1] e_2$. Its advection by u is the displayed f_{cross} because $J \zeta_2 \cdot e_2 = \sin s_2$. The remainder of that old-temperature advection is $(w_2 \Omega_2 / K_2) J \zeta_2 \cdot G_d F_2 = -w_2 a_{2,d} \Omega_2 F_2$; it cancels the coupling part of $w_2 \Theta'_2 F_2$. The new wave advects its own temperature and velocity by zero since $J \zeta_2 \cdot \zeta_2 = 0$. Its amplitude damping and its Laplacian give the two new-profile terms in the forces. The old wave pressure has gradient $T F_1(q_1) e_2$, so it cancels old wave buoyancy exactly. The new pressure has gradient $w_2 \Theta_2 \zeta_{2,2} \zeta_2 F_2$. The new momentum coupling vector before that pressure is $w_2 \Theta_2 (\zeta_{2,1} J \zeta_2 - e_2) F_2 = -w_2 \Theta_2 \zeta_{2,2} \zeta_2 F_2$, by resolving e_2 in the orthonormal basis $(\zeta_2, J \zeta_2)$. Hence it cancels as well. The base pressure supplies exactly the first vector-force term, as proved above, and divergence is zero by transversality. This verifies both full equations including all interactions.

For the sine profiles, $F_j + F_j'' = 0$ identically. The remaining scalar interaction force is then exactly

$$f_{\text{cross}} = -\frac{A_{1,d}e^{-qr}}{K_2}w_2\Omega_2 \sin s_2 [\cos(K_1x_2) - 1] \sin(q_2), \quad \|f_{\text{cross}}(r)\|_\infty \leq \frac{2A_{1,d}e^{-qr}}{K_2}|w_2\Omega_2| \sin s_2.$$

The bound follows from $|\cos u - 1| \leq 2$ and $|\sin v| \leq 1$. For the released linear-core profile the same explicit formulas hold, and $F_1' - 1 = 0$ on its original linear interval, so the interaction force vanishes there. Its old-profile diffusion defect is instead $-dK_1^2Tq_1$ on that interval, since $F_1(q_1) = q_1$ and $F_1''(q_1) = 0$. Its gradient is exactly $G_d'(r)$. Thus exponential damping of that released linear core requires the stated core force, and is not an unforced consequence of its zero Laplacian. The explicit forces, inverse solution maps, and convergent formulas above carry the feedback calculation to an actual spatial solution without conflating the sine state, its affine gradient, and the released profile. No compact-support or infinite-iteration claim is made in this section.

4 Reaching both original growth targets with retained diffusion

We now join the exact evolving-parent growth calculation to the direct second return theorem. Both stopping events are the original temperature thresholds. This changes the first growth time from the earlier fixed-inviscid-time variant in Section 3. The construction below specifies the frequencies and a nonempty interval of positive physical diffusivities, and proves that their actual entry states satisfy every inequality in Theorem 2. The spatial fields and their full forces are supplied by the compact construction following that theorem. In particular the amplitude damping on the released linear profile has its explicitly stated core force.

4.1 A finite choice within the original frequency schedule

Retain the original physical data

$$A_0 \geq 1, \quad \nu = \sqrt{A_0}, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil,$$

and all fixed source profiles, including $s_*, s_0, C_\ell, \eta, \eta_2, \Lambda_1$. Thus $0 < s_* \leq 1/8$, $C_\ell \geq 4$, $H = \|\eta_2\|_\infty \geq 1$, $\Lambda_1 \geq 6H$, and $3\Lambda_1 H \sin s_* \leq 1/4$. The first section's λ_1 denoted a source frequency; in this section we write that same quantity as $\hat{\lambda}_1$ and retain the physical frequencies separately as $K_j = \mu\hat{\lambda}_j$. Keep the source's $Q_* \geq 200$, $Q_j = Q_* + j$, its specified finite constants $K(k)$, and the exact choices

$$\begin{aligned} \hat{\lambda}_2 &= \hat{\lambda}_1^{Q_2}, & k_j &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_j, K(k) \leq \log \hat{\lambda}_j\}, \\ L_j &= (k_j + 41/8) \log \hat{\lambda}_j, & S_j &= \frac{A_0}{\mu} \hat{\lambda}_j^{-k_j-6}, \\ P_j^* &= \frac{A_0}{\mu} \hat{\lambda}_j^{-7/8} = S_j e^{L_j}. \end{aligned} \tag{71}$$

Here $K(k)$ is the source's prescribed parameter sequence, not the physical frequency K_j . The maximum is a nonempty finite set once $\log \hat{\lambda}_1 \geq K(0)$: $k = 0$ belongs to it and $k \leq \lfloor Q_j/120 \rfloor$. No smooth dependence of k_j on the frequencies is required.

The following fixed constants give one explicit sufficient frequency choice; their large numerical size is immaterial to finite existence:

$$\begin{aligned} C_A &= e^{2+\Lambda_1^{-1}}, & C_\sigma &= \sqrt{1+C_A}, & C_N &= 1 + 3C_\sigma/32, \\ C_L &= (\lfloor Q_2/120 \rfloor + 41/8)Q_2, \\ Y_0 &= 1 + \max \left\{ \begin{array}{l} 64, K(0), Q_2, 72 \cdot 16^3 H C_L^2 / \sqrt{e}, \quad 8 \log(4/e), \\ (16/47) \log(4C_\ell C_N / s_0), \quad (16/31) \log(2C_N / s_0) \end{array} \right\}. \end{aligned} \tag{72}$$

Choose any $\widehat{\lambda}_1 \geq e^{Y_0}$ and then use (71) literally. Write $L = L_2$, $\Gamma_1 = \nu \sin s_*$ and $C_{\text{post}} = 2 + \Lambda_1^{-1}$. Define the explicit positive number

$$d_{\max} = \min \left\{ \frac{\Gamma_1}{2K_1^2}, \frac{\Gamma_1 \log 2}{K_1^2 C_{\text{post}}}, \frac{\nu \log(3/2)}{16K_1^2}, \frac{L\nu \log 3}{12800K_2^2} \right\}. \quad (73)$$

Every entry in this minimum uses only the fixed original data and the chosen frequencies. In particular it does not require solving a circular inequality involving a yet unknown diffusive growth scale. All subsequent statements hold for each $0 < d \leq d_{\max}$, with physical viscosity and thermal diffusivity both equal to d .

4.2 First actual threshold and the changed first endpoint

Put $q = dK_1^2$ and $t_1^{\text{in}} = 1/\nu$. Growth and transition still have $\alpha = 0$. The signed first pair is exactly

$$\Theta_1(t) = -S_1 e^{(\Gamma_1 - q)(t - t_1^{\text{in}})}, \quad \Omega_1(t) = \frac{K_1}{\nu} \Theta_1(t). \quad (74)$$

Indeed its temperature coupling is $a_1 = A_0 \sin s_*/K_1$, its vorticity coupling is $b_1 = K_1 \sin s_*$, and $a_1 K_1/\nu = \Gamma_1$, $b_1 \nu/K_1 = \Gamma_1$. Differentiation verifies both damped equations and both original seed values. The linear integral equation proves uniqueness. Since $q \leq \Gamma_1/2$, the original first threshold is attained uniquely and transversely at

$$t_1^a = t_1^{\text{in}} + \frac{L_1}{\Gamma_1 - q}, \quad t_1^1 = t_1^a + \Gamma_1^{-1}, \quad t_1^b = t_1^1 + \Gamma_1^{-1}(1 + \Lambda_1^{-1}). \quad (75)$$

The first activation $\eta(\Gamma_1(t - t_1^{\text{in}}))$ is already one at t_1^a , since $L_1 \geq 1$.

Starting at t_1^a , take the original first transition and steering control with their displayed durations, translated to these actual times. The inverse map

$$(\Theta_1, \Omega_1)(t) = e^{-q(t - t_1^a)} (\Theta_{1,E}, \Omega_{1,E})(t)$$

relates their damped solution to the inviscid controlled solution with data

$$(\Theta_{1,E}, \Omega_{1,E})(t_1^a) = (-P_1^*, -K_1 P_1^*/\nu).$$

Differentiation gives the two damped equations, and multiplication by $e^{q(t - t_1^a)}$ gives the inverse equations and identical data. Both factors are positive. Thus the quotient, its entire trial family, and its unique first return root proved in Section 2 are unchanged on these translated intervals. At the selected endpoint

$$\alpha = s_*, \quad \zeta_1 = e_2, \quad \Omega_1 = 0, \quad A_d := -K_1 \Theta_1(t_1^b) = A_E e^{-qC_{\text{post}}/\Gamma_1}, \quad (76)$$

where the proved inviscid endpoint satisfies

$$2eA_0 \widehat{\lambda}_1^{1/8} \leq A_E \leq C_A A_0 \widehat{\lambda}_1^{1/8}. \quad (77)$$

This expression does not damp the inviscid growth duration: that duration has been replaced by the actual threshold time in (75). The first three diffusion exponents are exactly $qL_1/(\Gamma_1 - q)$ in growth, q/Γ_1 in transition, and $q(1 + \Lambda_1^{-1})/\Gamma_1$ in steering. The physical gains are respectively L_1 , $1 - q/\Gamma_1$, and the original selected steering gain minus the last exponent.

4.3 Actual second scale and all fixed design inequalities

The true inherited gradient at $t_2^{\text{in}} = t_1^b$ is $G_d = (A_0 \sin s_*, -A_0 \cos s_* - A_d)$. Evaluate the source scale at this changed endpoint, as required by its original rule:

$$\sigma^2 = |G_d| = \sqrt{A_0^2 + A_d^2 + 2A_0 A_d \cos s_*}, \quad s = s_2 = \frac{L\nu}{\sigma}, \quad a = \sin s, \quad \Gamma = \Gamma_2 = \sigma a. \quad (78)$$

Since $qC_{\text{post}}/\Gamma_1 \leq \log 2$, equations (77)–(78) imply

$$eA_0\hat{\lambda}_1^{1/8} \leq A_d \leq C_A A_0\hat{\lambda}_1^{1/8}, \quad \sqrt{e}\nu\hat{\lambda}_1^{1/16} \leq \sigma \leq C_\sigma\nu\hat{\lambda}_1^{1/16}. \quad (79)$$

For the lower scale bound use $\sigma^2 \geq A_d$, since the square of its left side differs from A_d^2 by a positive quantity. For the upper bound use the triangle inequality $\sigma^2 \leq A_0 + A_d \leq (1 + C_A)A_0\hat{\lambda}_1^{1/8}$. The choice $Y_0 > 8 \log(4/e)$ also gives $A_d > 4A_0$.

Let $y = \log \hat{\lambda}_1 \geq Y_0$. Then $64 < L \leq C_L y$. The exponential series with nonnegative terms proves $e^{y/16} \geq y^3/(6 \cdot 16^3)$; by (72) its right side is strictly greater than $12HC_L^2 y^2/\sqrt{e}$. Consequently

$$0 < s \leq \frac{L}{\sqrt{e} e^{y/16}} < \frac{1}{12HL} \leq \frac{1}{8}, \quad 0 < a \leq \frac{1}{8}, \quad 3LHa < \frac{1}{4}. \quad (80)$$

For $0 \leq s \leq 1/8$, integrating $\cos r \geq 1 - r^2/2 \geq 1/2$ on $[0, s]$ gives $s/2 \leq \sin s \leq s$. Thus

$$\frac{L\nu}{2} \leq \Gamma \leq L\nu. \quad (81)$$

The fixed base coefficients in the second-return coordinates obey

$$0 < C_0 = \frac{A_0 \cos s_*}{\sigma^2} < \frac{1}{4}, \quad 0 < C_1 = \frac{A_0 \sin s_*}{\sigma^2 a} \leq \frac{2\nu \sin s_*}{\sigma L} < \frac{1}{32}. \quad (82)$$

The last inequality uses $\sigma \geq \nu$, $L \geq 64$ and $\sin s_* < 1$; it has no missing comparison constant.

The same frequency choice proves the source support nesting for the entire second return and its finite hold. Set $N = 1 + (3 - \cos s)/a$. The exact matrix reconstruction in Theorem 2 gives both $\|M_2\|$ and $\|M_2^{-1}\|$ at most N , by the triangle inequality for its displayed shear. Equations (79) and $a \geq s/2$ imply $1/a \leq 2C_\sigma\hat{\lambda}_1^{1/16}/L$, whence $N \leq C_N\hat{\lambda}_1^{1/16}$. The final two logarithmic bounds in (72) give, strictly,

$$N\hat{\lambda}_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\hat{\lambda}_1^{-2} < s_0. \quad (83)$$

Indeed their respective left sides are at most $C_N e^{-47y/16}$ and $C_N e^{-31y/16}$, and our choices bound these by $s_0/(4C_\ell)$ and $s_0/2$. For $\ell_1 = s_0/(C_\ell\mu)$ and $\ell_2 = \hat{\lambda}_1^{-3}/\mu$, this places the entire second support inside the first envelope's constant region and its exact linear profile region: $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1\zeta_1 \cdot x| \leq K_1 N\ell_2 < s_0$. Every original factor of μ is present in these identities.

4.4 Second growth with the actual diffusing parent

Write $r = t - t_2^{\text{in}}$. The second growth and transition have $\alpha = s_*$, $\Omega_1 = 0$, $M_2 = I$, $\zeta_2 = e(s)$ and $A_1(r) = A_d e^{-qr}$. These claims follow from the closed subsystem: $\zeta_{1,1} = 0$ makes $\dot{\Omega}_1 = -q\Omega_1 = 0$; both controls F_1^{ctl} and F_2^{ctl} are then zero, and the phase matrix equation is $\dot{M}_2 = 0$. The first temperature equation is $\dot{A}_1 = -qA_1$. The second pair has the exact coefficients

$$a_2(r) = \frac{A_d e^{-qr} \sin s + A_0 \sin(s_* + s)}{K_2}, \quad b_2 = K_2 \sin s, \quad d_2^{\text{phys}} = dK_2^2, \quad (84)$$

and the exact original entry data $\Theta_2(0) = -S_2$, $\Omega_2(0) = \sqrt{b_2/a_2(0)}\Theta_2(0)$. This is precisely the system solved by the convergent fundamental functions in Section 3; the substitution here uses its displayed changed coefficients, not frozen values.

Put $a_i = a_2(0)$, $u_i = \sqrt{b_2/a_i}$, $\gamma_i = \sqrt{a_i b_2}$ and

$$n_i = \frac{\gamma_i^2}{\Gamma^2} = \frac{A_d + A_0 \cos s_*}{\sigma^2} + C_1 \cos s.$$

One has

$$\frac{4}{5} < n_i < \frac{33}{32} < 2. \quad (85)$$

The lower bound follows from $A_d/\sigma^2 \geq A_d/(A_d + A_0) > 4/5$. For the upper bound, square $\sigma^2 = \sqrt{(A_d + A_0 \cos s_*)^2 + A_0^2 \sin^2 s_*}$ to get $(A_d + A_0 \cos s_*)/\sigma^2 \leq 1$, then use (82). In particular $\gamma_i \geq \Gamma/\sqrt{2} \geq L\nu/(2\sqrt{2})$.

On the explicitly fixed horizon $0 \leq r \leq 16/\nu$, the third bound in (73) gives $qr \leq \log(3/2)$. Since both terms in a_2 are positive, it follows that

$$\frac{1}{2}a_i \leq a_2(r) \leq a_i. \quad (86)$$

To prove all signs and ratio estimates without assuming them, set $X = -e^{d_2^{\text{phys}} r} \Theta_2$ and $Y = -e^{d_2^{\text{phys}} r} \Omega_2$. They solve $X' = a_2 Y$, $Y' = b_2 X$ with positive initial values. Before any first zero, both derivatives are positive, excluding that zero by continuity. The linear integral equation, whose coefficient matrix is bounded on this horizon, gives existence there by its uniformly convergent iterated-integral series. Therefore $u = Y/X = \Omega_2/\Theta_2$ exists and satisfies $u' = b_2 - a_2 u^2$. The constant functions u_i and $\sqrt{2}u_i$ are respectively subsolution and supersolution on this horizon, because $b_2 - a_2 u_i^2 \geq 0$ and $b_2 - 2a_2 u_i^2 \leq 0$. Subtracting either comparison equation from the equation for u gives the linear difference coefficient $-a_2(u + u_{\text{comparison}})$; its integrating factor preserves the stated ordering. Hence

$$u_i \leq u \leq \sqrt{2}u_i, \quad \frac{1}{\sqrt{n_i}} \leq v := \frac{\sigma \Omega_2}{K_2 \Theta_2} \leq \sqrt{\frac{2}{n_i}}, \quad \frac{1}{2} < v < 2. \quad (87)$$

Here $\sigma u_i/K_2 = \Gamma/\gamma_i = 1/\sqrt{n_i}$; the equality follows by substituting the original a_i, b_2, K_2 .

For $P_2 = -\Theta_2$, the exact physical logarithmic derivative is

$$(\log P_2)' = a_2 u - d_2^{\text{phys}}.$$

The last bound in (73) gives $d_2^{\text{phys}} \leq L\nu \log 3/12800 < L\nu/16 \leq \gamma_i/4$. Thus throughout the fixed horizon

$$\frac{\gamma_i}{4} \leq (\log P_2)' \leq \sqrt{2}\gamma_i \leq 2\Gamma. \quad (88)$$

Indeed $a_2 u \geq a_i u_i/2 = \gamma_i/2$ and $a_2 u \leq \sqrt{2}a_i u_i = \sqrt{2}\gamma_i$. Consequently the original second target $P_2^* = S_2 e^L$ is reached at exactly one $r = r_a$, with nonzero derivative, and

$$\frac{L}{2\Gamma} \leq r_a \leq \frac{4L}{\gamma_i} \leq \frac{8\sqrt{2}}{\nu}, \quad r_1 = r_a + \Gamma^{-1} < \frac{16}{\nu}, \quad t_2^a = t_2^{\text{in}} + r_a, \quad t_2^1 = t_2^{\text{in}} + r_1. \quad (89)$$

For existence of the crossing, the upper candidate time is inside the fixed horizon; integrating the lower rate to that time gives at least L logarithmic gain. Strict monotonicity and the intermediate value theorem give existence and uniqueness. The transition length is at most $2/(L\nu) \leq 1/(32\nu)$, so the entire transition is also inside that already justified horizon. The lower bound on r_a ensures the second activation $\eta(\Gamma r)$ equals one before the growth threshold, since $L \geq 64$.

4.5 Closing every entry bound and the complete finite statement

At the actual steering entry t_2^1 , take $P_{1\text{in}} = A_d e^{-qr_1}/K_1$ and $P_{2\text{in}} = P_2(r_1)$. Equations (87) and (89) give the initial quotient interval in Theorem 2. Also

$$\frac{8}{15} < B_{\text{in}} = \frac{A_d e^{-qr_1}}{\sigma^2} \leq 1, \quad 0 < \delta_2 = \frac{dK_2^2}{\Gamma} \leq \frac{\log 3}{6400}, \quad 0 < \delta_1 = \frac{dK_1^2}{\Gamma} \leq \delta_2 < \frac{1}{16}. \quad (90)$$

The lower parent bound is $(4/5)(2/3)$; the upper is $A_d \leq \sigma^2$. The diffusion bounds follow from (73), (81) and $K_1 < K_2$. Equations (80), (82), (87), and (90) prove every entry/design hypothesis of the direct return theorem from the original seeds and the explicit choices (72)–(73).

Theorem 1. *For each fixed original source datum and profiles above, choose $\hat{\lambda}_1$ by (72), retain the source recursion (71), and choose any $0 < d \leq d_{\text{max}}$ in (73). The complete modified two-layer coefficient system exists through both original growth targets, both prescribed transitions, both selected vorticity returns, and a second holding interval of physical length $1/(32\Gamma_2)$. Its times, amplitudes, original coordinates, controls, and spatial support inequalities are the exact formulas above and the inverse formulas of Theorem 2. At the second return the new vorticity amplitude and horizontal phase component are zero, the first parent vorticity is strictly positive, and*

$$\log \frac{-\Theta_2(t)}{S_2} \geq L_2 + \frac{\gamma_i}{4\Gamma_2} + \frac{\log 3}{320} \quad \text{for } t_2^b \leq t \leq t_2^b + (32\Gamma_2)^{-1}. \quad (91)$$

The inherited parent is evolved throughout; no inviscid estimate is assumed to persist under diffusion. With the full profile/localization forces below these coefficients determine compact spatial solutions from the original initial datum and zero initial velocity.

Proof. The first stage was proved by the exact map at its actual threshold. The second growth and transition, their signs and all entry bounds were proved on a common explicit physical horizon. Theorem 2 therefore applies, constructs its entire trial family before taking the least root, and proves the return and nonstationary parent hold. The second growth contributes exactly L_2 to logarithmic gain. The transition contributes at least $\gamma_i/(4\Gamma_2)$ by (88). The steering and entire hold contribute at least $\log 3/320$ by that theorem and (90). Adding these three actual integrals proves (91). All finite join controls are smooth by flatness of the original step and interior support of its pulse; the solution of the smooth ODE therefore matches to every order at those joins. The spatial identities below prove the stated compact realization with every required force. $\square$

For use in those spatial bounds, throughout both stages and their hold one may use the following proved amplitude upper bounds:

$$\begin{aligned} \sup |\Theta_1| &\leq \max\{C_A P_1^*, 2\sigma^2/K_1\}, & \sup |\Omega_1| &\leq \max\{(K_1/\nu)C_A P_1^*, 9\sigma\}, \\ \sup |\Theta_2| &\leq e^{18} P_2^*, & \sup |\Omega_2| &\leq (3K_2/\sigma)e^{18} P_2^*. \end{aligned} \quad (92)$$

For layer one, use its first-stage positive multiplier bounds and the direct second-stage bounds $B \leq 2, W < 9$. For layer two, growth never exceeds its threshold; the transition has dimensionless coefficient $a_2 K_2/(\sigma\Gamma) = a_2 b_2/\Gamma^2 \leq n_i < 2$ and $v < 2$, so its logarithmic gain is at most 4. The selected steering gain is at most $12(1 + L^{-1}) < 14$, and holding only decreases temperature. These observations prove $e^{18} P_2^*$ as well as the vorticity bound from $0 \leq v \leq 3$ on the selected steering, $v < 2$ before it, and $v = 0$ on holding.

The exact second-stage diffusion exponents before steering are $dK_2^2 r_a$ and dK_2^2/Γ ; the first parent's exponent over these same intervals is $q(r_a + 1/\Gamma)$. The steering and holding exponents retain the evolving $|\zeta_2|^2$ inside the integrals in Theorem 2; together these are the complete two-stage amplitude diffusion costs.

Finally the proved viscosity range is explicitly frequency dependent. For fixed original data and Q_2 ,

$$0 < d_{\text{max}} \leq \frac{\nu C_L \log 3}{12800\mu^2} (\log \hat{\lambda}_1) \hat{\lambda}_1^{-2Q_2} \longrightarrow 0 \quad (\hat{\lambda}_1 \longrightarrow \infty). \quad (93)$$

To verify the limit, set $y = \log \hat{\lambda}_1$ and use $e^{2Q_2 y} \geq (2Q_2 y)^2/2$, giving $ye^{-2Q_2 y} \leq 1/(2Q_2^2 y) \rightarrow 0$. This is the dependence of the range proved here, not an assertion that every larger diffusivity fails or that a modified infinite construction is impossible. The actual finite result preserves both original thresholds and controls. A fixed-positive-diffusivity iteration would require further estimates or a changed scale design; it does not follow by iterating a range whose upper bound tends to zero.

4.6 An exact scaling restriction for this growth rule

The original datum ties the physical spatial scale to the initial amplitude: $\mu = \lceil \nu/2 \rceil$ with $\nu = \sqrt{A_0} \geq 1$. Consequently

$$\frac{\mu^2}{\nu} \geq \frac{1}{2}, \quad \hat{d} = d \frac{\mu^2}{\nu} \geq \frac{d}{2}, \quad (94)$$

where $\hat{d}$ is exactly the transformed diffusion coefficient in the original map $(X, T) = (\mu x, \nu t)$. To prove the first inequality, if $1 \leq \nu \leq 2$ then $\mu = 1$ and $\mu^2/\nu = 1/\nu \geq 1/2$. If $\nu > 2$, then $\mu \geq \nu/2$, whence $\mu^2/\nu \geq \nu/4 > 1/2$. Equality occurs at the original datum value $A_0 = 4$, $\nu = 2$, $\mu = 1$; this identifies the equality case and does not change the parameters of any constructed trajectory. The second inequality follows by multiplication by $d > 0$, using the exact diffusion operator map.

For the first modified growth rule proved here, attaining $P_1^* = S_1 e^{L_1} > S_1$ requires and is equivalent to $dK_1^2 < \Gamma_1$. This follows directly from (74): at equality the temperature amplitude is constant, and above it the amplitude strictly decreases for all growth times. Therefore every instance of this same growth rule must satisfy

$$\hat{d} \hat{\lambda}_1^2 < \sin s_*, \quad d \lambda_1^2 < 2 \sin s_*. \quad (95)$$

The first inequality is the identical physical inequality divided by ν , since $K_1 = \mu \hat{\lambda}_1$; the second follows from (94). Thus increasing A_0 while preserving the original spatial scaling cannot make $\hat{d}$ tend to zero at a fixed positive physical d . This necessary first-growth restriction is a result about the specified modified amplitude equation and original growth control. Any route past it must change an identified part of that dynamics or scale design and verify the corresponding full spatial forces. It is not an obstruction to all forced viscous flows.

5 A direct two-layer return theorem with positive physical diffusivity

This section proves a finite controlled return for the actual two-layer coefficient system, including the evolving first parent. It uses the released Alpöge–Buckmaster steering profile from Section 3.6.3, pp. 18–22, and its identification in Section 3.7.6, pp. 31–32. The estimates below are proved directly for the modified equations; no estimate for an inviscid trajectory is transferred to a diffusive one. The spatial realization of these amplitude equations requires the explicit profile and localization forces calculated in the companion sections. This section makes no assertion about an infinite sequence.

5.1 Original parameters, entry state, and control

Keep $A_0 \geq 1$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$, and the paper's time-scaling parameter $\nu_{AB} = \sqrt{A_0}$. The physical frequencies are $K_i = \mu \hat{\lambda}_i$, $i = 1, 2$. The prescribed source parameters s_* , $L = L_2$ and the source profiles η, η_2 are retained. Thus $0 < s_* \leq 1/8$, η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $\eta_2 \in C_c^\infty((0, 1))$ is nonnegative with $\int_0^1 \eta_2 = 1$. Put $H = \|\eta_2\|_\infty$. Physical viscosity and thermal diffusivity are both $d \geq 0$. They are distinct from ν_{AB} , μ , and the shooting parameter $m \in [0, 3]$.

Let t_1 be the actual second-layer steering-entry time. Its data are

$$\begin{aligned}\alpha(t_1) &= s_*, & \zeta_1(t_1) &= e_2, & \Omega_1(t_1) &= 0, & \Theta_1(t_1) &= -P_{1\text{in}} < 0, \\ M_2(t_1) &= I, & \zeta_2(t_1) &= e(s), & \Theta_2(t_1) &= -P_{2\text{in}} < 0, \\ \Omega_2(t_1) &= -\frac{K_2}{\sigma} P_{2\text{in}} v_{\text{in}}.\end{aligned}\tag{96}$$

Here $e(s) = (\sin s, \cos s)$, $\sigma > 0$ is the actual inherited scale, $s = L\nu_{\text{AB}}/\sigma$ is the actual source second angle, and $\Gamma = \sigma \sin s$. In particular σ, s, Γ are not reset to fixed values if the preceding diffusive stage changes them. The data $P_{1\text{in}}, P_{2\text{in}}, v_{\text{in}}$ are the actual values after growth and transition. Theorem 1 proves from both original seeds and both actual threshold crossings that these values satisfy the explicit entry margins below, for its displayed nonempty range of positive d . No coefficient comparison during steering is assumed.

For compact notation define the following exact coordinate change:

$$\begin{aligned}a &= \sin s, & h_0 &= \cos s = \sqrt{1 - a^2}, & B_{\text{in}} &= \frac{K_1 P_{1\text{in}}}{\sigma^2}, \\ C_0 &= \frac{A_0 \cos s_*}{\sigma^2}, & C_1 &= \frac{A_0 \sin s_*}{\sigma^2 a}, \\ \delta_1 &= \frac{dK_1^2}{\Gamma}, & \delta_2 &= \frac{dK_2^2}{\Gamma}, & \tau &= \Gamma(t - t_1).\end{aligned}\tag{97}$$

Every factor used as a divisor in this change is displayed and positive; physical time is recovered by $t = t_1 + \tau/\Gamma$. Set $T = 1 + L^{-1}$, $T_H = T + 1/32$, and use the exact control

$$z_m(\tau) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -mL\eta_2(L(\tau - 1)), & 1 \leq \tau \leq T, \\ 0, & T \leq \tau \leq T_H. \end{cases} \quad Z_m(t) = az_m(\Gamma(t - t_1)).\tag{98}$$

Flatness and interior support make these definitions smooth at both joins. This is exactly the source pulse, with source pulse amplitude $mL \sin s$, entry first component a , and entry phase length one.

5.2 Exact closed equations and their inverse map

Write $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The two phase gradients have the exact signed determinant $\det(\zeta_1, \zeta_2) = -a$. Since $|\zeta_1| = 1$, put

$$\begin{aligned}h &= \zeta_1 \cdot \zeta_2, & D_* &= h^2 + a^2 = |\zeta_2|^2, & Y &= \sqrt{D_* - a^2 z_m^2}, \\ B &= \frac{-K_1 \Theta_1}{\sigma^2}, & W &= \frac{\Omega_1}{\sigma}, & n &= B + C_0 + C_1 h.\end{aligned}$$

The symbol D_* is a positive scalar, not the parent velocity matrix. The square root selects the positive second component of ζ_2 . The complete older/geometry system in τ is

$$\begin{aligned}h' &= W, & h(0) &= h_0, \\ W' &= B \frac{Y - h z_m}{D_*} - \delta_1 W, & W(0) &= 0, \\ B' &= -C_1 W - \delta_1 B, & B(0) &= B_{\text{in}}, \\ n' &= -\delta_1 B, & n(0) &= B_{\text{in}} + C_0 + C_1 h_0.\end{aligned}\tag{99}$$

The last equation is a proved consequence of the preceding ones, not an additional independent constraint. Define $k = n/D_*$. The newest pair is recovered from

$$\begin{aligned}v' &= z_m - kv^2, & v(0) &= v_{\text{in}}, \\ P_2 &= P_{2\text{in}} \exp\left(\int_0^\tau (kv - \delta_2 D_*) dr\right), \\ \Theta_2 &= -P_2, & \Omega_2 &= -\frac{K_2}{\sigma} P_2 v.\end{aligned}\tag{100}$$

Here P_2 is recovered only on an interval where v has been proved finite. The proof below constructs that interval before using this map.

For clarity, the full geometric inverse and the older physical state are

$$\begin{aligned}\zeta_2 &= (az_m, Y), & \zeta_1 &= \left(\frac{haz_m - aY}{D_*}, \frac{hY + a^2z_m}{D_*} \right), \\ \phi_2 &= \arcsin(az_m/\sqrt{D_*}), & \phi_1 &= \phi_2 - \arctan(a/h), & \alpha &= s_* - \phi_1, \\ M_1 &= R_\alpha, & M_2 &= R_{\alpha-s_*} \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}, & \Theta_1 &= -\sigma^2 B/K_1, & \Omega_1 &= \sigma W.\end{aligned}\tag{101}$$

The expression for M_1 uses its original first-activation datum $M_1 = I$ when $\alpha = 0$ and $p_1 = e(s_*)$; it is not an artificial reset of the first deformation at t_1 .

Proof of the exact equations and inverse. The actual parent is

$$G = -A_0 R_\alpha e_2 - K_1 P_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1, \quad P_1 = -\Theta_1.$$

The first coefficient is $a_1 = A_0 \sin s_*/K_1$, because $\phi_1 + \alpha = s_*$. Therefore the diffusive first pair gives $\dot{A}_1 = -A_0 \sin s_* \Omega_1 - dK_1^2 A_1$ and $\dot{\Omega}_1 = -A_1 \sin \phi_1 - dK_1^2 \Omega_1$, where $A_1 = K_1 P_1$. The phase equation gives $\dot{h} = a\Omega_1$. The determinant representation is $\zeta_2 = h\zeta_1 - aJ\zeta_1$; solving its two components gives (101) and $-\sin \phi_1 = a(Y - hz_m)/D_*$. Substitution and division by the nonzero physical factors in (97) prove the first three equations in (99). Differentiating $B + C_0 + C_1 h$ proves the exact cancellation $n' = -\delta_1 B$.

The full second coefficient retains the original base term:

$$a_2 = \frac{aA_1 + A_0(a \cos s_* + h \sin s_*)}{K_2 D_*} = \frac{\sigma^2 a n}{K_2 D_*}, \quad b_2 = K_2 a z_m.$$

In particular the numerator changes when the first temperature diffuses; it has not been held fixed. The equations $\dot{\Theta}_2 = a_2 \Omega_2 - dK_2^2 D_* \Theta_2$ and $\dot{\Omega}_2 = b_2 \Theta_2 - dK_2^2 D_* \Omega_2$ give exactly (100) by differentiation. Conversely differentiating its exponential and product proves these two physical equations, so the coordinate map and its inverse preserve the complete dynamics.

For the matrices, write $S = \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}$. Then $\dot{S} = \Omega_1 (J e_2 \otimes e_2) S$ because $\dot{h} = a\Omega_1$. Differentiating $R_{\alpha-s_*} S$ proves $\dot{M}_2 = D_{<2} M_2$ and $M_2(t_1) = I$. Furthermore $S^{-T}(a, h_0) = (a, h)$, proving $M_2^{-T} e(s) = \zeta_2$ with its original oriented datum. Both matrices have determinant one. Differentiating $\zeta_{2,1} = Z_m$ in its phase equation proves the exact control

$$\dot{\alpha} = F_2^{\text{ctl}} - \frac{\dot{Z}_m}{Y}, \quad F_2^{\text{ctl}} = -\frac{\Omega_1 \zeta_{1,1} (J \zeta_1 \cdot \zeta_2)}{Y} = \frac{a \Omega_1 \zeta_{1,1}}{Y}.$$

Thus no extra angular control has been substituted for the source feedback. All divisions and inverse trigonometric branches are justified by the bounds proved next. $\square$

5.3 An explicit finite parameter range

Theorem 2. *Suppose the numerical entry and design parameters in (96)–(97) satisfy*

$$\begin{aligned}0 < s &\leq \arcsin(1/8), & 0 < a &\leq \frac{1}{8}, & L &\geq 64, & 3LHa &\leq \frac{1}{4}, \\ \frac{1}{2} &\leq B_{\text{in}} \leq 2, & 0 < C_0 &\leq \frac{1}{4}, & 0 < C_1 &\leq \frac{1}{32}, \\ \frac{1}{2} &\leq v_{\text{in}} \leq 2, & 0 &\leq \delta_1 \leq \frac{1}{16}.\end{aligned}\tag{102}$$

These are explicit inequalities on actual entry data and fixed design numbers, not hypotheses on a subsequently unknown trajectory. The displayed range of s fixes the original positive-cosine branch;

the inequality on $\sin s$ alone would not fix that branch. Then the entire family $m \in [0, 3]$ exists through $\tau = T$, with $P_2 > 0$, and its endpoint map $m \mapsto v(T; m)$ is continuous. It satisfies the quantitative endpoint margins

$$v(T; 0) \geq \frac{1}{7}, \quad v(T; 3) \leq -\frac{1}{480}. \quad (103)$$

The zero set is a nonempty compact subset of $(0, 3)$ and hence has a least element m_* . Every zero satisfies

$$\frac{13}{96} \leq m_* \leq 3 - \frac{1}{480}, \quad W(T; m_*) \geq \frac{13h_0}{3840} \exp(-\delta_1/L) > 0. \quad (104)$$

No monotonicity of the endpoint map or uniqueness of the root is claimed.

The selected state continues through the nonzero holding interval $[T, T_H]$, with $\zeta_{2,1} = \Omega_2 = 0$ and with the first parent still evolving according to (99). On the whole selected interval $[0, T_H]$,

$$\begin{aligned} h_0 \leq h < 3, \quad 0 \leq W < 9, \quad \frac{191}{512} < B \leq 2, \quad \frac{1}{4} < n < 3, \\ \frac{63}{64} \leq D_* < 10, \quad Y \geq \frac{\sqrt{59}}{8}, \quad \zeta_{1,2} > \frac{1}{16}, \quad \frac{1}{40} < k < 4. \end{aligned} \quad (105)$$

The same geometry bounds hold throughout every trial through T . Every trial has $|v| < 6$ through T , so its linear amplitude pair never loses the negative sign of Θ_2 .

The exact selected logarithmic gain at every $\tau_f \in [T, T_H]$ is

$$\log \frac{P_2(\tau_f)}{P_{2\text{in}}} = \int_0^T kv \, d\tau - \delta_2 \int_0^{\tau_f} D_* \, d\tau, \quad \frac{\log 3}{160} - 10\delta_2\tau_f \leq \log \frac{P_2(\tau_f)}{P_{2\text{in}}} \leq 12T. \quad (106)$$

In particular the strictly positive physical range

$$0 < d \leq \min \left\{ \frac{\Gamma}{16K_1^2}, \frac{\Gamma \log 3}{6400K_2^2} \right\} \quad (107)$$

gives $\log(P_2(\tau_f)/P_{2\text{in}}) \geq \log 3/320 > 0$. The interval is expressed in actual physical units. Theorem 1 supplies an explicit positive $d_{\max}$ using only the original data and chosen frequencies, and proves (102) and (107) for every $0 < d \leq d_{\max}$ while retaining the dependence of the actual entry data and Γ on d .

Proof. First construct geometry without using the newest amplitude pair. On an initial interval where $B > 0$, the bracket $Y - hz_m$ is nonnegative. If $0 \leq z_m \leq 1$, this follows by squaring: $Y^2 - h^2 z_m^2 = D_*(1 - z_m^2) \geq 0$; if $z_m \leq 0$, it follows from $Y, h > 0$. Consequently the integrating-factor formula for W gives $W \geq 0$. Therefore $h \geq h_0$ and $B' = -C_1 W - \delta_1 B \leq 0$, giving $B \leq B_{\text{in}}$. The control bound is $|az_m| \leq 1/4$, so

$$Y^2 \geq h_0^2 - \frac{1}{16} \geq \frac{59}{64} > 0.$$

For $0 \leq \tau \leq 1$, $Y - hz_m \leq \sqrt{D_*}$, hence $W' \leq B_{\text{in}}/h_0$ and $h \leq h_0 + B_{\text{in}}\tau^2/(2h_0)$. On the negative pulse the same argument gives

$$W' \leq \frac{B_{\text{in}}}{h_0} \{1 + mL\eta_2(L(\tau - 1))\}.$$

During holding its right side is B_{in}/h_0 . Integration on any interval up to T_H therefore yields

$$W \leq \frac{B_{\text{in}}}{h_0} (T_H + 3) \leq \frac{26}{3} < 9,$$

because $h_0 \geq 15/16$, $T_H \leq 17/16$, and $B_{\text{in}} \leq 2$. The duration after the first part is at most $L^{-1} + 1/32 \leq 3/64$, so

$$h \leq h_0 + \frac{B_{\text{in}}}{2h_0} + \frac{B_{\text{in}}}{h_0}(T_H + 3)\frac{3}{64} \leq 1 + \frac{16}{15} + \frac{13}{32} = \frac{1187}{480} < 3.$$

No quotient variable has entered these estimates.

The exact integrating-factor identity for the first temperature is

$$B(\tau) = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr.$$

Since $W \geq 0$, $h' = W$, $\delta_1 T_H \leq 17/256$, and $e^{-x} \geq 1 - x$ for $x \geq 0$, it follows that

$$B(\tau) \geq \frac{1}{2} \left(1 - \frac{17}{256} \right) - \frac{1}{32}(h(\tau) - h_0) > \frac{239}{512} - \frac{3}{32} = \frac{191}{512} > \frac{1}{4}.$$

This strict estimate excludes a first exit through $B = 0$. All square-root denominators stay uniformly positive and h, W, B remain bounded. The smooth finite-dimensional vector field is therefore bounded with bounded first derivatives on a compact neighborhood of every such solution. Local existence, uniqueness and the integral equation extend the solution through T_H ; the same neighborhood can be chosen uniformly for $m \in [0, 3]$. Differentiating that integral equation and applying the elementary Gronwall estimate proves continuous dependence on m on this interval. This constructs the entire older/geometry trial family before the newest pair.

Now $n = B + C_0 + C_1 h > 1/4$, and $n' = -\delta_1 B \leq 0$, so $n \leq B_{\text{in}} + C_0 + C_1 h_0 \leq 73/32 < 3$. The asserted D_* and k bounds follow. For the first phase, $|az_m| \leq 1/4$, $a \leq 1/8$, $h \geq 15/16$ and $Y > 7/8$ give

$$\zeta_{1,2} = \frac{hY + a^2 z_m}{D_*} > \frac{(15/16)(7/8) - 1/32}{10} = \frac{101}{1280} > \frac{1}{16}.$$

Thus the branches and control denominators in (101) stay in their specified domains. This proves all geometry claims.

Construct the Riccati variable next. On $[0, 1]$, its forcing $z_m = 1 - \eta$ is nonnegative. The reciprocal differential inequality $(1/v)' \leq 4$, initially while $v > 0$, gives

$$\frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} \leq v(\tau) \leq v_{\text{in}} + \tau \leq 3. \quad (108)$$

The positive lower bound rules out a first zero, and the upper bound follows by integrating $v' \leq 1$. These bounds also exclude finite escape. Put $v_d = v(1)$. In particular $v_d \geq 1/6$. Moreover

$$\int_0^1 k v^2 d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}^2}{(1 + 4v_{\text{in}}\tau)^2} d\tau = \frac{v_{\text{in}}^2}{40(1 + 4v_{\text{in}})} \geq \frac{1}{480}.$$

The last inequality follows because $x^2/(1 + 4x)$ is increasing for $x > 0$, as its derivative is $(2x + 4x^2)/(1 + 4x)^2 > 0$. Integrating the Riccati equation proves

$$\frac{1}{6} \leq v_d \leq 3 - \frac{1}{480}. \quad (109)$$

On the pulse put $V(y) = v(1 + y/L)$ and $M_2(y) = \int_0^y \eta_2(r) dr$. Its exact integral equation is

$$V(y) = v_d - m M_2(y) - \frac{1}{L} \int_0^y k(1 + r/L) V(r)^2 dr.$$

Under the temporary bound $|V| \leq 6$, it gives $V \leq v_d < 3$ and $V \geq -3 - 144/L \geq -21/4 > -6$. The bound is strict initially and strictly improves itself at every possible first exit, so $|V| < 6$ on the full pulse. The quotient has therefore been constructed with a uniform finite bound, and

the exponential formula (100) constructs a negative Θ_2 solving the entire original linear pair. Its uniqueness follows by subtracting two solutions of that linear pair and applying the integral Gronwall inequality to their zero initial difference. In particular the construction agrees with the unique linear-pair solution and has not assumed absence of a temperature zero.

For $m = 0$, the pulse solves $v' = -kv^2$, so $v(T; 0) \geq v_d/(1 + 4v_d/L) \geq 1/(6 + 4/L) > 1/7$. For $m = 3$, the integral equation and (109) give $v(T; 3) \leq v_d - 3 \leq -1/480$. The uniformly finite quotient and geometry imply continuous dependence of $v(T; m)$ on m by their integral equations. The intermediate value theorem proves existence of a zero. Its zero set is closed in $[0, 3]$ and excludes both endpoints, proving the stated compactness and least-root selection.

At any selected zero, $v' = -mL\eta_2 - kv^2 \leq 0$ throughout the pulse. Thus $0 \leq v \leq v_d$ there, and

$$m = v_d - \int_1^T kv^2 d\tau \in \left[v_d - \frac{4v_d^2}{L}, v_d \right] \subseteq \left[\frac{13}{96}, 3 - \frac{1}{480} \right].$$

For the lower inclusion use $v_d \leq 3$, $L \geq 64$ and $v_d \geq 1/6$. On the pulse,

$$W' + \delta_1 W = B \frac{Y + hmL\eta_2}{D_*} \geq \frac{h_0}{40} mL\eta_2 (L(\tau - 1)).$$

Its integrating-factor formula, $W(1) \geq 0$ and the unit integral of η_2 prove (104). On holding, $z_m = 0$ and $v(T) = 0$, so uniqueness gives $v = 0$. The older geometry continues by the already established bounds, and (100) gives the exact remaining temperature decay.

Finally, (108) gives

$$\int_0^1 kv d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} d\tau = \frac{1}{160} \log(1 + 4v_{\text{in}}) \geq \frac{\log 3}{160}.$$

The selected pulse adds nonnegative gain, the hold adds zero gain, and $v \leq 3$, $k < 4$ bound the gain above by $12T$. The exact diffusion exponent is at most $10\delta_2\tau_f$. This proves (106); substitution of (107) gives its positive lower bound. $\square$

5.4 Returned state, holding dynamics, and diffusion costs

At the selected physical return time $t_b = t_1 + T/\Gamma$ one has

$$\zeta_2(t_b) = (0, \sqrt{D_*(T)}), \quad \Omega_2(t_b) = 0, \quad \zeta_1(t_b) = \frac{(-a, h(T))}{\sqrt{D_*(T)}}, \quad \alpha(t_b) = s_* + \arctan(a/h(T)),$$

and $\Theta_1(t_b) = -\sigma^2 B(T)/K_1 < 0$, $\Omega_1(t_b) = \sigma W(T) > 0$. The first parent is therefore neither frozen nor returned to zero. Its full motion creates the specified returned state of the second wave. The controlled holding interval has physical duration $1/(32\Gamma)$. On it the complete system is

$$h' = W, \quad W' = B/\sqrt{h^2 + a^2} - \delta_1 W, \quad B' = -C_1 W - \delta_1 B, \\ P_2(\tau) = P_2(T) \exp\left(-\delta_2 \int_T^\tau (h(r)^2 + a^2) dr\right).$$

These formulas retain the evolving phase length inside the diffusion integral. They also prove the exact stationarity of $\zeta_{2,1}, \Omega_2$ throughout this finite hold.

The returned parent vorticity also gives positive scalar curl of the full local velocity at the origin. Indeed $z_m = z'_m = 0$ on holding, so the proved control and inverse give the exact physical formulas

$$\dot{\alpha} = -\frac{a^2 \Omega_1}{h^2 + a^2}, \quad \omega(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2}\right) \geq \frac{31}{32} \Omega_1 > 0. \quad (110)$$

For the inequality use $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a^2 \leq 1/64$. The positive forcing in the holding equation for W gives $W(\tau) \geq W(T)e^{-\delta_1(\tau-T)} > 0$ by its integrating factor. These statements

include the return endpoint because the original pulse is identically zero near that endpoint. The compact spatial construction has constant envelopes and exact linear profiles near the origin, so its curl there is precisely the quantity in (110); no cutoff derivative enters it.

The total first-wave damping exponents on the ramp, pulse and hold are δ_1 , δ_1/L and $\delta_1/32$, respectively. The corresponding second-wave exponents are exactly

$$\delta_2 \int_0^1 D_* d\tau, \quad \delta_2 \int_1^T D_* d\tau, \quad \delta_2 \int_T^{T_H} D_* d\tau.$$

Each exponent lies between $(63/64)\delta_2$ and $10\delta_2$ times its displayed interval length. In original physical coordinates these equal $dK_2^2 \int |\zeta_2(t)|^2 dt$, with $K_2 = \mu \hat{\lambda}_2$ and $dt = d\tau/\Gamma$; none of the factors μ , ν_{AB} , K_i or Γ has been dropped. The exact first-amplitude identity in the proof also records the additional parent-response term $-C_1 \int e^{-\delta_1(\tau-r)} W(r) dr$; a pure common-damping multiplier would miss that term.

For the parent affine fields the altered gradient is not independently prescribed. With $B_1 = -\sigma^2 B_{\zeta_1}$, direct differentiation gives

$$\dot{G} + D_{<2}^T G = -dK_1^2 B_1, \quad C = 2\dot{\alpha} + \Omega_1, \quad \dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1.$$

Thus $G \cdot x$, $D_{<2} x$ have scalar force $(-dK_1^2 B_1) \cdot x$ and full vector force $\frac{1}{2}(2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1) Jx$ with pressure $-\frac{1}{2}x^T \text{sym}(\dot{D}_{<2} + D_{<2}^2 - e_2 \otimes G)x$. Their Laplacians vanish. These affine formulas are local exact fields; the compact original-profile realization and its additional spatial forces are supplied by the accompanying finite-profile construction. They must not be discarded when interpreting the amplitude result as a statement about the full spatial equations.

Here is a direct verification of the affine force and pressure, including their signs. Put $D = D_{<2}$, $u(x, t) = D(t)x$ and $\theta(x, t) = G(t) \cdot x$. Both summands in D have zero trace, so $\text{div } u = 0$ and $D^2 = -\det(D)I$ by multiplication of a general trace-free 2×2 matrix. The scalar residual is exactly $(\dot{G} + D^T G) \cdot x$. In that derivative the term $K_1 \dot{\Theta}_1 \zeta_1$ cancels $(\Omega_1 J \zeta_1 \otimes \zeta_1)^T (-A_0 R_\alpha e_2)$, apart from $-dK_1^2 B_1$; the angular terms cancel because $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$ and $\partial_t(R_\alpha e_2) = \dot{\alpha} J R_\alpha e_2$. Also $\zeta_1 \cdot J \zeta_1 = 0$ eliminates the self-interaction. For momentum let $\mathcal{A} = \dot{D} + D^2 - e_2 \otimes G$. Then $\partial_t u + (u \cdot \nabla)u - \theta e_2 = \mathcal{A}x$, whereas the displayed pressure contributes $-\text{sym}(\mathcal{A})x$. Their sum is $\text{skew}(\mathcal{A})x$. The scalar curl of Dx is $C = D_{21} - D_{12}$; since D^2 is scalar and the curl of $(G \cdot x)e_2$ is G_1 ,

$$\text{skew}(\mathcal{A}) = \frac{\dot{C} - G_1}{2} J.$$

Finally $\dot{\Omega}_1 = -K_1 P_1 \zeta_{1,1} - dK_1^2 \Omega_1$ and $G_1 = A_0 \sin \alpha - K_1 P_1 \zeta_{1,1}$ give precisely the force stated above. Both affine Laplacians are zero by direct second differentiation. Thus these are the full vector equations with the specified pressure, not merely a vorticity residual.

6 Compact spatial realization of the changed two-stage trajectory

We now realize the actual damped two-stage trajectory, including its changed first threshold, inherited scale, second threshold, selected second return, and evolving finite hold, in the full spatial equations. The coefficients used here are those constructed in the direct return theorem and Theorem 1. The present calculation supplies their entire spatial realization; in particular it does not interpret a modal damping equation as an unforced equation for the released profile. The physical equations and orientation are

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta \theta + f_\theta, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, & \text{div } u &= 0, & p &= 0, \\ J &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & e_2 &= (0, 1), & x &\in \mathbb{R}^2. \end{aligned}$$

Both physical coefficients equal the explicitly selected $d > 0$. The original parameters remain

$$A_0 \geq 1, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad \nu = \sqrt{A_0}, \quad K_j = \mu \lambda_j \quad (j = 1, 2).$$

Thus d is not the source time-scaling parameter ν . Here λ_j denotes exactly the source frequency written $\hat{\lambda}_j$ in the preceding entry calculation.

6.1 The actual coefficients and the nested material regions

Write $e(s) = (\sin s, \cos s)$, $e_0 = R_\alpha e_2$, $p_1 = e(s_*)$, and $p_2 = e(s)$, where $s = L_2 \nu / \sigma$, $a = \sin s$, $h_0 = \cos s$, and $\Gamma = \sigma a$ use the actual diffusive inherited scale σ . The activation functions are

$$w_1 = \eta(\Gamma_1(t - t_{01})), \quad w_2 = \eta(\Gamma(t - t_{02})), \quad \Gamma_1 = \nu \sin s_*, \quad t_{01} = 1/\nu.$$

The source step η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$. All inactive increments below are defined to be zero. The finite coefficient construction has the exact equations

$$G_{<j} = -A_0 e_0 + \sum_{i < j} w_i K_i \Theta_i \zeta_i, \quad D_{<j} = \dot{\alpha} J + \sum_{i < j} \frac{w_i \Omega_i}{r_i^2} J \zeta_i \otimes \zeta_i, \quad (111)$$

$$\begin{aligned} \dot{M}_j &= D_{<j} M_j, & M_j(t_{0j}) &= I, & \zeta_j &= M_j^{-T} p_j, \quad r_j^2 = |\zeta_j|^2, \\ \dot{\Theta}_j &= -\frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j - d K_j^2 r_j^2 \Theta_j, & \dot{\Omega}_j &= K_j \zeta_{j,1} \Theta_j - d K_j^2 r_j^2 \Omega_j. \end{aligned} \quad (112)$$

The common angle is the selected original first control followed by the exact second feedback control. In particular the first threshold time is $t_{01} + L_1/(\Gamma_1 - d K_1^2)$, not the inviscid threshold time. The source first transition and pulse are translated to that actual threshold. During second growth and transition, $\alpha = s_*$, $M_2 = I$, $\Omega_1 = 0$, and Θ_1 diffuses. The second activation time is exactly $t_{02} = t_{b1}$, the actual first return time, with no intervening gap. During second steering and holding the exact inverse is

$$\begin{aligned} M_1 &= R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - h_0)/a \\ 0 & 1 \end{pmatrix}, \quad \Omega_1 = \sigma W, & \Theta_1 &= -\sigma^2 B / K_1, \\ h' &= W, \quad W' = B \frac{Y - hz}{h^2 + a^2} - \delta_1 W, & B' &= -C_1 W - \delta_1 B, \quad Y = \sqrt{h^2 + a^2 - a^2 z^2}, \end{aligned} \quad (113)$$

where $\tau = \Gamma(t - t_1)$, a prime denotes $d/d\tau$, $\delta_1 = d K_1^2 / \Gamma$, $C_1 = A_0 \sin s_*/(\sigma^2 a)$, and z is the selected source ramp, pulse, and zero hold. The proved bounds give

$$h_0 \leq h < 3, \quad 0 \leq W < 9, \quad 1/4 < B \leq 2, \quad h_0 \geq \sqrt{63}/8, \quad r_1 = 1, \quad h_0 \leq r_2 < \sqrt{10}.$$

These are actual changing coefficients, not independently chosen affine backgrounds. Every $D_{<j}$ has trace zero: the rotation has trace zero and $\text{tr}(J \zeta_i \otimes \zeta_i) = \zeta_i \cdot J \zeta_i = 0$. Jacobi's determinant formula therefore gives $\det M_j = 1$. The phase equation follows by differentiating the inverse transpose: $\dot{\zeta}_j = -D_{<j}^T \zeta_j$.

On $[-\pi, \pi]$ define the released profile by the following formula and then extend it periodically to $\mathbb{R}$:

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad P' = F, \quad \int_{-\pi}^{\pi} P(s) ds = 0.$$

Here χ is even, supported in $(-\pi, \pi)$, and equals one on $[-s_0, s_0]$, with $0 < s_0 < 1$. Oddness and periodicity give a unique even periodic primitive P with the specified mean. In particular $F(s) = s$ on $|s| \leq s_0$; its primitive there is $P(0) + s^2/2$, with its original constant $P(0)$ retained. Let $f \in C_c^\infty(B_1)$ be radial, $0 \leq f \leq 1$, and $f = 1$ on $B_{1/2}$. Retain the source radii

$$\ell_1 = \frac{s_0}{C_\ell \mu}, \quad \ell_2 = \frac{\lambda_1^{-3}}{\mu}, \quad C_\ell \geq 4.$$

Define the material coordinates and envelopes

$$y_j = M_j^{-1}x, \quad g_j(x, t) = f(y_j/\ell_j), \quad s_j = K_j \zeta_j \cdot x.$$

With $N = 1 + (3 - h_0)/a$, the matrix expression (113) and $\|I + qE_{12}\| \leq 1 + |q|$ show

$$\|M_1\| = \|M_1^{-1}\| = 1, \quad \|M_2\|, \|M_2^{-1}\| \leq N. \quad (114)$$

The second inequality also holds before steering, where $M_2 = I$. The entry construction, (83), selects λ_1 with the explicit strict inequalities

$$N\lambda_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\lambda_1^{-2} < s_0. \quad (115)$$

Consequently every point of the second support satisfies $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1\zeta_1 \cdot x| \leq K_1N\ell_2 < s_0$. It therefore lies in the exact affine cell of the first wave for the whole construction. The first support lies in B_{ℓ_1} ; since $\mu\ell_1 = s_0/C_\ell < s_0$, the base is exactly affine there for every angle. Strictness leaves an open neighborhood of each support on which the corresponding affine identities hold.

6.2 Original datum, fields, and complete residual forces

Retain the original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, equal to one on B_2 , and the exact datum

$$\theta_{\text{in}}(x) = -\frac{A_0}{\mu} \sin(\mu x_2) \chi_0(x), \quad u_{\text{in}} = 0.$$

Choose the radial $H_0 \in C_c^\infty(B_{R_H})$ used in the base construction, where $R_H > R_0 + 1$ and $H_0 = |x|^2/2$ on B_{R_0+1} . Put

$$\begin{aligned} \theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x) \chi_0(x), & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H_0, & u_b &= J\nabla \psi_b. \end{aligned} \quad (116)$$

The angle is zero before $1/\nu$; the initial interpolation is therefore exact and the initial velocity is zero. Define

$$\begin{aligned} T_j &= w_j \Theta_j, & Q_j &= \frac{w_j \Omega_j}{K_j^2 r_j^2}, & V_j &= T_j F(s_j) g_j, & \Psi_j &= Q_j P(s_j) g_j, \\ v_j &= J\nabla \Psi_j, & \theta &= \theta_b + V_1 + V_2, & u &= u_b + v_1 + v_2. \end{aligned} \quad (117)$$

The flat activation functions make the zero extensions of the increments smooth, with all spatial and temporal derivatives, at $t = t_{0j}$. The velocity is divergence free because mixed partial derivatives commute.

Hereafter fix a layer and temporarily suppress its index. Write $k = K\zeta$, $\rho = |k|^2 = K^2 r^2$, $D = D_{<j}$, $G = G_{<j}$, and $b = k_1$. All functions of one variable in the formulas below are evaluated at $s = k \cdot x$. Define

$$Q_* = \frac{\dot{w}\Omega + wb\Theta}{\rho} - Q \frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D^T k. \quad (118)$$

Thus $\dot{Q} = Q_* - d\rho Q$, directly from (112), without a constant-phase-length assumption. The complete added scalar and vector forces are

$$\begin{aligned} E_\theta &= \dot{w}\Theta Fg + QP(J\nabla g \cdot G) + QT(F^2 - PF')g(Jk \cdot \nabla g) \\ &\quad - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F\Delta g\}, \end{aligned} \quad (119)$$

$$E_u = 2Dv + Q_*J\{kFg + P\nabla g\} + (\nabla v)v - TFge_2 - d(\rho v + \Delta v). \quad (120)$$

These are full vector forces at the declared pressure $p = 0$. For completely componentwise evaluation one uses

$$v = QJ\{kFg + P\nabla g\},$$

$$\nabla v = QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \quad (121)$$

$$\Delta V = T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (122)$$

$$\Delta v = QJ\{\rho kF''g + \rho F'\nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + Fk\Delta g + P\nabla\Delta g\}. \quad (123)$$

Thus every phase, envelope, and mixed Laplacian term is displayed. The base forces are

$$f_{\theta,b} = \nu\eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}) - d\Delta\theta_b,$$

$$f_{u,b} = \ddot{\alpha}J\nabla H_0 + \dot{\alpha}^2(\nabla(J\nabla H_0))J\nabla H_0 - \theta_b e_2 - d\dot{\alpha}J\nabla\Delta H_0. \quad (124)$$

The requested total forces are exactly

$$f_\theta = f_{\theta,b} + E_{\theta,1} + E_{\theta,2}, \quad f_u = f_{u,b} + E_{u,1} + E_{u,2}. \quad (125)$$

Proof of the full spatial equations. For the material derivative $\mathcal{D} = \partial_t + Dx \cdot \nabla$, differentiating $M^{-1}x$ gives $\mathcal{D}(M^{-1}x) = 0$. The phase equation gives $\mathcal{D}(k \cdot x) = 0$ as well. Consequently $\mathcal{D}g = 0$, $\mathcal{D}V = \dot{T}Fg$, and $\mathcal{D}\Psi = \dot{Q}Pg$. On this support the actual older temperature and velocity are $G \cdot x$ and Dx by (115); in particular their gradient and Jacobian are exactly G, D , even while these change. The temperature advection increment is therefore $\dot{T}Fg + v \cdot G + v \cdot \nabla V$. Using $Jk \cdot k = 0$ and $J\nabla g \cdot \nabla g = 0$ gives

$$v \cdot \nabla V = QT(F^2 - PF')g(Jk \cdot \nabla g), \quad v \cdot G = QFg(Jk \cdot G) + QP(J\nabla g \cdot G).$$

Since $\dot{\Theta} = -(Jk \cdot G)\Omega/\rho - d\rho\Theta$, the first term in $v \cdot G$ cancels the coupled part of $\dot{T}Fg$ exactly. Subtraction of $d\Delta V$, computed by the product rule, proves (119).

For every trace-free two by two matrix, direct multiplication gives $DJ + JD^T = 0$. Differentiation of a gradient under $\mathcal{D}$ therefore gives

$$\mathcal{D}(J\nabla\Psi) = DJ\nabla\Psi + J\nabla(\mathcal{D}\Psi).$$

The other linear advection increment is $(v \cdot \nabla)(Dx) = Dv$. Adding it to this formula, adding the self-advection $(\nabla v)v$, and subtracting $Ve_2 + d\Delta v$ proves (120), after using $\dot{Q} = Q_* - d\rho Q$. Differentiating $\Delta\Psi = Q(\rho F'g + 2Fk \cdot \nabla g + P\Delta g)$ and applying J proves all six groups in (123).

The base phase $\mu e_0 \cdot x$ is transported by $\dot{\alpha}Jx$, and its radial cutoff is also transported. Before $1/\nu$ its velocity is zero. This proves the scalar base formula. Differentiating $\dot{\alpha}J\nabla H_0$ proves the vector base formula directly on all of $\mathbb{R}^2$. Finally, adding increments in the order 1, 2 accounts exactly for both directions of each interaction. In the second increment, $\mathcal{D}$ contains $u_b + v_1$, and the terms $v_2 \cdot G_{<2}$ and $D_{<2}v_2$ contain the reverse interactions with that full older state. Outside an increment's support all its derivatives vanish. Thus the residual identities telescope globally to (125), with no omitted interaction and no inverse curl or pressure reconstruction. $\square$

The material-coordinate form makes the original metric equally explicit. Set $C = M^{-1}M^{-T}$ and $\tilde{g}(y) = f(y/\ell)$. The pullback of the spatial derivative and Laplacian on a function of (s, y) are

$$\mathcal{P} = K\zeta\partial_s + M^{-T}\nabla_y, \quad \mathcal{L} = K^2r^2\partial_s^2 + 2K(Cp \cdot \nabla_y)\partial_s + \sum_{l,m} C_{lm}\partial_{y_l}\partial_{y_m}. \quad (126)$$

Indeed $s = Kp \cdot y$, $\nabla_x = M^{-T}\nabla_y$ on a material function, and $|M^{-T}p|^2 = p^T Cp = r^2$; multiplying the two derivative operators proves (126). For example $k \cdot \nabla_x g = K(Cp) \cdot \nabla_y \tilde{g}$. This proves an exact coordinate correspondence for every displayed mixed derivative and keeps the source radii in their original units.

6.3 What the affine cells and sine regions actually satisfy

On $g = 1$ and $|s| < s_0$ the wave is exactly

$$V = Tk \cdot x, \quad v = QJk \otimes kx = \frac{w\Omega}{r^2} J\zeta \otimes \zeta x.$$

Both Laplacians vanish, but their damping contributions to the forces (119)–(120) are

$$-d\rho V, \quad -d\rho v. \quad (127)$$

They do not vanish unless the corresponding wave vanishes. For the first wave the scalar core force is $-dK_1^2(w_1K_1\Theta_1\zeta_1) \cdot x$ after activation, and the curl of its damping vector force is $-dK_1^2w_1\Omega_1$. Thus the evolving affine parent itself has the force required by its changed amplitude equations. On an uncut region where $F(s) = \sin s$, instead, $F'' = -F$ and the two damping contributions are zero: $\rho V + \Delta V = 0$ and $\rho v + \Delta v = 0$. The non-damping vector terms in (120) are still present; this cancellation is not a claim of unforced momentum at $p = 0$. In cutoff regions every term in (122)–(123) must also be kept. These restrictions prove exactly how the damped coefficient dynamics map into the released spatial profile. They differ from retaining an inviscid state and adding only $-d\Delta$ of that state: the present amplitudes, parent, phases and controls have all changed according to (112).

6.4 Explicit bounds at every spatial derivative order

For a scalar, vector, or tensor field U , let $|U|_m = \sup_x \|D_x^m U(x)\|$, using Euclidean multilinear operator norms and $m \geq 0$. For a one-variable profile W write $[W]_m = \sup_s |W^{(m)}(s)|$, and put $F_m^f = \sup_y \|D^m f(y)\|$. Set $N_1 = 1$, $N_2 = N$, $r_{1,-} = r_{1,+} = 1$, $r_{2,-} = h_0$, $r_{2,+} = \sqrt{10}$. Define finite explicit numbers

$$\begin{aligned} \kappa_j^* &= K_j r_{j,+}, & e_{j,m} &= (N_j/\ell_j)^m F_m^f, \\ \mathcal{B}_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [W]_q e_{j,m-q}, & \mathcal{C}_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [P]_q e_{j,m-q+1}. \end{aligned} \quad (128)$$

Here κ_j^* is merely a frequency bound, not a diffusivity. Repeated differentiation and the chain rule give $|g_j|_m \leq e_{j,m}$, $|W(s_j)g_j|_m \leq \mathcal{B}_{j,m}(W)$, and $|P(s_j)\nabla g_j|_m \leq \mathcal{C}_{j,m}(P)$. To prove the middle inequality, assign each of the m derivatives either to the profile or to the envelope. There are $\binom{m}{q}$ assignments with q profile derivatives, each bounded by the corresponding summand. The other inequalities follow by the same argument. This proof applies to all directions in the defined operator norm, not only coordinate derivatives.

The actual threshold amplitudes are $P_j^* = (\nu^2/\mu)\lambda_j^{-7/8}$. The finite coefficient calculation, (92), supplies the following global bounds through the hold:

$$\begin{aligned} U_{\Theta,1} &= \max\{e^{2+\Lambda_1^{-1}} P_1^*, 2\sigma^2/K_1\}, & U_{\Omega,1} &= \max\{(K_1/\nu)e^{2+\Lambda_1^{-1}} P_1^*, 9\sigma\}, \\ U_{\Theta,2} &= e^{18} P_2^*, & U_{\Omega,2} &= 3K_2 U_{\Theta,2}/\sigma. \end{aligned} \quad (129)$$

In particular these preserve the actual source frequencies and the changed inherited scale. Put

$$A_1 = \sup_{[0,T_f]} |\dot{\alpha}|, \quad A_2 = \sup_{[0,T_f]} |\ddot{\alpha}|, \quad D_1^* = A_1, \quad D_2^* = A_1 + U_{\Omega,1}, \quad G_1^* = A_0, \quad G_2^* = A_0 + K_1 U_{\Theta,1}.$$

Explicit profile bounds for A_1, A_2 are given below. Set $W_{1,1} = \Gamma_1[\eta']_0$ and $W_{2,1} = \Gamma[\eta']_0$. Then

$$\begin{aligned} Q_j^* &= \frac{U_{\Omega,j}}{K_j^2 r_{j,-}^2}, \\ Q_{*,j}^* &= \frac{W_{j,1} U_{\Omega,j} + \kappa_j^* U_{\Theta,j} + 2D_j^* U_{\Omega,j}}{K_j^2 r_{j,-}^2}. \end{aligned} \quad (130)$$

The bounds $|Q_j| \leq Q_j^*$ and $|Q_{*,j}| \leq Q_{*,j}^*$ follow from (118) and $|\dot{\rho}/\rho| \leq 2\|D\|$. Define

$$\begin{aligned}\mathcal{D}_{\theta,j,m} &= dU_{\Theta,j}\{(\kappa_j^*)^2 \mathcal{B}_{j,m}(F) + 2\mathcal{B}_{j,m+2}(F)\}, \\ \mathcal{D}_{u,j,m} &= dQ_j^*\{(\kappa_j^*)^2 \mathcal{B}_{j,m+1}(P) + 2\mathcal{B}_{j,m+3}(P)\}.\end{aligned}\tag{131}$$

These bound the complete damping-plus-diffusion terms in (119)–(120). In particular they include the core damping and every envelope derivative. The two in front of $\mathcal{B}_{m+2}$ or $\mathcal{B}_{m+3}$ is the sum of the two coordinate Laplacian terms; J has operator norm one. The whole added forces satisfy the explicit inequalities

$$\begin{aligned}|E_{\theta,j}|_m &\leq W_{j,1}U_{\Theta,j}\mathcal{B}_{j,m}(F) + Q_j^*G_j^*\mathcal{C}_{j,m}(P) + \mathcal{D}_{\theta,j,m} \\ &\quad + Q_j^*U_{\Theta,j}\sum_{q=0}^m \binom{m}{q} \mathcal{B}_{j,q+1}(P)\mathcal{B}_{j,m-q+1}(F),\end{aligned}\tag{132}$$

$$\begin{aligned}|E_{u,j}|_m &\leq (2D_j^*Q_j^* + Q_{*,j}^*)\mathcal{B}_{j,m+1}(P) + U_{\Theta,j}\mathcal{B}_{j,m}(F) + \mathcal{D}_{u,j,m} \\ &\quad + (Q_j^*)^2\sum_{q=0}^m \binom{m}{q} \mathcal{B}_{j,q+1}(P)\mathcal{B}_{j,m-q+2}(P).\end{aligned}\tag{133}$$

For the scalar nonlinear term it is enough to use its exact alternative expression $v_j \cdot \nabla V_j$; differentiation of this product proves the displayed sum. For the vector nonlinear term differentiate $(\nabla v_j)v_j$ and interchange the summation index to obtain the final sum. These arguments prove all four bounds directly from the exact formulas.

The base has equally explicit bounds. Put $c_m = |\chi_0|_m$, $h_m = |H_0|_m$, and

$$b_m(W) = \sum_{q=0}^m \binom{m}{q} \mu^q [W]_q c_{m-q}, \quad B_m^0 = \frac{A_0}{\mu} \max\{b_m(F), b_m(\sin)\}.$$

The interpolation is a convex combination when $\alpha = 0$; afterwards $|\mu e_0| = \mu$. Hence $|\theta_b|_m \leq B_m^0$. The product rule and (124) give

$$\begin{aligned}|f_{\theta,b}|_m &\leq \frac{\nu A_0}{\mu} [\eta']_0 \{b_m(F) + b_m(\sin)\} + 2dB_{m+2}^0, \\ |f_{u,b}|_m &\leq A_2 h_{m+1} + B_m^0 + 2dA_1 h_{m+3} + A_1^2 \sum_{q=0}^m \binom{m}{q} h_{q+2} h_{m-q+1}.\end{aligned}\tag{134}$$

All total force norms are bounded by adding their base and two increment bounds. Integrating any bound over its interval multiplies its constant by that interval's physical length. For example

$$\int_0^{T_f} |\text{wave damping-plus-diffusion force}_j|_m dt \leq (T_f - t_{0j}) \mathcal{D}_{\bullet,j,m}.\tag{135}$$

Here $T_f = t_1 + (1 + L_2^{-1} + 1/32)/\Gamma$ is the actual end of the finite hold and the dot denotes the corresponding scalar or vector bound. Every frequency, envelope radius and diffusion coefficient is explicit. No estimate in this section asserts a bound uniform over a sequence $K_j \rightarrow \infty$.

6.5 Control derivatives and every mixed derivative order

The constants A_1, A_2 in the spatial force costs can be bounded without differentiating an unknown shooting root. The root is a fixed number in $[0, 3]$, and differentiation below is in time. For a compression L put

$$Z_0(L) = \max\{1, 3L[\eta_2]_0\}, \quad Z_b(L) = \max\{[\eta^{(b)}]_0, 3L^{b+1}[\eta_2^{(b)}]_0\} \quad (b \geq 1).$$

These bound the derivatives of the source ramp and pulse z ; the hold has zero derivatives. During the first control, $\alpha = s_* - \arcsin(\sin s_* z)$ and $|\sin s_* z| \leq 1/4$. Twice differentiating this displayed formula gives

$$A_{1,\text{first}} = \frac{4}{\sqrt{15}} \Gamma_1 \sin s_* Z_1(\Lambda_1),$$

$$A_{2,\text{first}} = \Gamma_1^2 \left\{ \frac{4}{\sqrt{15}} \sin s_* Z_2(\Lambda_1) + \frac{64}{15\sqrt{15}} \sin^3 s_* Z_0(\Lambda_1) Z_1(\Lambda_1)^2 \right\}.$$

During second steering and holding write $z_b = Z_b(L_2)$ and $y_- = \sqrt{59}/8$. The exact control is

$$\alpha_\tau = \frac{W\zeta_{1,1} - az'}{Y}, \quad Y' = \frac{hW - a^2 z z'}{Y}, \quad \zeta_{1,\tau} = \alpha_\tau J \zeta_1.$$

Define the explicit constants

$$U_\alpha = (9 + az_1)/y_-, \quad U_Y = (27 + az_1/4)/y_-,$$

$$U_{W'} = \frac{128}{63}(\sqrt{10} + 3z_0) + \frac{9}{16},$$

$$U_{\alpha'} = (U_{W'} + 9U_\alpha + az_2)/y_- + (9 + az_1)U_Y/y_-^2.$$

The bounds $Y \geq y_-$, $|az| \leq 1/4$, $h < 3$, $W < 9$, $B \leq 2$, $h^2 + a^2 \geq 63/64$, and $\delta_1 \leq 1/16$ give $|\alpha_\tau| \leq U_\alpha$, $|Y'| \leq U_Y$ and $|W'| \leq U_{W'}$. Differentiating the quotient for α_τ gives $|\alpha_{\tau\tau}| \leq U_{\alpha'}$. Thus one can use

$$A_1 \leq \max\{A_{1,\text{first}}, \Gamma U_\alpha\}, \quad A_2 \leq \max\{A_{2,\text{first}}, \Gamma^2 U_{\alpha'}\}.$$

On intervening growth and transition intervals the angle is constant.

For completeness the following exact finite recursion evaluates any mixed derivative, rather than treating spatial estimates as time estimates. For a fixed layer put $A = M^{-1}/\ell$ and introduce independent variables t, x, s, y . Define operators on smooth expressions in these variables by

$$\mathfrak{T} = \partial_t + (\dot{k} \cdot x) \partial_s + (\dot{A}x) \cdot \nabla_y, \quad \mathfrak{X}_i = \partial_{x_i} + k_i \partial_s + \sum_l A_{li} \partial_{y_l}. \quad (136)$$

Here ∂_t differentiates every time-dependent coefficient and ∂_{x_i} differentiates the explicitly occurring x . If $\iota^* R = R(t, x, k(t) \cdot x, A(t)x)$, the chain rule proves

$$\partial_t^b \partial_x^\gamma (\iota^* R) = \iota^* (\mathfrak{T}^b \mathfrak{X}_1^{\gamma_1} \mathfrak{X}_2^{\gamma_2} R). \quad (137)$$

Induction proves this identity for all nonnegative b, γ_i . For example, for $R = Q(t)P(s)f(y)$ it gives every streamfunction jet; applying one more $\mathfrak{X}$ and J gives every velocity jet. The commutators vanish: in $[\mathfrak{T}, \mathfrak{X}_i]$ the derivative $\dot{k}_i$ is canceled by differentiating $\dot{k} \cdot x$, and $\dot{A}_{li}$ by differentiating $(\dot{A}x)_l$. The $\mathfrak{X}_i$ commute by their displayed coefficients. This checks the order consistency of the recursion.

Apply (137) to each summand of (119)–(124), using a separate pair (s, y) for each layer when necessary. After expansion the answer is a finite sum of terms of the form

$$c(t)x^\beta \prod_{l=1}^q W_l^{(n_l)}(s_j) \prod_{l=1}^r \partial^{\gamma_l} f(y_j),$$

or the analogous base terms with χ_0, H_0 . An explicit bound for each term is obtained by replacing it by

$$\sup_{[0, T_f]} |c(t)| (N_j \ell_j)^{|\beta|} \prod_{l=1}^q [W_l]_{n_l} \prod_{l=1}^r F_{|\gamma_l|}^f,$$

and summing; for a base term use $R_H^{|\beta|}$. Each coefficient jet in c is computed by differentiating (112), the phase equations, (113), and the explicit controls. This is a finite recursion using only addition,

multiplication, the stated profile derivatives and the nonzero denominators a , K_j , Γ , r_j^2 , Y . Their positive lower bounds have already been proved. Induction on the requested order, applying the product and quotient rules to those equations, therefore supplies finite bounds for every coefficient jet from the preceding-order bounds. This specifies every mixed force and diffusion norm by an exact finite procedure, including the time variation of the material cutoff. It does not replace the cutoff by a stationary radial function for layer two.

6.6 Smooth finite slab, compact forces, and exact returned state

All fields and forces are smooth and supported in B_{R_H} on $0 \leq t \leq T_f$. Each velocity is the rotated gradient of a compact streamfunction, so its energy is finite. It also has the usual exact Biot–Savart relation. Indeed for $N_0(x) = (2\pi)^{-1} \log |x|$ and compact smooth $\psi = \psi_b + \Psi_1 + \Psi_2$, distributional integration by parts gives $N_0 * \Delta\psi = (\Delta N_0) * \psi = \psi$; applying $J\nabla$ gives $u = J\nabla N_0 * \omega$ with $\omega = \Delta\psi$. All integrations are legitimate because ψ is compact and N_0 is locally integrable. Odd F , even P , and even cutoffs give odd θ, u, f_θ, f_u and even ω . The pressure $p = 0$ has $p(0, t) = 0$.

At the selected second return and throughout its physical hold of duration $1/(32\Gamma)$, the coefficient calculation proves $\zeta_{2,1} = \Omega_2 = 0$, $w_2 = 1$, and

$$\Theta_2(t) = -P_2(t_b) \exp \left[-dK_2^2 \int_{t_b}^t |\zeta_2(r)|^2 dr \right].$$

Thus $v_2 = 0$ everywhere on this hold, while its temperature and the first parent continue changing. In the inner second affine cell the total fields have exactly

$$\theta = (-A_0 e_0 + K_1 \Theta_1 \zeta_1 + K_2 \Theta_2 \zeta_2) \cdot x, \quad u = (\dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1) x.$$

The second holding shear is zero, but the first holding shear is positive at return and evolves by the displayed system. In particular these exact full spatial fields have the origin vorticity calculated in (110):

$$\text{curl } u(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2} \right) \geq \frac{31}{32} \Omega_1 > 0 \quad (t_b \leq t \leq T_f). \quad (138)$$

Indeed on holding $\alpha = s_* + \arctan(a/h)$ and $\dot{h} = a\Omega_1$, so $\dot{\alpha} = -a^2\Omega_1/(h^2 + a^2)$. Moreover $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a \leq 1/8$ prove the numerical lower bound. Positivity of Ω_1 at return persists because the holding equation has positive forcing B/r_2 . These are the actual final states realized by the full spatial equations.

There is also an honest compact extension of the forces in time. For $t < 0$ extend the initial state by its constant values, with $\alpha = 0$ and both increments zero. Flatness of η makes the formulas smooth at zero. To the right, continue $z = 0$ and the exact holding ODE for a collar of physical length $\epsilon_R = 1/(128\Gamma)$. Its extension is quantitative: the total scaled duration is at most $17/16$ since $L_2 \geq 64$; the same integrating-factor estimates give

$$0 \leq W \leq 26/3, \quad h \leq 1 + 16/15 + (26/3)(7/128) = 813/320 < 3, \quad B > 191/512 > 1/4.$$

For the last bound use $B_{\text{in}} \geq 1/2$, $\delta_1 \leq 1/16$, $C_1 \leq 1/32$ and $B = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr$. These strict bounds close the positivity bootstrap and keep all denominators away from zero; the smooth finite-dimensional vector field therefore continues through this collar. The support bounds continue as well. Extend the newest temperature by its exact hold exponential, and the angle and matrices by their exact inverse. In particular no nonstationary endpoint is frozen.

Choose $\epsilon_L = 1/\nu$ and multiply the extended residual forces, not the fields on their required slab, by

$$\beta(t) = \eta(1 + t/\epsilon_L) \eta(1 + (T_f - t)/\epsilon_R).$$

This function is one on $[0, T_f]$, zero outside $(-\epsilon_L, T_f + \epsilon_R)$, and flat at the two external endpoints. The resulting forces belong to $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$ and still give exactly the solution already proved on

$[0, T_f]$. On the collar the same mixed-derivative recursion applies, and the product rule includes the explicit additional factors $\epsilon_L^{-b}[\eta^{(b)}]_0$ and $\epsilon_R^{-b}[\eta^{(b)}]_0$. No equation for the continued fields outside the required slab is claimed after this force cutoff. This completes the compact finite spatial realization and its full derivative and support costs.

7 Exact coupled feedback and the scope of common damping

Retain the source layer frequencies λ_j , and write $K_j = \mu\lambda_j$ for physical frequencies. Let Θ_j, Ω_j denote physical amplitudes. For each activated layer j , let $\zeta_j \neq 0$, $r_j = |\zeta_j|$, $e_j = \zeta_j/r_j$, $S_j = Je_j \otimes e_j$, and $A_j = -K_j r_j \Theta_j$. The exact physical counterpart of source (3.3)–(3.5) is

$$\begin{aligned} G_{<q} &= -A_0 e_0 + \sum_{j<q} w_j K_j \Theta_j \zeta_j, & D_{<q} &= \dot{\alpha} J + \sum_{j<q} w_j \Omega_j S_j, \\ \dot{M}_q &= D_{<q} M_q, \quad M_q(t_q^{\text{in}}) = I, & \zeta_q &= M_q^{-T} p_q, \\ \dot{\zeta}_q &= -D_{<q}^T \zeta_q, & \dot{\Theta}_q &= -\frac{J \zeta_q \cdot G_{<q}}{K_q r_q^2} \Omega_q, \quad \dot{\Omega}_q = K_q \zeta_{q,1} \Theta_q. \end{aligned} \quad (139)$$

Each sum begins at $j = 1$, and contains finitely many terms at every finite preterminal time. The rotating base coefficient is the original A_0 , not a reset unit value. The physical growth-scale definition is $\sigma_q^2 = |G_{<q+1}(t_q^b)|$, so $\sigma_q = \nu \tilde{\sigma}_q$.

For $\zeta_{j,2} \neq 0$ define

$$\mathcal{F}_j = -\frac{\sum_{i<j} w_i \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}, \quad \mathcal{F}_0 = 0. \quad (140)$$

Transposing S_i gives $S_i^T \zeta_j = e_i (J e_i \cdot \zeta_j)$, and the first component of $-\dot{\alpha} J^T \zeta_j$ is $-\dot{\alpha} \zeta_{j,2}$. Hence

$$\dot{\zeta}_{j,1} = (\mathcal{F}_j - \dot{\alpha}) \zeta_{j,2}. \quad (141)$$

It follows by direct differentiation that $\dot{\alpha} = \mathcal{F}_j - \dot{Z}/\zeta_{j,2}$ gives $\dot{\zeta}_{j,1} = \dot{Z}$; equality of initial values gives $\zeta_{j,1} = Z$ on its entire interval of definition. This proves the control map exactly, including its denominator and the laboratory component that drives buoyancy.

There is no omitted parent evolution in (139). For example, differentiation of the definitions gives

$$\dot{G}_{<q} = -A_0 \dot{\alpha} J e_0 + \sum_{j<q} K_j \{(\dot{w}_j \Theta_j + w_j \dot{\Theta}_j) \zeta_j - w_j \Theta_j D_{<j}^T \zeta_j\}, \quad (142)$$

$$\dot{D}_{<q} = \ddot{\alpha} J + \sum_{j<q} \{(\dot{w}_j \Omega_j + w_j \dot{\Omega}_j) S_j + w_j \Omega_j \dot{S}_j\}, \quad (143)$$

$$\dot{S}_j = -\frac{J D_{<j}^T \zeta_j \otimes \zeta_j + J \zeta_j \otimes D_{<j}^T \zeta_j}{r_j^2} + \frac{2 \zeta_j \cdot D_{<j}^T \zeta_j}{r_j^2} S_j. \quad (144)$$

To verify the last line, use $S_j = J \zeta_j \otimes \zeta_j / r_j^2$, $\dot{\zeta}_j = -D_{<j}^T \zeta_j$, and $(r_j^2)' = -2 \zeta_j \cdot D_{<j}^T \zeta_j$. These three identities give exactly the three displayed terms by the quotient rule. Equations (142)–(144), together with the amplitude rows, are exact equations for both inherited backgrounds, including activation and the common angular control.

7.1 Damped amplitude equations as an exact controlled realization

For a sine profile on an affine background, retaining physical diffusion gives the amplitude rows

$$\begin{aligned} \dot{\Theta}_j &= -\kappa K_j^2 r_j^2 \Theta_j - \frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j + c_{\theta,j}, \\ \dot{\Omega}_j &= -\varepsilon K_j^2 r_j^2 \Omega_j + K_j \zeta_{j,1} \Theta_j + c_{\omega,j}. \end{aligned} \quad (145)$$

For an actual finite family these equations are coupled to the phase rows, D, G , and the control; the symbols $c_{\theta,j}$ and $c_{\omega,j}$ here explicitly record added amplitude forcing. Inserting them in (142)–(143) produces the exact diffusion contributions

$$\begin{aligned}(\dot{G}_{<q})_{\text{amplitude diffusion}} &= -\kappa \sum_{j<q} w_j K_j^3 r_j^2 \Theta_j \zeta_j, \\(\dot{D}_{<q})_{\text{amplitude diffusion}} &= -\varepsilon \sum_{j<q} w_j K_j^2 r_j^2 \Omega_j S_j,\end{aligned}\tag{146}$$

and additional terms $\sum w_j K_j c_{\theta,j} \zeta_j$ and $\sum w_j c_{\omega,j} S_j$, respectively. The phase geometry and feedback then use those changing backgrounds, rather than their inviscid values.

For the released general F , (145) has a different precise meaning: after substitution into the exact uncut scalar and vorticity residuals, the wave forces are

$$\begin{aligned}R_{\theta,j} &= c_{\theta,j} F - \kappa K_j^2 r_j^2 \Theta_j (F + F''), \\R_{\omega,j} &= c_{\omega,j} F' - \varepsilon K_j^2 r_j^2 \Omega_j (F' + F''').\end{aligned}\tag{147}$$

These formulas follow because $\Delta[\Theta_j F(K_j \zeta_j \cdot x)] = K_j^2 r_j^2 \Theta_j F''$ and $\Delta[\Omega_j F'(K_j \zeta_j \cdot x)] = K_j^2 r_j^2 \Omega_j F'''$. On an affine core they are $c_{\theta,j} s - \kappa K_j^2 r_j^2 \Theta_j s$ and $c_{\omega,j} - \varepsilon K_j^2 r_j^2 \Omega_j$. Thus damping these amplitudes for this profile requires force *inside* the affine core. Taking $c_{\theta,j} = \kappa K_j^2 r_j^2 \Theta_j$ and $c_{\omega,j} = \varepsilon K_j^2 r_j^2 \Omega_j$ restores exactly the original coupled amplitude rows and leaves instead $-\kappa K_j^2 r_j^2 \Theta_j F''$ and $-\varepsilon K_j^2 r_j^2 \Omega_j F'''$. These vanish in the core and are precisely the compensation used in (11)–(12), with all cutoff terms additionally retained there.

7.2 The exact defect of multiplying all old amplitudes by damping

To quantify a proposed map, let a finite inviscid coupled trajectory be decorated with E , and suppose for this calculation that $\kappa = \varepsilon = \delta > 0$. Keep its phases and angle and put

$$d_j(t) = \exp \left[-\delta K_j^2 \int_{t_j^{\text{in}}}^t r_j(s)^2 ds \right], \quad \Theta_j^d = d_j \Theta_j^E, \quad \Omega_j^d = d_j \Omega_j^E.$$

The backgrounds computed from these candidate fields differ by

$$\Delta G_{<q} = \sum_{j<q} w_j K_j (d_j - 1) \Theta_j^E \zeta_j, \quad \Delta D_{<q} = \sum_{j<q} w_j (d_j - 1) \Omega_j^E S_j.\tag{148}$$

Differentiating the candidate fields and using the inviscid equations gives the exact defects

$$\dot{\zeta}_q + (D_{<q}^d)^T \zeta_q = \Delta D_{<q}^T \zeta_q,\tag{149}$$

$$\dot{\Theta}_q^d + \delta K_q^2 r_q^2 \Theta_q^d + \frac{J \zeta_q \cdot G_{<q}^d}{K_q r_q^2} \Omega_q^d = \frac{J \zeta_q \cdot \Delta G_{<q}}{K_q r_q^2} \Omega_q^d,\tag{150}$$

$$\dot{\Omega}_q^d + \delta K_q^2 r_q^2 \Omega_q^d - K_q \zeta_{q,1} \Theta_q^d = 0.\tag{151}$$

For instance the derivative of d_q cancels the diagonal diffusion term, leaving the difference of the two temperature couplings; this is precisely (150), with its positive sign. Also the feedback computed at these same phases has the exact difference

$$\mathcal{F}_j^d - \mathcal{F}_j^E = -\frac{\sum_{i<j} w_i (d_i - 1) \Omega_i^E e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}.\tag{152}$$

Changing $\dot{\alpha}$ by this expression cancels the first component of (149) for that controlled index, as follows from (141); it does not in general cancel its second component or the equations for other

phases. Equations (149)–(152) are an exact relation between the two candidate systems, rather than an assertion that the systems are unrelated.

The defects are nonzero even in a particularly transparent actual holding geometry. After the first stage, $\zeta_1 = e_2$, $\Omega_1 = 0$, and $\Theta_1 < 0$. At the start of the next growth interval $\zeta_2 = e(s_2)$, $s_2 > 0$. For the damping candidate with $d_1 < 1$,

$$J\zeta_2 \cdot \Delta G_{<2} = (d_1 - 1)K_1\Theta_1 \sin s_2 \neq 0.$$

All factors on the right are nonzero with the indicated signs. Thus even when the old shear vanishes exactly on a hold, its decaying temperature changes the next layer’s amplification coefficient. The common scalar damping identity for a fixed prescribed D, G has not proved a conjugacy of this coupled system.

For the compact finite stage of Section 1, the stronger exact forced relation is available: retain the original coupled fields and add the explicitly calculated Laplacian forces. It preserves both backgrounds and all controls, and quantifies the entire cost. The next section records precisely why this particular relation has no smooth-force extension along the unchanged infinite source trajectory.

8 The exact cost along the released infinite trajectory

The finite construction above proves no infinite viscous singularity. There is a precise calculation for the particular proposed extension that retains the released infinite fields and changes only the forces by their Laplacians. We record both that calculation and its scope.

The source trajectory in this section means the specific construction of Alpöge–Buckmaster, including their compact corrections. Its existence and infinite inviscid conclusions are the source’s Theorems 3.2, 7.3, 8.2, and 1.1, not new conclusions of this note. We use the following exact properties established in those proofs: at the growth crossing t_q^a , activation is complete and

$$\Theta_q(t_q^a) = -\frac{\nu^2}{\mu}\lambda_q^{-7/8}, \quad r_q(t_q^a) \geq r_* > \frac{1}{2}, \quad \left| \frac{\Omega_q(t_q^a)}{\Theta_q(t_q^a)} \right| \geq \frac{K_q}{2\sigma_{q-1}}. \quad (153)$$

The first is the definition of the stopping time, source (3.8) and item (i) after (3.11), p. 10. Here $r_* := 1 - C_r\epsilon_{\text{sch}} > 1/2$ is the improved phase-length bound, with the fixed constant C_r from the source estimate; it is not the separate bootstrap constant called ζ_- in source (3.14). The second is source (3.18), p. 13 and the tolerance choice on p. 12. The last is the tracking estimate in the complete growth proof, pp. 29–30, between (3.63) and (3.64), with the low-order tolerance decreased within its permitted fixed range. For clarity the extraction of a numerical factor does not presume equality with the reference growing eigenline. With $u = \Omega/\Theta$, $u_* = (b/a)^{1/2}$, and $z = u/u_*$, direct differentiation gives

$$z' = \sqrt{ab}(1 - z^2) - z(\log u_*)'.$$

The source seed is chosen on the growing eigenline, so $z(t_q^{\text{in}}) = 1$. For $q = 1$ the eigenline and the ratio are exact throughout growth, as proved in the finite-stage section; the following tracking computation applies to later stages. If the source error bound is at most $\delta \leq 1/4$, its Riccati barrier gives $z \geq 1 - \delta$: at $z = 1 - \delta$, $z' \geq \sqrt{ab}\{1 - (1 - \delta)^2 - \delta(1 - \delta)\} = \delta\sqrt{ab} > 0$. At $z = 1 + \delta$ the corresponding upper derivative is $-\delta\sqrt{ab} < 0$. A first exit is therefore impossible. To extract the coefficient ratio, use the exact source (3.55)–(3.56): during growth the preceding direction is vertical, $a_q = \mathcal{G}_q/(K_q r_q)$ and $b_q = K_q d_q$, where $d_q = r_q \sin(\phi_q - \phi_{q-1})$ and $\mathcal{G}_q = \sum_{i < q} w_i A_i \sin(\phi_q - \phi_i)$ includes the base term with A_0 and $w_0 = 1$. Hence $u_* = K_q \sqrt{d_q r_q / \mathcal{G}_q}$. The three source estimates are $r_q \geq 1 - \delta$, $d_q / \sin s_q \geq 1 - \delta$, and $\mathcal{G}_q / (\sigma_{q-1}^2 \sin s_q) \leq 1 + \delta$. Together they give $u_* \geq (K_q / \sigma_{q-1})(1 - \delta) / \sqrt{1 + \delta}$. Their product is at least $K_q / (2\sigma_{q-1})$, because $(3/4)^2 / \sqrt{5/4} > 1/2$. This proves the displayed extraction from the source’s explicit tracking estimates without assigning a sign to an uncontrolled error.

The source also gives

$$\sigma_{q-1} \leq \nu \sqrt{C_\sigma} \lambda_{q-1}^{1/16}, \quad \lambda_q = \lambda_{q-1}^{Q_q}, \quad Q_q = Q_* + q, \quad Q_* \geq 200, \quad (154)$$

by (3.7),(3.18). The material radius is $\ell_q^{\text{phys}} = \lambda_{q-1}^{-3}/\mu$ for $q \geq 2$. Each older layer and the base are affine on the full new support (Lemma 4.3, pp. 37–38), while all corrections to the newest layer vanish on its entire material plateau (Proposition 6.1, pp. 54–55). These spatial statements are essential to the following pointwise computation: a bound on a central affine ball alone would not provide the points used here.

Choose one fixed $s_\dagger \in (-\pi, \pi)$ with $F''(s_\dagger) \neq 0$. Such a point exists for the released profile. Otherwise F'' would vanish on the full period and by periodicity everywhere, so F would be an affine periodic function, hence constant; oddness would make it zero, contradicting $F'(0) = 1$. Fix this point once for every layer. At the growth crossing put

$$a_q = \frac{s_\dagger}{K_q} p_q, \quad x_q = M_q(t_q^a) a_q. \quad (155)$$

The activation vectors have $|p_q| = 1$ and $\zeta_q = M_q^{-T} p_q$, so $K_q \zeta_q \cdot x_q = s_\dagger$ exactly. The point belongs to the plateau as soon as

$$\frac{|s_\dagger|}{K_q} < \frac{\ell_q^{\text{phys}}}{2}, \quad \text{equivalently} \quad 2|s_\dagger| < \lambda_{q-1}^{Q_q-3}.$$

This holds for all sufficiently large q by (154). The entire neighborhood of this point in the plateau has constant envelope, zero correction fields, and affine older fields. No later layer is yet activated at t_q^a . Consequently at this point the Laplacians of the *full released fields*, not merely those of an isolated formal wave, are exactly

$$\begin{aligned} \Delta \theta_E(x_q, t_q^a) &= \Theta_q K_q^2 r_q^2 F''(s_\dagger), \\ \Delta u_E(x_q, t_q^a) &= \Omega_q K_q J \zeta_q F''(s_\dagger). \end{aligned} \quad (156)$$

For the second line use $u_q = \Omega_q J \zeta_q F(s)/(K_q r_q^2)$ on the plateau; applying the Laplacian contributes $K_q^2 r_q^2$ and cancels the displayed denominator. This verifies both the vector direction and every frequency and phase-length factor.

Inserting (153) gives

$$|\kappa \Delta \theta_E(x_q, t_q^a)| \geq \kappa \nu^2 \mu r_*^2 |F''(s_\dagger)| \lambda_q^{9/8}, \quad (157)$$

$$|\varepsilon \Delta u_E(x_q, t_q^a)| \geq \frac{\varepsilon \nu \mu r_*}{2\sqrt{C_\sigma}} |F''(s_\dagger)| \lambda_q^{9/8} \lambda_{q-1}^{-1/16}. \quad (158)$$

For the second line first use $|\Omega_q| \geq |\Theta_q| K_q / (2\sigma_{q-1})$ and $|J \zeta_q| = r_q$, and then (154). For fixed positive κ the first right-hand side tends to infinity. For fixed positive ε the second does also, since

$$\lambda_q^{9/8} \lambda_{q-1}^{-1/16} = \lambda_{q-1}^{9Q_q/8-1/16} \rightarrow \infty.$$

In particular even the case $\kappa = 0, \varepsilon > 0$ has an explicit diverging vector diffusion cost along these unchanged fields.

The original forces extend smoothly through the source terminal time and have fixed compact support, and hence are bounded; this is Proposition 9.4(b),(c), pp. 72–73, proved using Lemma 9.3, pp. 71–72. For the specific exact extension

$$f_\theta^{\kappa, \varepsilon} = f_\theta^E - \kappa \Delta \theta_E, \quad f_u^{\kappa, \varepsilon} = f_u^E - \varepsilon \Delta u_E, \quad p^{\kappa, \varepsilon} = p_E,$$

the reverse triangle inequality and (157)–(158) therefore prove unbounded forces at the corresponding positive coefficient, before the terminal time. They cannot have a continuous, and thus cannot have a smooth, terminal extension. This proves failure of the specified unchanged-pressure

map. There is a stronger, pressure-independent conclusion for these same fields. Choose $s_{\dagger}$ with $F'''(s_{\dagger}) \neq 0$. It exists: otherwise F'' would be constant, and periodicity would successively force $F'' = F' = 0$, contradicting $F'(0) = 1$. Use the plateau point in (155) with this phase. The vorticity there equals a spatially constant older vorticity plus $\Omega_q F'(s)$, so

$$\Delta\omega_E = \Omega_q K_q^2 r_q^2 F'''(s_{\dagger}), \quad |\varepsilon\Delta\omega_E| \geq \frac{\varepsilon\nu\mu^2 r_*^2}{2\sqrt{C_\sigma}} |F'''(s_{\dagger})| \lambda_q^{17/8} \lambda_{q-1}^{-1/16} \longrightarrow \infty.$$

For any smooth pressure p_{new} and any vector force producing the same u_E, θ_E in the viscous equation, subtraction of the inviscid momentum equation and application of curl give the exact identity

$$\text{curl } f_{u,\text{new}} = \text{curl } f_u^E - \varepsilon\Delta\omega_E,$$

because mixed derivatives of $p_{\text{new}} - p_E$ commute. The source curl force is bounded. The displayed divergence therefore excludes a terminal extension with bounded first spatial derivatives for *every* pressure and force that retain these exact state fields. In fact the failure is local at the origin and terminal time. Source Theorem 3.2(iv) gives $\|M_q\| \leq \sqrt{C_\sigma} \lambda_{q-1}^{1/16}$, so the test points satisfy

$$|x_q| \leq \frac{|s_{\dagger}| \sqrt{C_\sigma}}{\mu} \lambda_{q-1}^{1/16 - Q_q} \longrightarrow 0, \quad t_q^a \longrightarrow T_*.$$

A continuous derivative on a neighborhood of $(0, T_*)$ is bounded on a smaller compact neighborhood, contradicting the computed curl divergence there. If a vector field has bounded first spatial derivatives then its curl is bounded by their sum, which proves this implication explicitly. This still does not exclude a modified coupled trajectory, a different state construction, or viscous blowup; none of those claims follows from evaluating the unchanged source state.

The completed result of this note is the actual finite forced stage proved above and below, including its original datum, control, background feedback, return, hold, support and every diffusion term. The released infinite inviscid sequence is reconstructed as source mathematics. A new infinite viscous sequence with forces smooth through its terminal time is not established by these calculations.

Source and scope. The source is Levent Alpöge and Tristan Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, the supplied 76-page released PDF. Page numbers below are the printed PDF page numbers. The companion source receipt identifies the exact local PDF by SHA-256. The equations and proofs here reconstruct the finite-dimensional system in Section 3, especially pp. 8–10 and 26–33, and its exact affine realization in Section 4, pp. 37–39. The source's local comparison bounds are identified as source results where mentioned; the new claims proved below are the coordinate, operator, feedback, and aggregate-background identities. No infinite retained-viscosity construction is asserted.

Original data and the distinct parameters. Keep the original data of source (1.4), p. 2:

$$A_0 \geq 1, \quad \lambda_0 = \left\lceil \frac{\sqrt{A_0}}{2} \right\rceil, \quad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0} \sin(\lambda_0 x_2) \chi_0(x), \quad u_{\text{in}} = 0.$$

Here $\chi_0 \in C_c^\infty(B_{R_0})$ is radial, $R_0 > 2$, $0 \leq \chi_0 \leq 1$, and $\chi_0 = 1$ on B_2 . Write

$$\mu = \lambda_0, \quad \nu_{\text{AB}} = \sqrt{A_0}, \quad X = \mu x, \quad T = \nu_{\text{AB}} t.$$

The symbol ν_{AB} is precisely the time-scaling parameter called ν in source Lemma 1.2, pp. 6–7. It is not physical viscosity. Physical momentum viscosity is $\varepsilon_\nu \geq 0$ and thermal diffusivity is $\kappa_{\text{th}} \geq 0$. The source uses μ again locally for its shooting amplitude; we display that independent parameter as μ_p , and never identify it with $\mu = \lambda_0$.

Use a hat for a source Section 3 quantity, which is expressed there in (X, T) . The exact dictionary, keeping both presentations, is

$$\begin{aligned}\theta(x, t) &= \frac{\nu_{AB}^2}{\mu} \widehat{\theta}(X, T), & u(x, t) &= \frac{\nu_{AB}}{\mu} \widehat{u}(X, T), & p(x, t) &= \frac{\nu_{AB}^2}{\mu^2} \widehat{p}(X, T), \\ \omega(x, t) &= \nu_{AB} \widehat{\omega}(X, T), & \psi(x, t) &= \frac{\nu_{AB}}{\mu^2} \widehat{\psi}(X, T), \\ f_\theta(x, t) &= \frac{\nu_{AB}^3}{\mu} \widehat{f}_\theta(X, T), & f_u(x, t) &= \frac{\nu_{AB}^2}{\mu} \widehat{f}_u(X, T).\end{aligned}\quad (159)$$

The transformed diffusivities are

$$\widehat{\kappa}_{th} = \kappa_{th} \frac{\mu^2}{\nu_{AB}}, \quad \widehat{\varepsilon}_v = \varepsilon_v \frac{\mu^2}{\nu_{AB}}. \quad (160)$$

These statements follow directly from $\partial_t = \nu_{AB} \partial_T$, $\nabla_x = \mu \nabla_X$, and $\Delta_x = \mu^2 \Delta_X$: both scalar transport terms have factor ν_{AB}^3/μ , both momentum transport terms and gravity have factor ν_{AB}^2/μ , and the respective Laplacian terms have factors $\kappa_{th} \nu_{AB}^2 \mu$ and $\varepsilon_v \nu_{AB} \mu$. The divergence scales by ν_{AB} , curl by ν_{AB} , and $J \nabla_x \psi = (\nu_{AB}/\mu) J \nabla_X \widehat{\psi}$. Thus the laboratory gravity vector $e_2 = (0, 1)$ is unchanged. The inverse transformations divide by the displayed nonzero factors and evaluate at $(x, t) = (X/\mu, T/\nu_{AB})$, so these are bijections of the corresponding smooth fields, with support radii divided by μ and time intervals divided by ν_{AB} .

Layer dictionary. For $j \geq 1$ retain the original source $\widehat{\lambda}_j$, and define its actual spatial frequency $\lambda_j = \mu \widehat{\lambda}_j$. The remaining dictionary is

$$\begin{aligned}M_j(t) &= \widehat{M}_j(\nu_{AB} t), & \zeta_j(t) &= \widehat{\zeta}_j(\nu_{AB} t), & p_j &= \widehat{p}_j, \\ \Theta_j(t) &= \frac{\nu_{AB}^2}{\mu} \widehat{\Theta}_j(\nu_{AB} t), & \Omega_j(t) &= \nu_{AB} \widehat{\Omega}_j(\nu_{AB} t), & \alpha(t) &= \widehat{\alpha}(\nu_{AB} t), \\ A_j(t) &= \nu_{AB}^2 \widehat{A}_j(\nu_{AB} t), & \sigma_j &= \nu_{AB} \widehat{\sigma}_j, & \Gamma_j &= \nu_{AB} \widehat{\Gamma}_j, \\ D_{<j}(t) &= \nu_{AB} \widehat{D}_{<j}(\nu_{AB} t), & G_{<j}(t) &= \nu_{AB}^2 \widehat{G}_{<j}(\nu_{AB} t), & \ell_j &= \widehat{\ell}_j/\mu.\end{aligned}\quad (161)$$

In particular $\sigma_0 = \nu_{AB}$, $A_0 = \nu_{AB}^2$, and the original $\lambda_0 = \mu$ has not been reset. All unadorned derivatives below are in original physical time t .

The exact coupled variables. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad e(\phi) = (\sin \phi, \cos \phi), \quad e_0 = R_\alpha e_2 = e(-\alpha), \quad w_0 = 1.$$

For $j \geq 1$ set $r_j = |\zeta_j|$, $e_j = \zeta_j/r_j = e(\phi_j)$, $P_j = -\Theta_j$, $A_j = \lambda_j r_j P_j$, and $B_j = \lambda_j \Theta_j \zeta_j = -A_j e_j$. The positive-amplitude variables are used only on intervals where their positivity has been established. Put $B_0 = -A_0 e_0$, and

$$\begin{aligned}G_{<j} &= B_0 + \sum_{1 \leq i < j} w_i B_i, & D_{<j} &= \dot{\alpha} J + \sum_{1 \leq i < j} w_i S_i, & S_i &= \Omega_i J e_i \otimes e_i, \\ \dot{M}_j &= D_{<j} M_j, & M_j(t_j^{\text{in}}) &= I, & \zeta_j &= M_j^{-T} p_j, \\ \dot{\Theta}_j &= a_j \Omega_j, & a_j &= -\frac{J \zeta_j \cdot G_{<j}}{\lambda_j r_j^2}, & \dot{\Omega}_j &= b_j \Theta_j, \quad b_j = \lambda_j \zeta_{j,1}.\end{aligned}\quad (162)$$

These are source (3.2)–(3.3), p. 8, in original units. To check their scaling, $a_j = (\nu_{AB}^2/\mu) \widehat{a}_j$ and $b_j = \mu \widehat{b}_j$; consequently both amplitude equations have their correct factors ν_{AB}^3/μ and ν_{AB}^2 . Since $\text{tr } J = \text{tr}(J e_i \otimes e_i) = 0$, Jacobi's determinant formula gives $\det M_j = 1$ throughout every finite existence interval.

Derivation of the wave equations. Let F be the exact odd smooth periodic profile of source (4.1), and let $\mathcal{P}$ be its even mean-zero primitive. Put $s = \lambda_j \zeta_j \cdot x$ and

$$\vartheta_j = \Theta_j F(s), \quad \psi_j = \frac{\Omega_j}{\lambda_j^2 r_j^2} \mathcal{P}(s), \quad v_j = \frac{\Omega_j}{\lambda_j r_j^2} J \zeta_j F(s), \quad \varpi_j = \Omega_j F'(s).$$

Differentiating $M_j^{-T} p_j$ gives $\dot{\zeta}_j = -D_{<j}^T \zeta_j$, hence $(\partial_t + D_{<j} x \cdot \nabla) s = 0$. Moreover $J \zeta_j \cdot \zeta_j = 0$, so $v_j \cdot \nabla \vartheta_j = v_j \cdot \nabla \varpi_j = 0$. On an affine earlier state $G_{<j} \cdot x, D_{<j} x$, the added scalar residual is $(\dot{\Theta}_j + \Omega_j J \zeta_j \cdot G_{<j} / (\lambda_j r_j^2)) F$, and the added vorticity residual is $(\dot{\Omega}_j - \lambda_j \zeta_{j,1} \Theta_j) F'$. This proves the exact map from (162) to the unactivated local wave equations, including the fixed sign and gravity component.

The control maps and every stage. Source (3.4), p. 9, defines

$$F_j^{\text{ctl}} = -\frac{\sum_{1 \leq i < j} w_i \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}, \quad F_0^{\text{ctl}} = 0. \quad (163)$$

The superscript distinguishes the paper's feedback F_j from its periodic profile F . Transposing $D_{<j}$ in (162) and taking component one gives

$$\dot{\zeta}_{j,1} = (F_j^{\text{ctl}} - \dot{\alpha}) \zeta_{j,2}.$$

Thus $\dot{\alpha} = F_j^{\text{ctl}} - \dot{Z}/\zeta_{j,2}$ implies $\zeta_{j,1}(t) = Z(t)$ whenever equality holds initially and the second component is nonzero. This proves the feedback map; the second component is not replaced by a constant.

Here is the complete source schedule (3.7)–(3.12), pp. 9–10, preserving its units. Fix $\beta = 1/8$, $Q_* \geq 200$, $Q_j = Q_* + j$, and

$$\begin{aligned} \hat{\lambda}_1 &\gg 1, & \hat{\lambda}_j &= \hat{\lambda}_{j-1}^{Q_j} \quad (j \geq 2), \\ \hat{\ell}_1 &= s_0/C_\ell, \quad C_\ell \geq 4, & \hat{\ell}_j &= \hat{\lambda}_{j-1}^{-3} \quad (j \geq 2), \\ \hat{\Pi}_1 &= (\log \hat{\lambda}_1)^{10}, & \hat{\Pi}_j &= \hat{\lambda}_{j-1}^4 (\log \hat{\lambda}_j)^{10} \quad (j \geq 2), \\ k_j &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_j, K(k) \leq \log \hat{\lambda}_j\}, \\ J_j &= 2k_j + 8, & L_j &= (k_j + 5 + \beta) \log \hat{\lambda}_j. \end{aligned} \quad (164)$$

The prescribed seed is $\Theta_j^{\text{seed}} = -(\nu_{\text{AB}}^2/\mu) \hat{\lambda}_j^{-k_j-6}$, and its gain target is

$$P_j(t_j^a) = \frac{\nu_{\text{AB}}^2}{\mu} \hat{\lambda}_j^{-7/8}, \quad \frac{P_j(t_j^a)}{|\Theta_j^{\text{seed}}|} = e^{L_j}.$$

In particular the physical frequency recursion is $\lambda_j = \mu(\lambda_{j-1}/\mu)^{Q_j}$, not $\lambda_{j-1}^{Q_j}$. Let η be smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$; let $\eta_2 \in C_c^\infty((0, 1))$, $\eta_2 \geq 0$, $\int_0^1 \eta_2 = 1$, and $H_2 = \|\eta_2\|_\infty$. The fixed design is

$$c_p = 3, \quad \bar{v} = 2, \quad \Lambda_1 \geq 2c_p H_2, \quad 0 < s_* \leq 1/8, \quad c_p \Lambda_1 H_2 \sin s_* \leq 1/4.$$

Set $t_1^{\text{in}} = 1/\nu_{\text{AB}}$, $s_1 = s_*$, $\Gamma_1 = \nu_{\text{AB}} \sin s_*$, and for $j \geq 2$

$$t_j^{\text{in}} = t_{j-1}^b, \quad s_j = L_j \frac{\sigma_{j-2}}{\sigma_{j-1}}, \quad \Gamma_j = \sigma_{j-1} \sin s_j, \quad \Lambda_j = L_j.$$

At activation $p_j = e(s_j)$, $\Theta_j = \Theta_j^{\text{seed}}$, $\Omega_j = \sqrt{b_j/a_j} \Theta_j$, where source Theorem 3.2 proves $a_j, b_j > 0$ on the constructed growth interval. The activation multiplier is $w_j(t) = \eta(\Gamma_j(t - t_j^{\text{in}}))$.

1. Growth: $\dot{\alpha} = F_{j-1}^{\text{ctl}}$ until the first amplitude threshold t_j^a displayed above.
2. Transition: $t_j^1 = t_j^a + \Gamma_j^{-1}$, $\eta_b = \eta(\Gamma_j(t - t_j^a))$, and $\dot{\alpha} = (1 - \eta_b)F_{j-1}^{\text{ctl}} + \eta_b F_j^{\text{ctl}}$.
3. Steering: write $z_1 = \zeta_{j,1}(t_j^1)$ and $\chi_1 = r_j(t_j^1)$. For $0 \leq h \leq h_j^{\max} = c_p \Lambda_j \sin s_j$ define

$$Z_j(t) = z_1[1 - \eta(\Gamma_j(t - t_j^1))] - h\chi_1\eta_2(\Lambda_j\Gamma_j(t - t_j^1 - \Gamma_j^{-1})),$$

and $\dot{\alpha} = F_j^{\text{ctl}} - \dot{Z}_j/\zeta_{j,2}$. The fixed endpoint is $t_j^b = t_j^1 + \Gamma_j^{-1}(1 + \Lambda_j^{-1})$. Select the least h in the compact zero set $\Omega_j(t_j^b; h) = 0$ proved nonempty in the source. No uniqueness is asserted for $j \geq 2$.

4. Holding: use $\dot{\alpha} = F_j^{\text{ctl}}$ through t_{j+1}^a , concurrently with next-layer growth; evaluate $\sigma_j^2 = |G_{<j+1}(t_j^b)|$ only after selecting the stage. Since $\zeta_{j,1} = \Omega_j = 0$, both $\dot{\Omega}_j = 0$ and $\dot{\Theta}_j = 0$. This proves stationarity of this amplitude pair, while the older state and M_j may continue to evolve.

All these control maps are in physical time; $F_j^{\text{ctl}} = \nu_{\text{AB}} \hat{F}_j^{\text{ctl}}$, and Z_j is dimensionless. The source's smooth time joins follow from flatness of η , interior support of η_2 , and uniqueness for the stationary endpoint pair.

Exact aggregate parent evolution. The following identities explicitly give the background feedback rather than prescribing D, G independently.

Proposition 1. *For every finite activated system (162), put $G = G_{<q}$, $D = D_{<q}$, and $C = D_{21} - D_{12} = 2\dot{\alpha} + \sum_{i<q} w_i \Omega_i$. Then*

$$\dot{G} + D^T G = \sum_{1 \leq i < q} \dot{w}_i B_i, \quad (165)$$

$$\dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin \alpha + \sum_{1 \leq i < q} \dot{w}_i \Omega_i, \quad (166)$$

$$\begin{aligned} \dot{D} = \ddot{\alpha} J + \sum_{1 \leq i < q} (\dot{w}_i \Omega_i + w_i b_i \Theta_i) J e_i \otimes e_i \\ + \sum_{1 \leq i < q} w_i \Omega_i (\dot{\alpha} + \Sigma_i) (J e_i \otimes J e_i - e_i \otimes e_i), \\ \Sigma_i = \sum_{1 \leq k < i} w_k \Omega_k \sin^2(\phi_i - \phi_k). \end{aligned} \quad (167)$$

Proof. Differentiate $B_i = \lambda_i \Theta_i \zeta_i$ and use (162):

$$\dot{B}_i + D_{<i}^T B_i = -\Omega_i e_i (J e_i \cdot G_{<i}) = -S_i^T G_{<i}.$$

Also $\dot{B}_0 + (\dot{\alpha} J)^T B_0 = 0$. Expand $\dot{G} + D^T G$, insert these equalities, and keep all cross terms. Its nonactivation terms are

$$\sum_i w_i S_i^T B_0 - \sum_i w_i S_i^T \left(B_0 + \sum_{k < i} w_k B_k \right) + \sum_i w_i \sum_{k \geq i} w_k S_k^T B_i.$$

The B_0 terms cancel. The terms with $k > i$ in the last sum cancel the preceding double sum after exchanging its indices. Its diagonal terms are $w_i^2 S_i^T B_i = 0$, because $J e_i \cdot e_i = 0$. This proves (165) with every activation derivative retained. Next $\dot{C} = 2\ddot{\alpha} + \sum_i \dot{w}_i \Omega_i + \sum_i w_i \lambda_i \zeta_{i,1} \Theta_i$, while $G_1 = B_{0,1} + \sum_i w_i \lambda_i \Theta_i \zeta_{i,1}$ and $B_{0,1} = A_0 \sin \alpha$. Subtraction proves (166). Finally direct projection of $\dot{\zeta}_i = -D_{<i}^T \zeta_i$ onto e_i and $J e_i$ gives $\dot{e}_i = (\dot{\alpha} + \Sigma_i) J e_i$. Differentiate $S_i = \Omega_i J e_i \otimes e_i$, use $J^2 = -I$, and sum with the activation derivatives. This proves (167). $\square$

For completeness these affine dynamics have an exact momentum realization. On an affine region set $u = Dx$, $\theta = G \cdot x$, $H = D' + D^2 - e_2 \otimes G$, and

$$p = -\frac{1}{2}x^T(\text{sym } H)x, \quad f_\theta = (\dot{G} + D^T G) \cdot x, \quad f_u = \frac{1}{2}(\dot{C} - G_1)Jx.$$

Indeed $H - \text{sym } H$ is the antisymmetric matrix with entry difference $H_{21} - H_{12} = \dot{C} - G_1$; here $D^2 = -\det(D)I$ for a trace-free 2×2 matrix, verified by direct matrix multiplication. Thus $u_t + u \cdot \nabla u + \nabla p - \theta e_2 = f_u$. Both Laplacians $\Delta \theta$ and Δu are identically zero, so these same fields and forces solve the retained-viscosity equations on that region for every $\kappa_{\text{th}}, \varepsilon_v \geq 0$.

Relative geometry and its proof. Set $\rho_j = \log r_j$, $d_{ij} = \phi_j - \phi_i$, $\phi_0 = -\alpha$, and

$$\mathcal{G}_j = \sum_{0 \leq i < j} w_i A_i \sin d_{ij}.$$

Projection of the phase equation, using $Je_i \cdot e_j = -\sin d_{ij}$ and $e_i \cdot e_j = \cos d_{ij}$, gives

$$\dot{\rho}_j = \sum_{1 \leq i < j} w_i \Omega_i \sin d_{ij} \cos d_{ij}, \quad \dot{\phi}_j = -\dot{\alpha} - \Sigma_j, \quad \dot{d}_{ij} = \Sigma_i - \Sigma_j, \quad (168)$$

$$a_j = \mathcal{G}_j / (\lambda_j r_j), \quad \dot{\Omega}_j = -A_j \sin \phi_j, \quad (\log P_j)' = -\mathcal{G}_j \Omega_j / A_j,$$

$$F_j^{\text{ctl}} = -\Sigma_j + \dot{\rho}_j \tan \phi_j = \frac{\sum_{1 \leq i < j} w_i \Omega_i \sin \phi_i \sin d_{ij}}{\cos \phi_j}.$$

The last equality follows from $\sin \phi_i = \sin \phi_j \cos d_{ij} - \cos \phi_j \sin d_{ij}$. These are source (3.6), p. 9, and (3.55), p. 26, with A_0 retained.

For adjacent phases $\vartheta_j = \phi_j - \phi_{j-1}$, the matrix difference is $D_{<j} - D_{<j-1} = w_{j-1} \Omega_{j-1} J e_{j-1} \otimes e_{j-1}$. Its transpose annihilates ζ_{j-1} , so both ζ_{j-1} and ζ_j solve the same transport matrix equation. Its trace is zero, hence the derivative of their determinant is zero. At activation $\zeta_{j-1} = r_{j-1}(t_j^{\text{in}})e_2$ and $\zeta_j = e(s_j)$, and the determinant in that order is $-r_{j-1}(t_j^{\text{in}}) \sin s_j$. Therefore, with the sign of this orientation kept,

$$d_j^{\text{adj}} := r_j \sin \vartheta_j = \sin s_j e^{-\rho_{j-1}(t) + \rho_{j-1}(t_j^{\text{in}})}. \quad (169)$$

This is source (3.56), p. 26; its symbol d_j^{adj} denotes the source's separate $\mathfrak{d}_j$ rather than an angle d_{ij} .

For $q \geq 2$, all older activation weights are one on stage q . Define

$$E_2 = \sum_{i=1}^{q-2} \Omega_i \sin d_{iq} \cos d_{iq}, \quad E_3 = \dot{\rho}_{q-1}, \quad E_1 = \sin \vartheta_q \sum_{i=1}^{q-2} \Omega_i \sin(d_{iq} + d_{i,q-1}),$$

and $R_q = \sum_{i=0}^{q-2} A_i \sin d_{iq}$. Then

$$\dot{\vartheta}_q = -E_1 - \Omega_{q-1} \sin^2 \vartheta_q, \quad \dot{\rho}_q = E_2 + \Omega_{q-1} \sin \vartheta_q \cos \vartheta_q, \quad \mathcal{G}_q = A_{q-1} \sin \vartheta_q + R_q. \quad (170)$$

The first equality follows by subtracting the two angle equations and using $\sin^2 a - \sin^2 b = \sin(a-b)\sin(a+b)$; the other two are the corresponding split sums. At $q = 2$ the shear sums E_1, E_2, E_3 vanish, but $R_2 = A_0 \sin d_{02}$ is retained. The source makes this exceptional base term explicit on p. 28.

Exact later-stage model, including parent response. Use $\sigma = \sigma_{q-1}$, $s = s_q$, $\Gamma = \sigma \sin s$, $L = L_q$, $\tau = \Gamma(t - t_q^1)$, $\chi = r_q$, $\chi_1 = r_q(t_q^1)$, $W = \Omega_{q-1}/\sigma$, and $\mu_p = h/(L \sin s)$. Write $d = d_q^{\text{adj}}$, $d_1^* = d(t_q^1)$, and $e_4 = z_1/d_1^* - 1$. To avoid collision with the error coefficient d_1 , the asterisk here marks the fixed adjacent determinant factor at steering entry. Then source (3.66), p. 31, is exactly

$$c_1 = (d/\sin s) \cos \vartheta_q, \quad d_1 = \chi E_2 / \Gamma,$$

$$\begin{aligned}
c_2 = c_5 &= (A_{q-1}/\sigma^2)(d/\sin s), & c_3 &= \sin s/d, \\
d_3 &= 1 - (d_1^*/d)(1 + e_4), & c_4 &= (d_1^*/\sin s)(1 + e_4) = z_1/\sin s, \\
d_4 &= \frac{\chi_1 R_q}{\chi \sigma^2 \sin s}, & d_2 &= \frac{A_{q-1}}{\sigma^2 \sin s} \frac{d(\cos \phi_q - 1) - Z(\cos \vartheta_q - 1)}{\chi}.
\end{aligned} \tag{171}$$

Define $\tilde{k} = c_5 \chi_1 / \chi^2 + d_4$ and

$$\begin{aligned}
g(\tau) &= \begin{cases} \eta(\tau) + d_3(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ 1 + \mu_p c_3 L \chi_1 \eta_2(L(\tau - 1)), & 1 \leq \tau \leq 1 + L^{-1}, \end{cases} \\
f(\tau) &= \begin{cases} (c_4/\chi_1)(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ -\mu_p L \eta_2(L(\tau - 1)), & 1 \leq \tau \leq 1 + L^{-1}. \end{cases}
\end{aligned}$$

On every interval where $\Theta_q \neq 0$, put $v = \sigma \Omega_q / (\lambda_q \chi_1 \Theta_q)$. The exact system is

$$\chi' = c_1 W + d_1, \quad W' = c_2 g / \chi + d_2, \quad v' = f - \tilde{k} v^2, \quad (\log P_q)' = \tilde{k} v, \tag{172}$$

where these primes mean τ derivatives.

Proof. From (170), $\dot{r}_q = r_q E_2 + \Omega_{q-1} d \cos \vartheta_q$, proving the first equation. Next

$$\dot{\Omega}_{q-1} = A_{q-1} \sin(\vartheta_q - \phi_q) = \frac{A_{q-1}}{\chi} (d \cos \phi_q - Z \cos \vartheta_q).$$

Split its last parentheses as $d - Z + d(\cos \phi_q - 1) - Z(\cos \vartheta_q - 1)$. The profile gives $1 - Z/d = g$ on each of its two intervals, proving the second equation including the full d_2 . Also

$$\frac{a_q \lambda_q \chi_1}{\sigma \Gamma} = \frac{\chi_1}{\sigma^2 \sin s} \left(\frac{A_{q-1} d}{\chi^2} + \frac{R_q}{\chi} \right) = \tilde{k}, \quad \frac{\sigma b_q}{\Gamma \lambda_q \chi_1} = \frac{Z}{\chi_1 \sin s} = f.$$

Differentiate the quotient and $\log P_q$ to obtain the last equations. This proves the exact substitution without freezing $A_{q-1}, \Omega_{q-1}, \rho_{q-1}$, or any older layer. $\square$

Where source selection is proved. The source first proves interval-uniform bounds for (172) in Lemma 3.9, pp. 18–21, then excludes zeros of Θ_q in Proposition 3.10, pp. 21–22. For the actual stage it first constructs the closed older/geometry subsystem on a common compact subset of

$$\det M_j > 1/2, \quad |M_j^{-T} p_j| > 1/4, \quad \zeta_{q-1,2} > z_*/2, \quad \zeta_{q,2} > z_*/2,$$

by the simultaneous bounds in Sections 3.7.2–3.7.6, pp. 26–32. In original units the pair data and coefficient sizes change by the dictionary above, while this geometric domain is unchanged. The newest pair is absent from that subsystem: F_q^{ctl} uses only older vorticities and phases, and M_q uses only $D_{<q}$. One then solves the newest linear pair over the constructed subsystem. This order matters for the exact map (172). The source's $c_i = 1 + O(\epsilon_{\text{sch}})$, $d_i = O(\epsilon_{\text{sch}})$ estimates are consequences of its coupled induction, not independent prescriptions available after changing those dynamics.

For reference, source (3.67), p. 32, gives on the selected actual stage

$$\frac{\Omega_{q-1}(t_q^b)}{\sigma} \in [1/3 - C\epsilon_{\text{sch}}, 1 + \bar{v} + C\epsilon_{\text{sch}}], \quad \log \frac{P_q(t_q^b)}{(\nu_{\text{AB}}^2/\mu) \hat{\lambda}_q^{-7/8}} \in [1 + I_0 - C\epsilon_{\text{sch}}, 1 + \bar{v} + C\epsilon_{\text{sch}}],$$

where $I_0 = \int_0^1 [(1 + \tau^2/2)^2(1 + \tau)]^{-1} d\tau \in [2/9, \log 2]$. Source (3.58), p. 27, gives the separate first-stage gain interval $[1 + \log 2, 2 + \Lambda_1^{-1}]$ and exact duration $(L_1 + 2 + \Lambda_1^{-1})/(\nu_{\text{AB}} \sin s_*)$. These are source bounds, not claims that the diffusive evolution below satisfies them without further estimates.

An explicit growth-end vorticity lower bound. The source growth proof is Section 3.7.5, pp. 29–30, including (3.63)–(3.64), not the later steering quotient alone. Its identities admit the following explicit extraction from the source’s choice of low-order tolerance. Put $u = \Omega_q/\Theta_q$, $u_* = (b_q/a_q)^{1/2}$, $\gamma = (a_q b_q)^{1/2}$, and $z = u/u_*$. The source chooses its fixed error tolerance before its frequency threshold; require the following finite additional strict comparison when making that same choice: all the growth errors used here have absolute value at most $\delta \leq 1/4$. Thus its proved geometric and coefficient estimates give

$$r_q \geq 1 - \delta, \quad \frac{d_q^{\text{adj}}}{\sin s_q} \geq 1 - \delta, \quad \frac{\mathcal{G}_q}{\sigma_{q-1}^2 \sin s_q} \leq 1 + \delta, \quad |(\log u_*)'| \leq \delta \gamma.$$

The original source’s error rates in (3.63) are $C\epsilon_{\text{sch}}\Gamma_q/L_q$, while $\gamma = (1 + O(\epsilon_{\text{sch}}))\Gamma_q$, so this comparison is obtained by the same fixed-data choice of ϵ_{sch} ; it imposes no new order-dependent frequency condition.

Here is the complete quotient proof used for the extraction. Direct differentiation before any possible zero of Θ_q gives

$$u' = b_q - a_q u^2, \quad z' = \gamma(1 - z^2) - z(\log u_*)', \quad z(t_q^{\text{in}}) = 1.$$

At $z = 1 + \delta$, the right side is at most $\gamma[-2\delta - \delta^2 + \delta(1 + \delta)] = -\delta\gamma < 0$; at $z = 1 - \delta$ it is at least $\gamma[2\delta - \delta^2 - \delta(1 - \delta)] = \delta\gamma > 0$. The first-exit argument proves $1 - \delta \leq z \leq 1 + \delta$. This bound also excludes a first zero of Θ_q : at such a zero it would force $\Omega_q = 0$, contradicting backward uniqueness of the continuous-coefficient linear system from its nonzero entry datum. Consequently the bound holds through t_q^a . On growth the preceding phase is vertical, so $b_q = \lambda_q d_q^{\text{adj}}$ and $a_q = \mathcal{G}_q/(\lambda_q r_q)$. It follows exactly that

$$u_* = \lambda_q \sqrt{\frac{d_q^{\text{adj}} r_q}{\mathcal{G}_q}}, \quad \frac{|\Omega_q|}{|\Theta_q|} = z u_* \geq \frac{\lambda_q}{\sigma_{q-1}} \frac{(1 - \delta)^2}{\sqrt{1 + \delta}} \geq \frac{\lambda_q}{2\sigma_{q-1}}.$$

For the last inequality the function $(1 - \delta)^2/\sqrt{1 + \delta}$ is decreasing on $[0, 1/4]$ by direct logarithmic differentiation, and its value at $1/4$ is $9/(8\sqrt{5}) > 1/2$ because $81 > 80$. Thus the exact physical growth threshold satisfies

$$|\Omega_q(t_q^a)| \geq \frac{\lambda_q}{2\sigma_{q-1}} \frac{\nu_{\text{AB}}^2}{\mu} \hat{\lambda}_q^{-7/8} = \frac{\nu_{\text{AB}}^2}{2\sigma_{q-1}} \hat{\lambda}_q^{1/8}, \quad r_q(t_q^a) \geq 3/4. \quad (173)$$

For $q = 1$ the exact growing eigenline instead gives equality $|\Omega_1/\Theta_1| = \lambda_1/\nu_{\text{AB}}$ and $r_1 = 1$, which is stronger than the displayed lower bound. No unproved viscous tracking assertion is used in this extraction.

The actual profile and retained viscosity. The source profile, (4.1), p. 37, is

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad |s| \leq \pi,$$

periodically continued, where χ is even, smooth, compactly supported in $(-\pi, \pi)$ and one on $[-s_0, s_0]$. Thus $F = s$ on that interval. Direct differentiation of the local waves gives the retained-diffusion residual increments

$$\begin{aligned} \mathcal{R}_\theta &= (\dot{\Theta}_j - a_j \Omega_j) F(s) - \kappa_{\text{th}} \lambda_j^2 r_j^2 \Theta_j F''(s), \\ \mathcal{R}_\omega &= (\dot{\Omega}_j - b_j \Theta_j) F'(s) - \varepsilon_v \lambda_j^2 r_j^2 \Omega_j F'''(s). \end{aligned} \quad (174)$$

Consequently the actual fixed-profile, finite-stage retained-viscosity realization uses the original coupled system and the explicit additional forces $-\kappa_{\text{th}}\Delta\theta$ and $-\varepsilon_v\Delta u$. The signs in (174) are the signs of these forces when diffusion is placed on the right of the PDE. They are not positive damping terms in the forcing.

This repair preserves the exact actual D, G in Proposition 1: on every older affine region, $F'' = F''' = 0$, all envelope derivatives vanish, and the source's finite corrections vanish on that plateau. Hence the repaired force increments vanish in each affine core, and both Laplacians of the full older affine fields are zero. In physical coordinates the affine core of layer j contains

$$B_{r_j^{\text{aff}}(t)}, \quad r_j^{\text{aff}}(t) = \min \left\{ \frac{\ell_j}{2\|M_j^{-1}\|}, \frac{s_0}{\lambda_j r_j} \right\}.$$

The exact source nesting criterion is $K_j K_q \ell_q < \ell_j/2$ and $\lambda_j r_j K_q \ell_q < s_0$ for every older layer $j < q$; scaling by μ preserves both inequalities. The full-vector repair and the unchanged support follow because differentiation of a smooth compactly supported field does not enlarge its support. Quantitative force costs are provided separately in the parent calculation; the algebraic identity alone gives no uniform terminal force bound.

The exact feedback terms in a scalar-damped comparison. For transparency, compute also the altered finite-dimensional equations obtained from Fourier eigenprofiles. They are not the released fixed profile. Let $d_j^\theta = \kappa_{\text{th}} \lambda_j^2 r_j^2$ and $d_j^\omega = \varepsilon_v \lambda_j^2 r_j^2$, and replace only the two amplitude equations by

$$\dot{\Theta}_j = a_j \Omega_j - d_j^\theta \Theta_j, \quad \dot{\Omega}_j = b_j \Theta_j - d_j^\omega \Omega_j, \quad (175)$$

while computing all D, G, F_j^{ctl} from the altered amplitudes and phases. For a sine profile these are precisely the homogeneous local wave equations, since $F'' = -F$ and $F''' = -F'$. Repeating the proof of Proposition 1 gives exactly

$$\begin{aligned} \dot{G} + D^T G &= \sum_{i < q} \dot{w}_i B_i - \sum_{i < q} w_i d_i^\theta B_i, \\ \dot{C} - G_1 &= 2\ddot{\alpha} - A_0 \sin \alpha + \sum_{i < q} \dot{w}_i \Omega_i - \sum_{i < q} w_i d_i^\omega \Omega_i. \end{aligned} \quad (176)$$

In (167), replace $b_i \Theta_i$ by $b_i \Theta_i - d_i^\omega \Omega_i$. The geometric equations (168), (169), and (170) retain their formulas but now use altered amplitudes. The exact gradient-amplitude drift becomes

$$\frac{\dot{A}_i}{A_i} = \dot{\rho}_i - \frac{\mathcal{G}_i \Omega_i}{A_i} - d_i^\theta. \quad (177)$$

This follows by differentiating $A_i = \lambda_i r_i P_i$ and (175). In particular

$$\begin{aligned} \dot{\mathcal{G}}_j &= A_0 \cos d_{0j} (\Sigma_0 - \Sigma_j) + \sum_{1 \leq i < j} \dot{w}_i A_i \sin d_{ij} \\ &\quad + \sum_{1 \leq i < j} w_i A_i \left[\left(\dot{\rho}_i - \frac{\mathcal{G}_i \Omega_i}{A_i} - d_i^\theta \right) \sin d_{ij} + \cos d_{ij} (\Sigma_i - \Sigma_j) \right], \quad \Sigma_0 = 0. \end{aligned}$$

No base term or activation derivative has been discarded.

The feedback derivative is also explicit. With $N_{ij} = w_i \Omega_i \sin \phi_i \sin d_{ij}$,

$$\begin{aligned} \dot{F}_j^{\text{ctl}} &= \frac{1}{\cos \phi_j} \sum_{i < j} [(\dot{w}_i \Omega_i + w_i (b_i \Theta_i - d_i^\omega \Omega_i)) \sin \phi_i \sin d_{ij} \\ &\quad + w_i \Omega_i \cos \phi_i (-\dot{\alpha} - \Sigma_i) \sin d_{ij} + w_i \Omega_i \sin \phi_i \cos d_{ij} (\Sigma_i - \Sigma_j)] + F_j^{\text{ctl}} \tan \phi_j (-\dot{\alpha} - \Sigma_j). \end{aligned}$$

The direct damping term is therefore $-\sum_{i < j} w_i d_i^\omega \Omega_i \sin \phi_i \sin d_{ij} / \cos \phi_j$, in addition to all changes in the evolving state and the other terms displayed above.

For the later-stage variables and coefficients (171), direct differentiation gives the exact altered model

$$\chi' = c_1 W + d_1, \quad W' = c_2 g / \chi + d_2 - \frac{d_{q-1}^\omega}{\Gamma} W, \quad (178)$$

$$v' = f - \tilde{k}v^2 + \frac{d_q^\theta - d_q^\omega}{\Gamma}v, \quad (\log P_q)' = \tilde{k}v - \frac{d_q^\theta}{\Gamma}.$$

To prove the extra terms, $W' = \dot{\Omega}_{q-1}/(\sigma\Gamma)$ supplies its damping term; differentiating Ω_q/Θ_q supplies $(d_q^\theta - d_q^\omega)\Omega_q/\Theta_q$; differentiating $\log(-\Theta_q)$ supplies $-d_q^\theta$. The previous derivation supplies all remaining terms. These are identities on every interval where their displayed denominators are nonzero, and do not assert a new existence theorem from assumed coefficient bounds.

Even when $\kappa_{\text{th}} = \varepsilon_v$, only the explicit damping term in the quotient equation cancels. The preceding-layer term $-d_{q-1}^\omega W/\Gamma$ remains, A_{q-1} and R_q obey the altered gradient drifts, and hence c_2, c_5, d_4 evolve differently. On holding, $\zeta_{q,1} = \Omega_q = 0$ still solves those two equations, but

$$P_q(t) = P_q(t_q^b) \exp \left(-\kappa_{\text{th}} \lambda_q^2 \int_{t_q^b}^t r_q(s)^2 ds \right)$$

is no longer stationary if $\kappa_{\text{th}} > 0$. This follows by solving $\dot{P}_q = -d_q^\theta P_q$; r_q retains its evolving value inside the integral.

Finite-stage gain and exact diffusion integrals. For the damped comparison, (178) proves the exact identity

$$\log \frac{P_q(t_b)}{P_q(t_a)} = \int_{t_a}^{t_b} \left(-\frac{\mathcal{G}_q \Omega_q}{A_q} \right) dt - \kappa_{\text{th}} \lambda_q^2 \int_{t_a}^{t_b} r_q^2 dt.$$

For a stage partitioned into growth, transition, positive steering, negative steering, and holding, the last integral is the sum over all five actual intervals. Its original-unit coefficient is

$$\kappa_{\text{th}} \lambda_q^2 \int r_q^2 dt = \frac{\kappa_{\text{th}} \mu^2}{\nu_{\text{AB}}} \hat{\lambda}_q^2 \int \hat{r}_q^2 dT.$$

On any finite interval of length H with proved bounds $m \leq r_q \leq M$, its exact value is bracketed by $\kappa_{\text{th}} \lambda_q^2 m^2 H$ and $\kappa_{\text{th}} \lambda_q^2 M^2 H$. The inequality is simply the integrated pointwise inequality $m^2 \leq r_q^2 \leq M^2$. It does not permit retaining the original source gain integral when the coupled parent trajectory has changed. Conversely, in the actual fixed-profile force repair the trajectory and the gain integral remain exactly those of the source, and the diffusion cost is paid by the additional explicit force, not by a damping loss in that trajectory.

Status of the infinite sequence. The source constructs its inviscid infinite sequence after fixing low-order design bounds, then all coefficient/correction constants, then the nondecreasing majorant K of (7.21), p. 63, and finally one initial frequency threshold in (8.1)–(8.6), pp. 66–67. It proves the coupled stages before evaluating each new σ_q , and it proves full-lifetime bounds before applying the finite corrections. The retained-viscosity finite-stage force repair proved here commutes exactly with every finite activated sum. Extending its force series smoothly to the limiting time requires estimates for the additional Laplacian terms; no such terminal estimates have been established in this document. The exact Fourier comparison (178) does not supply them, and its altered parent terms prevent importing the source's inviscid coefficient bounds without a new calculation. Neither statement is a disproof of another retained-viscosity construction. This calculation concerns the precise periodic profile in Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, released 76-page source, Sections 3.1 and 4.1, equations (3.1), (3.2), (4.1)–(4.4). It proves exact identities and a profile evolution for a prescribed affine background, including the exact force that keeps the original profile. It also identifies the change to the affine region when that force is omitted. No conclusion about an infinite viscous controlled sequence follows from these profile identities alone.

9 The original profile and all coefficients

Let $0 < s_0 < 1$ and let $\chi \in C_c^\infty((-\pi, \pi))$ be even and equal to one on $[-s_0, s_0]$. The actual source profile is the 2π -periodic extension of

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad -\pi < s < \pi. \quad (179)$$

It is smooth because it equals $\sin s$ near the endpoints of this period; it is odd because χ is even. Hence its period mean is zero. It equals s on $[-s_0, s_0]$, in particular $F'(0) = 1$. Let P be its unique mean-zero periodic primitive. Indeed $\tilde{P}(s) = \int_0^s F(r) dr$ is even and periodic, and $P = \tilde{P} - \langle \tilde{P} \rangle$ has the claimed properties. Here $\langle h \rangle = (2\pi)^{-1} \int_{-\pi}^{\pi} h(s) ds$.

Fix a closed finite time interval $I = [t_0, t_1]$, smooth $D : I \rightarrow \mathbb{R}^{2 \times 2}$ with $\text{tr } D = 0$, smooth $G : I \rightarrow \mathbb{R}^2$, a frequency $\lambda > 0$, and a nonzero initial vector $\zeta(t_0)$. Define $J(x_1, x_2) = (-x_2, x_1)$ and

$$\dot{\zeta} = -D^T \zeta, \quad s(x, t) = \lambda \zeta(t) \cdot x, \quad \kappa(t) = |\zeta(t)|^2, \quad K(t) = \lambda^2 \kappa(t). \quad (180)$$

The notation $\kappa = |\zeta|^2$ is the source's geometric notation. The physical temperature diffusivity is $\kappa_{\text{th}} \geq 0$, and the physical momentum viscosity is $\nu_{\text{phys}} \geq 0$; neither is the paper's time-scaling parameter named ν . These constants are not set to one. This calculation does not rescale A_0, λ_0, μ , the physical coordinates, time, or laboratory gravity $e_2 = (0, 1)$.

The nonvanishing of ζ follows directly from the invertible fundamental matrix of $\dot{M} = DM$, $M(t_0) = \text{Id}$: differentiation gives $\zeta = M^{-T} \zeta(t_0)$, and differentiation of the determinant gives $\det M = \exp \int_{t_0}^t \text{tr } D = 1$. Write

$$a(t) = -\frac{J\zeta \cdot G}{\lambda \kappa}, \quad b(t) = \lambda \zeta_1. \quad (181)$$

In particular the buoyancy coefficient uses the first laboratory component ζ_1 .

For any smooth scalar θ and divergence-free velocity u , with $\omega = \partial_1 u_2 - \partial_2 u_1$, define the full retained-diffusion residuals by

$$\mathcal{R}_\theta(\theta, u) = \partial_t \theta + u \cdot \nabla \theta - \kappa_{\text{th}} \Delta \theta, \quad \mathcal{R}_\omega(\theta, u) = \partial_t \omega + u \cdot \nabla \omega - \partial_1 \theta - \nu_{\text{phys}} \Delta \omega. \quad (182)$$

Thus these residuals are the required scalar force and curl of the required momentum force. On the region under consideration the older fields are $\theta_{\text{old}} = G \cdot x$, $u_{\text{old}} = Dx$; they need not have zero residual. All statements below subtract their actual residuals.

10 Exact residual and failure of constant-profile damping

Proposition 1. *For arbitrary smooth real amplitudes Θ, Ω , define the source wave*

$$\vartheta = \Theta F(s), \quad \psi = \frac{\Omega}{\lambda^2 \kappa} P(s), \quad v = \nabla^\perp \psi = \frac{\Omega}{\lambda \kappa} J \zeta F(s), \quad \varpi = \Delta \psi = \Omega F'(s). \quad (183)$$

The scalar and curl residual increments are exactly

$$\mathcal{R}_\theta(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\theta(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Theta} - a\Omega)F - \kappa_{\text{th}} K \Theta F'', \quad (184)$$

$$\mathcal{R}_\omega(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\omega(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Omega} - b\Theta)F' - \nu_{\text{phys}} K \Omega F'''. \quad (185)$$

Every profile in these formulas is evaluated at $s(x, t)$.

Proof. Equation (180) gives $(\partial_t + Dx \cdot \nabla)s = \lambda(\dot{\zeta} + D^T \zeta) \cdot x = 0$. The orthogonality $J\zeta \cdot \zeta = 0$ gives both $v \cdot \nabla \vartheta = 0$ and $v \cdot \nabla \varpi = 0$. The older vorticity $D_{21} - D_{12}$ is spatially constant, so $v \cdot \nabla \omega_{\text{old}} = 0$. Moreover $v \cdot G = -a\Omega F$, $\partial_1 \vartheta = b\Theta F'$, $\Delta \vartheta = K\Theta F''$, and $\Delta \varpi = K\Omega F'''$. Expanding (182), canceling the older residual, and retaining all these terms proves (184)–(185). The displayed velocity and vorticity in (183) follow by differentiating $P' = F$ twice; their signs follow from the specified J . $\square$

The profile derivatives, including their transition-region terms, are

$$F' = \cos s + \chi'(s)(s - \sin s) + \chi(s)(1 - \cos s), \quad (186)$$

$$F'' = (\chi - 1) \sin s + 2\chi'(1 - \cos s) + \chi''(s - \sin s), \quad (187)$$

$$F''' = (\chi - 1) \cos s + 3\chi' \sin s + 3\chi''(1 - \cos s) + \chi'''(s - \sin s). \quad (188)$$

These follow by the product rule on $(-\pi, \pi)$ and extend smoothly across the period endpoints.

Proposition 2. *There is no constant $c \in \mathbb{R}$ such that $F'' = cF$ on the circle, and no constant $c \in \mathbb{R}$ such that $F''' = cF'$ on the circle. Consequently, at a time with $\kappa_{\text{th}} > 0$ and $\Theta \neq 0$, no choice of scalar $\dot{\Theta} - a\Omega$ makes (184) vanish for every phase. At a time with $\nu_{\text{phys}} > 0$ and $\Omega \neq 0$, no choice of scalar $\dot{\Omega} - b\Theta$ makes (185) vanish for every phase.*

Proof. On $(-s_0, s_0)$, $F(s) = s$ and $F'' = 0$. An identity $F'' = cF$ therefore implies $cs = 0$ for every such s , so $c = 0$. Then $F'' = 0$ everywhere would make F globally affine. Its periodicity would force its slope to be zero, contradicting $F'(0) = 1$. If $F''' = cF'$, evaluation inside that same interval gives $0 = c$, since $F' = 1$ there. Then $F''' = 0$ everywhere makes F a quadratic polynomial on $\mathbb{R}$. Periodicity forces both its quadratic and linear coefficients to vanish, again contradicting $F'(0) = 1$. Since $K > 0$, each asserted residual cancellation with a nonzero amplitude would imply precisely one of these excluded constant-coefficient identities, by division by $\kappa_{\text{th}}K\Theta$ or $\nu_{\text{phys}}K\Omega$. $\square$

For example a proposed common damping equation $\dot{\Theta} = a\Omega - d\Theta$, $\dot{\Omega} = b\Theta - d\Omega$ gives $-d\Theta F - \kappa_{\text{th}}K\Theta F''$ and $-d\Omega F' - \nu_{\text{phys}}K\Omega F'''$ as its two residuals. On the exact affine core they are $-d\Theta s$ and $-d\Omega$. This proves failure of that fixed-profile amplitude prescription, not failure of a viscous controlled construction.

11 An exact evolving-profile map for all diffusivities

We now construct the profile evolution instead of replacing the missing profile dynamics by scalar damping. Let $T(s, t), V(s, t)$ be smooth, mean-zero periodic functions, and let $Q_s = V$, $\langle Q \rangle = 0$. Set

$$\vartheta = T(s, t), \quad \psi = \frac{Q(s, t)}{\lambda^2 \kappa}, \quad v = \frac{J\zeta}{\lambda \kappa} V(s, t), \quad \varpi = V_s(s, t). \quad (189)$$

The time derivative T_t here and below holds phase s fixed.

Proposition 3. *The residual increments for (189) are*

$$T_t - aV - \kappa_{\text{th}}KT_{ss}, \quad \partial_s(V_t - bT - \nu_{\text{phys}}KV_{ss}). \quad (190)$$

Their vanishing is equivalent in the mean-zero class to

$$T_t = aV + \kappa_{\text{th}}KT_{ss}, \quad V_t = bT + \nu_{\text{phys}}KV_{ss}. \quad (191)$$

For the initial data $T(s, t_0) = \Theta_0 F(s)$, $V(s, t_0) = \Omega_0 F(s)$, this system has a unique smooth periodic solution on I given by the following exact Fourier evolution. Write

$$F(s) = \sum_{n \geq 1} f_n \sin(ns), \quad f_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(s) \sin(ns) ds. \quad (192)$$

For each $n \geq 1$, let $U_n(t, t_0)$ solve

$$\partial_t U_n = \begin{pmatrix} -\kappa_{\text{th}}Kn^2 & a \\ b & -\nu_{\text{phys}}Kn^2 \end{pmatrix} U_n, \quad U_n(t_0, t_0) = \text{Id}. \quad (193)$$

Then

$$\begin{pmatrix} T(s, t) \\ V(s, t) \end{pmatrix} = \sum_{n \geq 1} f_n U_n(t, t_0) \begin{pmatrix} \Theta_0 \\ \Omega_0 \end{pmatrix} \sin(ns). \quad (194)$$

Proof. The phase and tangency computations in Proposition 1 remain valid, including $v \cdot \nabla T = v \cdot \nabla V_s = 0$. Now $v \cdot G = -aV$ and $\partial_1 T = bT_s$, which proves (190). The expression inside its second derivative has zero period mean. If its phase derivative vanishes, it is constant in s , and its zero mean forces this constant to be zero. This proves the exact equivalence to (191).

For completeness, each matrix solution in (193) can be defined by the convergent iterated-integral series $U_n = \text{Id} + \sum_{j \geq 1} \int_{t_0 < r_j < \dots < r_1 < t} A_n(r_1) \cdots A_n(r_j) dr_j \cdots dr_1$, where A_n is the displayed matrix. For a fixed n , its j th term is bounded by $(|I| \sup_I \|A_n\|)^j / j!$, so termwise differentiation gives the ODE. Uniqueness follows by integrating the difference and applying the iterated integral estimate, which tends to zero as $j \rightarrow \infty$. For its column solution $y = (y_1, y_2)$, direct differentiation gives

$$\frac{1}{2} \partial_t |y|^2 = -Kn^2(\kappa_{\text{th}} y_1^2 + \nu_{\text{phys}} y_2^2) + (a+b)y_1 y_2 \leq \frac{1}{2}(|a| + |b|)|y|^2.$$

Multiplication by $\exp(-\int_{t_0}^t (|a| + |b|))$ and differentiation therefore proves the bound, uniform in n ,

$$|y(t)| \leq e^{\frac{1}{2} \int_{t_0}^t (|a| + |b|)} |y(t_0)|.$$

Repeated differentiation of (193) gives, for each fixed j , $\sup_I |\partial_t^j y| \leq C_j(1+n)^{2j} |y(t_0)|$, by induction: the coefficient derivatives of A_n are bounded by constants times $(1+n)^2$, and the differentiated product has finitely many terms.

Integration by parts any prescribed number of times in (192) shows $|f_n| \leq C_N n^{-N}$ for every N , without endpoint terms by periodicity. Thus (194) and every mixed derivative converge uniformly on $\mathbb{T} \times I$, and termwise differentiation proves the equations and the prescribed initial values. The Fourier expansion indeed equals F : its absolutely convergent series has the same Fourier coefficients as F , and a continuous function with all Fourier coefficients zero is zero. To see this last assertion directly, convolve with the Fejér kernel $\mathfrak{F}_N(r) = N^{-1}(\sin(Nr/2)/\sin(r/2))^2$. It has integral 2π , is nonnegative, and its integral outside every fixed neighborhood of 0 tends to zero because its numerator is at most one and the denominator is bounded below there. Uniform continuity therefore gives uniform convergence of its convolution to the continuous function. When all Fourier coefficients are zero each such convolution is zero, proving the assertion. The same Fourier coefficient ODE and its uniqueness show uniqueness among smooth solutions of (191). $\square$

Proposition 4. *The map in Proposition 3 also realizes the full momentum equation. Let $S_s = T$, $\langle S \rangle = 0$, and retain $Q_s = V$, $\langle Q \rangle = 0$. For a solution of (191), the exact pressure increment is*

$$p^{\text{wave}}(x, t) = -\frac{2\zeta \cdot DJ\zeta}{\lambda^2 \kappa^2} Q(s, t) + \frac{\zeta_2}{\lambda \kappa} S(s, t). \quad (195)$$

With this pressure increment, adding (T, v) to any affine older state preserves its actual vector momentum residual, including retained viscosity ν_{phys} and the unchanged laboratory buoyancy θe_2 .

Proof. Put $c = J\zeta/(\lambda\kappa)$, a vector in this proof. The vector momentum residual increment before adding pressure is

$$(\dot{c} + Dc)V + c(V_t - \nu_{\text{phys}}KV_{ss}) - Te_2.$$

Indeed the transported phase has zero material derivative under Dx , $v \cdot \nabla v = 0$, $v \cdot \nabla(Dx) = Dv$, and $\Delta v = KcV_{ss}$. For every trace-free 2×2 matrix, $DJ = -JD^T$; multiplying its four entries verifies the identity. It gives $\dot{c} = Dc - c\dot{\kappa}/\kappa$ and $\dot{\kappa} = -2\zeta \cdot D\zeta$. The identity $(J\zeta) \cdot D(J\zeta) + \zeta \cdot D\zeta = \kappa \text{tr } D = 0$ shows that $\dot{c} + Dc$ is orthogonal to $J\zeta$. Its component along ζ is therefore

$$\dot{c} + Dc = \frac{2\zeta \cdot DJ\zeta}{\lambda \kappa^2} \zeta.$$

Also $bc - e_2 = -\zeta_2 \zeta / \kappa$, by the two explicit components of $J\zeta = (-\zeta_2, \zeta_1)$. Substituting the second profile equation reduces the residual to

$$\zeta \left[\frac{2\zeta \cdot DJ\zeta}{\lambda \kappa^2} V - \frac{\zeta_2}{\kappa} T \right].$$

The spatial gradient of (195) is its negative, because $\nabla s = \lambda\zeta$, $Q_s = V$ and $S_s = T$. This proves the full momentum statement. Mean-zero periodic primitives exist because the profiles have zero mean, by the same integration argument used for P above. $\square$

Proposition 5. *If $\kappa_{\text{th}} = \nu_{\text{phys}} = \delta \geq 0$, put*

$$\tau(t) = \delta\lambda^2 \int_{t_0}^t |\zeta(r)|^2 dr, \quad F_\tau(s) = \mathcal{H}_\tau F(s) = \sum_{n \geq 1} f_n e^{-n^2 \tau} \sin(ns). \quad (196)$$

Let $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$ with the given initial amplitudes. The exact solution in Proposition 3 is

$$T = \Theta F_\tau, \quad V = \Omega F_\tau, \quad Q = \Omega \mathcal{H}_\tau P, \quad \varpi = \Omega \partial_s F_\tau. \quad (197)$$

In particular this is an exact retained-diffusion map with the same prescribed D, G, ζ , all original frequency factors, and laboratory gravity. Each nonzero temperature mode has exact gain

$$\frac{T_n(t)}{T_n(t_0)} = \frac{\Theta(t)}{\Theta_0} e^{-n^2 \tau(t)} \quad (f_n \Theta_0 \neq 0). \quad (198)$$

For vorticity the corresponding ratio is $(\Omega(t)/\Omega_0)e^{-n^2 \tau(t)}$ when $nf_n\Omega_0 \neq 0$.

Proof. The matrix in (193) is $B(t) - \delta K(t)n^2 \text{Id}$, where $B = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$. If U solves $\dot{U} = BU$, differentiation of $e^{-n^2 \tau(t)} U(t, t_0)$ gives exactly (193), including its initial identity. Hence uniqueness proves this factorization and (197). Since derivatives commute termwise with the Fourier series, $(\mathcal{H}_\tau P)_s = \mathcal{H}_\tau F$ and its mean remains zero. This verifies the streamfunction as well as the scalar and vorticity profiles. Formula (198) is division of the actual mode coefficients, retaining its stated nonzero denominator. $\square$

The exact temperature phase norm is

$$\|T(\cdot, t)\|_{L^2(-\pi, \pi)}^2 = \pi \Theta(t)^2 \sum_{n \geq 1} f_n^2 e^{-2n^2 \tau(t)}. \quad (199)$$

This follows by integrating the uniformly convergent squared series and using the exact orthogonality identity

$$\int_{-\pi}^{\pi} \sin(ns) \sin(ms) ds = \pi \mathbf{1}_{n=m}.$$

Thus $\|F_\tau\|_2 \leq e^{-\tau} \|F\|_2$, and for any index with $f_n \neq 0$ one has $\|F_\tau\|_2 \geq \sqrt{\pi} |f_n| e^{-n^2 \tau}$. These are inequalities about phase profiles; an unlocalized affine wave need not be integrable over $\mathbb{R}^2$.

12 Positive heat time and the affine core

Proposition 6. *For every $\tau > 0$, $\mathcal{H}_\tau F$ has no affine restriction to any nonempty open interval of $\mathbb{R}$. In particular it cannot retain the source's nontrivial exact linear phase interval. Also F has infinitely many nonzero Fourier modes.*

Proof. For any $R > 0$ and any nonnegative integer m , the series of m th complex derivatives of $\sum_{n \geq 1} f_n e^{-n^2 \tau} \sin(nz)$ is bounded on $|\text{Im } z| \leq R$ by $\sum_{n \geq 1} |f_n| n^m e^{-n^2 \tau + nR} < \infty$. For instance boundedness of f_n and completion of the square make the tail bounded by a polynomial times $e^{-\tau n^2/2}$. The sum therefore defines an entire function $h(z)$ and has its termwise derivatives. If $h(s) = cs + d$ on a nonempty real open interval, the entire function $h(z) - cz - d$ vanishes identically. The identity principle used here follows from Taylor expansion: a nonzero holomorphic function has a first nonzero Taylor coefficient at each zero where its Taylor series is not identically zero, hence that zero is isolated; an accumulating set of zeros forces the Taylor series to vanish locally, and this local vanishing propagates through the connected domain. Periodicity now gives $2\pi c =$

$h(z + 2\pi) - h(z) = 0$. Oddness gives $d = 0$. Thus every Fourier coefficient $f_n e^{-n^2 \tau}$ would vanish, implying $f_n = 0$ for all n . The Fourier uniqueness argument in Proposition 3 would imply $F = 0$, contradicting $F'(0) = 1$. If F had only finitely many nonzero modes, the same entire-function argument, applied to its finite trigonometric sum at heat time zero, would contradict its interval $F(s) = s$. $\square$

There is nevertheless an exact estimate of the defect deep inside the original interval. Define, for $\tau > 0$ and $d > 0$,

$$\mathcal{E}(d, \tau) = \frac{2}{\sqrt{\pi}} \int_{d/(2\sqrt{\tau})}^{\infty} e^{-r^2} dr \leq \frac{2\sqrt{\tau}}{d\sqrt{\pi}} e^{-d^2/(4\tau)}. \quad (200)$$

The inequality follows from $r \geq d/(2\sqrt{\tau})$ and $\int_a^{\infty} r e^{-r^2} dr = e^{-a^2}/2$.

Proposition 7. For $0 < r_0 < s_0$, $d = s_0 - r_0$, and $|s| \leq r_0$,

$$|\partial_s F_\tau(s) - 1| \leq \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (201)$$

$$|F_\tau(s) - s| \leq |s| \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (202)$$

$$|\partial_s^m F_\tau(s)| \leq \|F^{(m)}\|_\infty \mathcal{E}(d, \tau) \quad (m \geq 2). \quad (203)$$

Proof. The periodized Gaussian kernel of $\mathcal{H}_\tau$ is $\sum_{k \in \mathbb{Z}} (4\pi\tau)^{-1/2} e^{-(r+2\pi k)^2/(4\tau)}$. To verify this representation, its n th Fourier multiplier is the Gaussian integral $\int_{\mathbb{R}} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} e^{-inr} dr = e^{-n^2 \tau}$. One direct verification of this integral differentiates it with respect to n , integrates the Gaussian derivative by parts, and obtains $I'(n) = -2\tau n I(n)$ with $I(0) = 1$; the last normalization is the usual Gaussian integral, obtained by squaring it and integrating in polar coordinates. Fourier uniqueness then proves the representation. For any periodic bounded function h vanishing on $[-s_0, s_0]$, its periodic lift satisfies $h(s - r) = 0$ whenever $|s| \leq r_0$ and $|r| < d$. The Gaussian representation on the line therefore gives

$$|\mathcal{H}_\tau h(s)| \leq \|h\|_\infty \int_{|r| \geq d} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} dr = \|h\|_\infty \mathcal{E}(d, \tau).$$

Apply this to $h = F' - 1$ and $h = F^{(m)}$ for $m \geq 2$, using commutation of derivatives with $\mathcal{H}_\tau$ and $\mathcal{H}_\tau 1 = 1$. Finally $F_\tau(0) = 0$ by oddness; integrating the first bound between 0 and s proves the second bound with the same d . $\square$

The source background in equation (3.3) is obtained because each older layer is exactly affine on the newer support, by equations (4.3)–(4.6). For a heat-evolved older layer with nonzero scalar or velocity amplitude, Proposition 6 removes this exact affine premise. Its gradient at the origin can still be computed, but its values away from the origin are not that gradient times x . The tail estimates give explicit defects rather than an identification of those two fields. Consequently Proposition 5 alone supplies no coupled feedback law for the heat-modified older background.

13 Exact preservation by force and the localized diffusion costs

For the original inviscid amplitude equations $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$, Proposition 1 shows that the exact additional scalar and curl forces are

$$f_{\theta, \text{diff}} = -\kappa_{\text{th}} K \Theta F'', \quad E_{\text{diff}} = -\nu_{\text{phys}} K \Omega F'''. \quad (204)$$

They vanish on the entire affine interval $|s| < s_0$. This retains exactly the original core, amplitudes, phase and hence its local affine feedback; it pays a nonzero diffusion force in the rest of the profile. In the unlocalized wave this force is periodic in phase and is not a compactly supported force on $\mathbb{R}^2$.

Here is the exact compact counterpart for the leading fields in source equation (4.2). Let M solve $\dot{M} = DM$, $M(t_0) = \text{Id}$, let $p = \zeta(t_0)$, and let $\ell > 0$. Fix the source's radial $f \in C_c^\infty(B_1)$, $f = 1$ on $B_{1/2}$, and put

$$\begin{aligned} a_{\text{mat}} &= M^{-1}x, & g(a_{\text{mat}}) &= f(a_{\text{mat}}/\ell), \\ \theta^L &= w\Theta F(s)g(a_{\text{mat}}), & \psi^L &= \frac{w\Omega}{\lambda^2\kappa}P(s)g(a_{\text{mat}}), & u^L &= \nabla^\perp \psi^L. \end{aligned} \quad (205)$$

The activation $w(t)$ remains the original smooth scalar function; no time differentiation of it is omitted from the existing inviscid residual. The following formulas give exactly the additional diffusion terms, with w retained. At a fixed time put

$$B = M^{-1}, \quad H = BB^T, \quad q = B\zeta, \quad \mathcal{L} = \lambda^2\kappa\partial_s^2 + 2\lambda q \cdot \nabla_a \partial_s + H : \nabla_a^2. \quad (206)$$

Although $s = \lambda p \cdot a_{\text{mat}}$ upon evaluation, s, a are independent variables in this differential operator.

Proposition 8. *For every smooth profile $A(s, a)$, $\Delta_x[A(\lambda\zeta \cdot x, Bx)] = (\mathcal{L}A)(\lambda\zeta \cdot x, Bx)$. Consequently the exact added forces for (205) are*

$$f_{\theta, \text{diff}}^L = -\kappa_{\text{th}}w\Theta[\lambda^2\kappa F''g + 2\lambda F'q \cdot \nabla g + F(H : \nabla^2g)], \quad (207)$$

$$f_{u, \text{diff}}^L = -\nu_{\text{phys}}\Delta u^L = -\nu_{\text{phys}}\nabla^\perp \Delta \psi^L, \quad E_{\text{diff}}^L = -\nu_{\text{phys}}\frac{w\Omega}{\lambda^2\kappa}\mathcal{L}^2(Pg). \quad (208)$$

Writing $\mathcal{H} = H : \nabla_a^2$, the last operator is explicitly

$$\begin{aligned} \mathcal{L}^2(Pg) &= \lambda^4\kappa^2F'''g + 4\lambda^3\kappa F''q \cdot \nabla g + 2\lambda^2\kappa F'q \cdot \nabla \mathcal{H}g + 4\lambda^2F'(q \cdot \nabla)^2g \\ &\quad + 4\lambda F(q \cdot \nabla)\mathcal{H}g + P\mathcal{H}^2g. \end{aligned} \quad (209)$$

All these forces are supported in $M\bar{B}_\ell$. They vanish in the exact affine region $MB_{\ell/2} \cap \{|s| < s_0\}$.

Proof. The chain rule gives $\partial_{x_i}A = \lambda\zeta_i A_s + \sum_j B_{ji}A_{a_j}$. Applying it a second time and summing in i gives the three terms in (206), since $\sum_i B_{ji}\zeta_i = (B\zeta)_j$ and $\sum_i B_{ji}B_{ki} = (BB^T)_{jk}$. Applying the same identity twice to Pg proves (208). Expanding the square of the commuting constant-in- (s, a) operators in (206), and using $P' = F$, $P'' = F'$, $P''' = F''$, $P'''' = F'''$, proves all six terms of (209). Spatial differentiation does not enlarge the support of a smooth compactly supported function, proving the support statement. On the plateau $g = 1$ all positive derivatives of g vanish. On the linear phase interval $F'' = F''' = 0$. Thus the scalar and curl formulas vanish on their intersection. There $\psi^L = (w\Omega/(\lambda^2\kappa))(P(0) + s^2/2)$ is quadratic in x , so u^L is linear and $\Delta u^L = 0$, proving the vector-force statement as well. Finally $\text{curl}(-\nu_{\text{phys}}\Delta u^L) = -\nu_{\text{phys}}\Delta^2\psi^L$, so the specified vector force realizes the curl force without any inverse-curl ambiguity. $\square$

For explicit quantitative costs let $C_j = \sup_a \|\nabla^j f(a)\|_{\text{HS}}$, with the Euclidean Hilbert–Schmidt tensor norm, and $C_0 = \|f\|_\infty$. Then $\|\nabla^j g\|_\infty = C_j \ell^{-j}$ by the chain rule. Write $h = \|H\|_{\text{HS}}$ and $q_* = |q|$. Cauchy–Schwarz on the tensor contractions in (207) and (209) proves

$$\|f_{\theta, \text{diff}}^L\|_\infty \leq \kappa_{\text{th}}|w\Theta|[\lambda^2\kappa\|F''\|_\infty C_0 + 2\lambda q_*\|F'\|_\infty C_1 \ell^{-1} + h\|F\|_\infty C_2 \ell^{-2}], \quad (210)$$

$$\begin{aligned} \|E_{\text{diff}}^L\|_\infty &\leq \frac{\nu_{\text{phys}}|w\Omega|}{\lambda^2\kappa}[\lambda^4\kappa^2\|F''' \|_\infty C_0 + 4\lambda^3\kappa q_*\|F''\|_\infty C_1 \ell^{-1} \\ &\quad + (2\lambda^2\kappa h + 4\lambda^2 q_*^2)\|F'\|_\infty C_2 \ell^{-2} + 4\lambda q_* h\|F\|_\infty C_3 \ell^{-3} + h^2\|P\|_\infty C_4 \ell^{-4}]. \end{aligned} \quad (211)$$

The same reasoning applies to every compact correction: for a specified smooth correction (θ^c, ψ^c) the extra scalar and vector forces are exactly $-\kappa_{\text{th}}\Delta\theta^c$ and $-\nu_{\text{phys}}\nabla^\perp \Delta\psi^c$, with curl $-\nu_{\text{phys}}\Delta^2\psi^c$. This is the exact linear map from an already constructed inviscid forced field to a retained-diffusion forced field:

$$(\theta, u, p, f_\theta, f_u) \mapsto (\theta, u, p, f_\theta - \kappa_{\text{th}}\Delta\theta, f_u - \nu_{\text{phys}}\Delta u). \quad (212)$$

Substitution in the scalar and momentum equations proves the map: adding $\kappa_{\text{th}}\Delta\theta$ or $\nu_{\text{phys}}\Delta u$ to the new right-hand side cancels its respective force decrement. Thus pressure, divergence, initial data and the entire trajectory are retained exactly. On each closed finite smooth stage these are smooth finite forces; where the original fields and their derivatives are compactly supported, the new terms have the same support. This statement supplies no uniform bound as the layer index tends to infinity. The explicit factors λ^2 , ℓ^{-2} , and ℓ^{-4} above must be retained in such an analysis.

Source and scope receipt

The primary source read here is the 76-page PDF *Blowup for the Boussinesq equations with smooth forcing*, Levent Alpöge and Tristan Buckmaster; source equations (3.1)–(3.6) are on pages 8–9 and (4.1)–(4.6) on pages 37–38. The exact file SHA-256 is

895a628d1783bcb039374686f50b895b5f450f53b8ef8aa173523487a7a4a21b.

The companion replay script checks the profile differentiation, full two-dimensional residual algebra, localized Laplacian square, common diffusivity matrix factorization, and eigenfunction obstruction identities. Its checks supplement the proofs above; they do not certify an infinite-stage existence or force-smoothness theorem.

14 The actual third growth target on the moving second hold

We continue from the actual second returned state, with its first vorticity strictly positive. The third wave sees both older temperature gradients and the changing first shear. This section proves attainment of its original growth threshold, with the original third insertion rule and retained positive physical diffusion. The second parent is held by its original feedback during this interval. The third transition changes that control and requires its own calculation; no third return is inferred from the growth result.

14.1 A further explicit choice in the same source family

Retain every original datum, profile, frequency and constant from Section 4. In particular $\nu = \sqrt{A_0}$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$ and $K_j = \mu \hat{\lambda}_j$ are still physical quantities. Write $\sigma_1 = \sigma$ for the actual second-stage scale in (78). Extend the exact source recursion to

$$\begin{aligned} Q_3 &= Q_* + 3, & \hat{\lambda}_3 &= \hat{\lambda}_2^{Q_3}, & k_3 &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_3, K(k) \leq \log \hat{\lambda}_3\}, \\ L_3 &= (k_3 + 41/8) \log \hat{\lambda}_3, & S_3 &= (A_0/\mu) \hat{\lambda}_3^{-k_3-6}, & P_3^* &= (A_0/\mu) \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}. \end{aligned} \quad (213)$$

The finite maximum exists for the same reason as the earlier two. Define fixed numbers

$$\begin{aligned} c_2 &= (41/8)Q_2, & c_3 &= (41/8)Q_2Q_3, \\ C_3 &= (\lfloor Q_3/120 \rfloor + 41/8)Q_2Q_3, \\ C_2 &= e^9 \sqrt{5\sqrt{10}/4}, & C_M &= \frac{256C_2}{15\sqrt{e}}. \end{aligned}$$

Then $L_2 \geq c_2 \log \hat{\lambda}_1$, $L_3 \geq c_3 \log \hat{\lambda}_1$, and $L_3 \leq C_3 \log \hat{\lambda}_1$ for all these exact choices. Using the already defined Y_0, C_σ, C_N , select

$$Y_3 = 1 + \max \left\{ \begin{array}{l} Y_0, \quad 1024/c_2, \quad 16384/c_3, \quad \frac{8}{Q_2-1} \log(12C_\sigma^2), \\ \frac{16}{Q_2-2} \log(2C_\sigma^2 C_3/c_2), \quad \frac{\log(290C_N)}{3Q_2-49/16}, \quad \frac{\log(145\sqrt{10}/s_0)}{2Q_2}, \\ \frac{\log(4C_N C_M)}{47Q_2/16-3}, \quad \frac{\log(\sqrt{11}C_M/s_0)}{(31Q_2+1)/16} \end{array} \right\}. \quad (214)$$

Choose $\widehat{\lambda}_1 \geq e^{Y_3}$. All denominators are positive because $Q_2 \geq 202$. These additional lower bounds retain the same source frequency family and integer selection; they do not replace any earlier scale by one. Set

$$\sigma_- = \sqrt{e} \nu \widehat{\lambda}_1^{1/16}, \quad d_{3,\max} = \min \left\{ d_{\max}, \frac{\sigma_- \log 2}{160 K_2^2}, \frac{L_3 \sigma_-}{128 K_3^2} \right\} > 0. \quad (215)$$

For the remainder choose any $0 < d \leq d_{3,\max}$ and construct the actual first and second stages already proved. Every number on the right of (215) is specified before selecting the second root or solving a new growth equation. The lower bound $\sigma_1 \geq \sigma_-$ follows from (79).

14.2 The actual third insertion scale

Let $t_b = t_2^b$ be the selected second return, and use a subscript b for evaluation there. Set $a = \sin s_2$, $h_b = h(t_b)$, $\chi_b = \sqrt{h_b^2 + a^2}$, and $P_{2,b} = -\Theta_2(t_b) > 0$. Define the original second gradient amplitude and next scale at this actual endpoint by

$$A_{2,b} = K_2 \chi_b P_{2,b}, \quad \sigma_2^2 = |G_{<3}(t_b)|, \quad s_3 = \frac{L_3 \sigma_1}{\sigma_2}, \quad \Gamma_3 = \sigma_2 \sin s_3. \quad (216)$$

In particular σ_2 is evaluated after the selected second control, rather than at an inviscid endpoint. The complete older gradient at this time is

$$G_{<3}(t_b) = -A_0 R_{\alpha_b} e_2 - A_{1,b} \zeta_{1,b} - A_{2,b} e_2, \\ A_{1,b} = -K_1 \Theta_1(t_b), \quad \zeta_{1,b} = \frac{(-a, h_b)}{\chi_b}, \quad \alpha_b = s_* + \arctan(a/h_b).$$

Theorem 2 gives $\chi_b \geq 1$, $P_{2,b} \geq P_2^*$, $A_{1,b} \leq 2\sigma_1^2$, and $A_0 \leq \sigma_1^2$. The new choice $L_2 > 1024$ and (80) imply

$$0 < a < \frac{1}{12HL_2} \leq \frac{1}{12288}. \quad (217)$$

Because $h_b \geq \cos s_2 \geq 15/16$, $0 < \alpha_b \leq 1/8 + (16/15)a < 1/4$. Both older terms in the displayed $G_{<3}$ have negative vertical component. Its norm is therefore at least $A_{2,b}$, while the triangle inequality bounds it above by $A_{2,b} + 3\sigma_1^2$. Moreover (214) and the original threshold give

$$A_{2,b} \geq A_0 \widehat{\lambda}_2^{1/8}, \quad \frac{A_{2,b}}{\sigma_1^2} \geq \frac{\widehat{\lambda}_1^{(Q_2-1)/8}}{C_\sigma^2} > 12.$$

Consequently

$$A_{2,b} \leq \sigma_2^2 \leq \frac{5}{4} A_{2,b}, \quad \sigma_2 > \sigma_1, \quad \sigma_2 \geq \nu \widehat{\lambda}_2^{1/16}, \quad \frac{4}{5} \leq \frac{A_{2,b}}{\sigma_2^2} \leq 1. \quad (218)$$

The actual third angle obeys an inequality stronger than a branch choice. Since $a \geq L_2 \nu / (2\sigma_1)$, the scale bounds imply

$$\frac{s_3}{a} \leq \frac{2L_3 \sigma_1^2}{L_2 \nu \sigma_2} \leq 2C_\sigma^2 \frac{C_3}{c_2} \widehat{\lambda}_1^{-(Q_2-2)/16} < 1.$$

Thus $0 < s_3 < a < 1/8$. The elementary inequality $\sin x \geq x - x^3/6 \geq (15/16)x$ for $0 \leq x \leq 1/8$ follows by integrating $\cos x \geq 1 - x^2/2$. It proves

$$\frac{15}{16} L_3 \sigma_1 \leq \Gamma_3 \leq L_3 \sigma_1. \quad (219)$$

No small-angle identity has replaced the actual sine in the controls.

14.3 Exact moving geometry and the changing coupling numerator

Use physical elapsed time $r = t - t_b$ and the original third growth control $\dot{\alpha} = F_2^{\text{ctl}}$. The second amplitude and its horizontal phase component remain zero in vorticity and phase, respectively, while both older temperatures and the first shear evolve:

$$\begin{aligned} \zeta_1 &= (-a, h)/\chi, & \zeta_2 &= (0, \chi), & \chi &= \sqrt{h^2 + a^2}, & \dot{h} &= a\Omega_1, \\ \dot{A}_1 &= -A_0 \sin s_* \Omega_1 - dK_1^2 A_1, & \dot{P}_2 &= -dK_2^2 \chi^2 P_2, & \dot{\alpha} &= -a^2 \Omega_1 / \chi^2. \end{aligned} \quad (220)$$

These are the previously proved holding equations in physical time. Let

$$\beta_b = \arctan(a/h_b), \quad b = \sin(s_3 + \beta_b), \quad g_b = \cos(s_3 + \beta_b), \quad c = bh_b - ag_b = \chi_b \sin s_3 > 0. \quad (221)$$

Here b is an oriented determinant magnitude, not an amplitude coefficient; the latter will carry its layer index below. Since $s_3 \leq a$ and $\beta_b \leq (16/15)a$, $0 < s_3 + \beta_b < 3a < 1/8$, so $g_b > 0$ and $0 < b < 3a$.

The third material matrix and phase have the exact expressions

$$\begin{aligned} M_3(t) &= M_2(t)M_2(t_b)^{-1}, & M_3(t_b) &= I, \\ g(t) &= g_b + (b/a)(h(t) - h_b), & \zeta_3(t) &= g\zeta_1 - bJ\zeta_1 = \frac{(c, hg + ab)}{\chi}, & R_3 &= |\zeta_3|^2 = g^2 + b^2. \end{aligned} \quad (222)$$

Here is a direct verification preserving the original datum $p_3 = e(s_3)$. Since $\Omega_2 = 0$, the actual matrices $D_{<3}$ and $D_{<2}$ are equal, so differentiation proves the matrix equation for M_3 with its stated initial value. In the basis $\zeta_1, J\zeta_1$, the third phase equation is $\dot{g} = b\Omega_1$, $\dot{b} = 0$; these follow by differentiating its scalar products and using $\dot{\zeta}_1 = \dot{\alpha}J\zeta_1$, $\dot{\zeta}_3 = \dot{\alpha}J\zeta_3 - \Omega_1\zeta_1(J\zeta_1 \cdot \zeta_3)$. Integration and $\dot{h} = a\Omega_1$ give (222). At birth, resolving $e(s_3)$ in this oriented basis gives precisely g_b, b in (221). The identity for c follows from $h_b = \chi_b \cos \beta_b$, $a = \chi_b \sin \beta_b$ and the sine subtraction formula. This verifies both components of the phase formula and its initial datum. Uniqueness of the linear phase equation identifies it with $M_3^{-T}p_3$.

The full inherited gradient and third amplitude coefficients are

$$\begin{aligned} G_{<3} &= -(A_1 + A_0 \cos s_* + K_2 P_2 h)\zeta_1 + (-A_0 \sin s_* + aK_2 P_2)J\zeta_1, \\ N_3 &= b(A_1 + A_0 \cos s_*) + gA_0 \sin s_* + cK_2 P_2, \\ a_3 &= \frac{N_3}{K_3 R_3}, & b_3 &= K_3 \zeta_{3,1} = \frac{K_3 c}{\chi}. \end{aligned} \quad (223)$$

To verify the first line, use $R_\alpha e_2 = \cos s_* \zeta_1 + \sin s_* J\zeta_1$ and $\zeta_2 = h\zeta_1 - aJ\zeta_1$ in the original sum $G_{<3} = -A_0 R_\alpha e_2 - A_1 \zeta_1 - K_2 P_2 \zeta_2$. The scalar product of $J\zeta_3 = gJ\zeta_1 + b\zeta_1$ with that sum is $-N_3$, proving the coefficient including the original gravity direction. Differentiating its full numerator gives

$$\dot{N}_3 = -d\{bK_1^2 A_1 + cK_2^3 \chi^2 P_2\}. \quad (224)$$

Indeed the terms $-bA_0 \sin s_* \Omega_1$ from $\dot{A}_1$ and $b\Omega_1 A_0 \sin s_*$ from $\dot{g}$ cancel exactly; the second temperature contributes the other displayed term. Thus the inviscid numerator would be exactly constant despite the changing shear. At positive d both parent losses are retained.

14.4 A common proved horizon before taking an amplitude quotient

Set $R_H = 16/\sigma_1$. Its duration in the second-stage time is $\Gamma_2 R_H = 16a < 1/32$, by (217). Thus the entire closed older system exists on $0 \leq r \leq R_H$ by its already proved second hold, uniformly for the current d . There A_1, P_2 are positive, $\Omega_1 < 9\sigma_1$, $\chi \geq \chi_b \geq 1$, and

$$\begin{aligned} 0 &\leq h - h_b \leq 144a, & 0 &\leq g - g_b \leq 144b, \\ 1 &\leq \sqrt{R_3} \leq 1 + 432a \leq 265/256 < \sqrt{2}, & 1 &\leq \frac{\chi}{\chi_b} \leq 259/256 < 2. \end{aligned} \quad (225)$$

The first inequality integrates $\dot{h} = a\Omega_1$. The second uses the exact relation (222). For the third apply the Euclidean triangle inequality to $(g, b) = (g_b, b) + (g - g_b, 0)$, whose first summand has norm one. The fourth follows similarly from $\sqrt{h^2 + a^2} \leq \chi_b + (h - h_b)$ and $\chi_b \geq 1$. The rational constants use $432/12288 = 9/256$ and $144/12288 = 3/256$; $265^2 < 2 \cdot 256^2$ proves the strict square-root bound without a numerical approximation.

All four terms in N_3 are positive on this horizon. Since $K_1 < K_2$ and $\chi^2 < 10$, (224) gives $-10dK_2^2 N_3 \leq \dot{N}_3 \leq 0$. Multiplication by $e^{10dK_2^2 r}$ and integration, followed by (215), proves

$$\frac{1}{2}N_i \leq N_3(r) \leq N_i, \quad \frac{1}{4}a_i \leq a_3(r) \leq a_i, \quad \frac{1}{2}b_i \leq b_3(r) \leq b_i, \quad (226)$$

where $N_i = N_3(t_b)$, $a_i = N_i/K_3$, and $b_i = K_3 \sin s_3$ are the actual birth values. Indeed $10dK_2^2 R_H = 160dK_2^2/\sigma_1 \leq \log 2$; the remaining bounds use (225) in the exact coefficient formulas. No value of a later temperature is assumed in these estimates.

For the actual initial rate $\gamma_i = \sqrt{a_i b_i}$ put $n_i = \gamma_i^2/\Gamma_3^2 = N_i/(\sigma_2^2 \sin s_3)$. The positive newest-parent contribution to its numerator is $cK_2 P_{2,b} = A_{2,b} \sin s_3$. Its remaining two terms satisfy

$$b(A_{1,b} + A_0 \cos s_*) + g_b A_0 \sin s_* \leq 4\sigma_1^2,$$

using $b, g_b \leq 1$, $A_{1,b} \leq 2\sigma_1^2$ and $A_0 \leq \sigma_1^2$. Equations (218) and (219) therefore give

$$\frac{4}{5} \leq n_i \leq 1 + \frac{64\sigma_1}{15L_3\sigma_2} \leq \frac{16}{15}, \quad \gamma_i \geq \frac{7}{8}\Gamma_3 \geq \frac{105}{128}L_3\sigma_1. \quad (227)$$

For the middle upper bound use $L_3 \geq 64$ and $\sigma_2 > \sigma_1$. The lower rate follows from $4/5 > (7/8)^2$.

The original third seed is exactly $\Theta_3(t_b) = -S_3$, $\Omega_3(t_b) = -S_3\sqrt{b_i/a_i}$. Solve its complete linear system with the changing coefficients:

$$\dot{\Theta}_3 = a_3\Omega_3 - dK_3^2 R_3\Theta_3, \quad \dot{\Omega}_3 = b_3\Theta_3 - dK_3^2 R_3\Omega_3. \quad (228)$$

Its coefficients and derivatives are finite on the common horizon, so the iterated integral series provides existence and uniqueness there. Define

$$E(r) = \exp\left(dK_3^2 \int_0^r R_3(t_b + u) du\right), \quad X = -E\Theta_3, \quad Y = -E\Omega_3.$$

Differentiation gives $\dot{X} = a_3Y$, $\dot{Y} = b_3X$. Both initial values are positive, and before a hypothetical first zero both derivatives are positive. This excludes the zero and proves $\Theta_3, \Omega_3 < 0$ throughout the horizon. The quotient may now be formed. With $u = \Omega_3/\Theta_3$ and $u_i = \sqrt{b_i/a_i}$, it satisfies $\dot{u} = b_3 - a_3u^2$. The bounds in (226) make $u_i/\sqrt{2}$ a constant subsolution and $2u_i$ a constant supersolution. The integrating-factor comparison for their differences proves

$$\frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad \sqrt{15/32} \leq v_3 := \frac{\sigma_2\Omega_3}{K_3\Theta_3} \leq \sqrt{5} < 3. \quad (229)$$

The exact identity $\sigma_2 u_i/K_3 = 1/\sqrt{n_i}$ and (227) give the latter interval.

14.5 The original third threshold and its full diffusion cost

The last bound in (215) and (227) imply $dK_3^2 \leq L_3\sigma_1/64 \leq \gamma_i/(16\sqrt{2})$. For the last comparison it suffices to use $\gamma_i \geq L_3\sigma_1/2$ and $\sqrt{2} \leq 2$. Thus the actual $P_3 = -\Theta_3$ has the positive physical rate

$$\frac{\gamma_i}{8\sqrt{2}} \leq (\log P_3)' = a_3u - dK_3^2 R_3 \leq 2\gamma_i < 3\Gamma_3. \quad (230)$$

The lower bound uses $a_3u \geq a_i u_i/(4\sqrt{2})$, $R_3 < 2$, and the displayed diffusion bound. The upper bound uses $a_3u \leq 2a_i u_i$ and $n_i \leq 16/15 < 9/4$.

Theorem 3. *For every fixed original datum and profiles, choose the source frequencies by (214) and any $0 < d \leq d_{3,\max}$ in (215). After the actual two returns already constructed, the original third growth control reaches its original target P_3^* exactly once, at $t_3^a = t_b + r_a$, with*

$$\frac{L_3}{2\gamma_i} \leq r_a \leq \frac{8\sqrt{2}L_3}{\gamma_i} \leq \frac{1024\sqrt{2}}{105\sigma_1} < \frac{14}{\sigma_1} < \frac{16}{\sigma_1}. \quad (231)$$

The crossing is transverse, its activation is already complete, and its initial and final physical amplitude ratios obey (229). Both inherited temperatures and the first shear evolve by (220) throughout. The exact logarithmic diffusion cost of the third amplitude is

$$dK_3^2 \int_0^{r_a} R_3(t_b + r) dr, \quad dK_3^2 r_a \leq dK_3^2 \int_0^{r_a} R_3(t_b + r) dr \leq 2dK_3^2 r_a. \quad (232)$$

Its total actual logarithmic gain is exactly L_3 .

Proof. The complete older and new systems were constructed on a fixed horizon before the threshold was chosen. The upper candidate time in (231) is within that horizon: squaring positive numbers gives $(1024\sqrt{2})^2 < 1470^2$. Integrating the lower rate in (230) to that candidate time gives at least L_3 . Continuity and strict increase therefore prove a unique crossing. Integrating the upper rate gives its lower time bound. Because $2\gamma_i < 3\Gamma_3$ and $L_3 \geq 64$, $\Gamma_3 r_a > L_3/3 > 1$, so the original activation $\eta(\Gamma_3 r)$ is one before this crossing. The rate lower bound proves transversality. The positive phase-length bounds in (225) give (232) by integration, with all physical factors retained. The target identity in (213) gives the exact gain L_3 . $\square$

For the two older amplitudes during this third growth, the exact diffusion exponents are $dK_1^2 r_a$ and $dK_2^2 \int_0^{r_a} \chi(t_b + r)^2 dr$. The second one is at most $10dK_2^2 r_a \leq \log 2$ on our common horizon. The additional first-temperature coupling $-A_0 \sin s_* \Omega_1$ is retained in (220); it is not a damping exponent. The numerator's exact total change is the integral of (224), including both parent contributions.

Exact third support nesting. The source third support also fits the changing exact affine cells. Indeed the relative matrix (222) is a product of two rotations and the shear with off-diagonal entry $-(h - h_b)/a$; hence $\|M_3\|, \|M_3^{-1}\| \leq 145$ on this horizon. The older bound is $\|M_2^{-1}\| \leq N \leq C_N \hat{\lambda}_1^{1/16}$. With the exact source radius $\ell_3 = \hat{\lambda}_2^{-3}/\mu$, the two growth-support conditions in (214) give

$$2 \cdot 145N \hat{\lambda}_2^{-3} < \hat{\lambda}_1^{-3}, \quad 145\sqrt{10}\hat{\lambda}_2^{-2} < s_0.$$

Thus $|M_2^{-1}x| < \ell_2/2$ and $|K_2\zeta_2 \cdot x| < s_0$ on the entire third support. The earlier proved nesting places it in the first and base affine cells as well. These explicit inequalities justify the complete compact third spatial increment and its full forces below.

This proves a further original growth event with retained diffusion, not a third return or an infinite iteration. The newly permitted range includes a factor K_3^{-2} and retains that dependence. The following transition must evolve both older vorticities and its prescribed blend of F_2^{ctl} and F_3^{ctl} ; the exact third-growth numerator identity identifies the parent losses that must remain in that calculation.

15 Compact spatial equations through the third threshold

The third growth starts at the actual second return $t_b = t_2^b$. The preceding coefficient construction proves its actual crossing time $t_{a3} := t_3^a > t_b$ in Theorem 3 of the original threshold. This section realizes that construction in the full equations, with the same physical diffusivity $d > 0$ in both equations and the declared pressure $p = 0$:

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta \theta + f_\theta, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, \quad \text{div } u = 0. \end{aligned}$$

Every field is defined on $\mathbb{R}^2$ in the original coordinate x . Retain

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A_0 \geq 1, \quad \nu = \sqrt{A_0}, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad K_j = \mu \hat{\lambda}_j \quad (1 \leq j \leq 3).$$

Here ν is the source time scale, not d . The hats distinguish the source frequencies from their physical multiples K_j . The released recursive frequency, source growth logarithm, inherited scales, angle, and activation are

$$\begin{aligned} \hat{\lambda}_3 &= \hat{\lambda}_2^{Q_3}, & L_3 &= (k_3 + 5 + 1/8) \log \hat{\lambda}_3, & s_3 &= L_3 \sigma_1 / \sigma_2, & \Gamma_3 &= \sigma_2 \sin s_3, \\ S_3 &= (\nu^2 / \mu) \hat{\lambda}_3^{-k_3-6}, & P_3^* &= S_3 e^{L_3} = (\nu^2 / \mu) \hat{\lambda}_3^{-7/8}, & w_3(t) &= \eta(\Gamma_3(t - t_b)). \end{aligned} \quad (233)$$

The source step η is smooth, nondecreasing, zero on $(-\infty, 0]$, and one on $[1, \infty)$. The actual coefficient proof, rather than an inviscid comparison, supplies the following bounds through t_{a3} and existence through the larger horizon $t_b + 16/\sigma_1$:

$$\begin{aligned} h_0 \leq h < 3, \quad 0 < \Omega_1 < 9\sigma_1, \quad 1/4 < B \leq 2, \quad a = \sin s_2 \in (0, 1/8], \quad h_0 = \cos s_2 \geq \sqrt{63}/8, \\ \chi = (h^2 + a^2)^{1/2}, \quad r_1 = 1, \quad 1 \leq \chi^2 < 10, \quad 1 \leq r_3^2 \leq 2, \quad 0 < -\Theta_3 \leq P_3^*, \quad |\Omega_3| \leq 3K_3 P_3^* / \sigma_2. \end{aligned} \quad (234)$$

The angle branch is the actual small source branch $0 < s_2, s_3 < 1/8$. The old amplitudes remain the actual ones:

$$\Theta_1 = -\sigma_1^2 B / K_1, \quad \Omega_2 = 0, \quad \Theta_2(t) = -P_{2b} \exp \left[-dK_2^2 \int_{t_b}^t \chi(r)^2 dr \right], \quad 0 < P_{2b} \leq e^{18} P_2^*.$$

These equations do not hold either old temperature fixed.

15.1 The exact third material flow and support inclusions

Put $\vartheta = \arctan(a/h)$ and write $h_b = h(t_b)$, $\vartheta_b = \vartheta(t_b)$. The actual held second feedback is

$$\begin{aligned} \alpha &= s_* + \vartheta, & \dot{h} &= a\Omega_1, & \dot{\alpha} &= -a^2\Omega_1/\chi^2, \\ \zeta_1 &= (-a, h)/\chi, & \zeta_2 &= (0, \chi), \\ G_{<3} &= -A_0 R_\alpha e_2 + K_1 \Theta_1 \zeta_1 + K_2 \Theta_2 \zeta_2, \\ D_{<3} &= \dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1. \end{aligned} \quad (235)$$

The second velocity increment is zero, but its scalar increment contributes the entire last term in $G_{<3}$. The closed older amplitude equations on this interval are

$$\begin{aligned} \dot{\Theta}_1 &= \frac{A_0 \sin s_*}{K_1} \Omega_1 - dK_1^2 \Theta_1, & \dot{\Omega}_1 &= -\frac{aK_1}{\chi} \Theta_1 - dK_1^2 \Omega_1, \\ \dot{\Theta}_2 &= -dK_2^2 \chi^2 \Theta_2, & \dot{\Omega}_2 &= 0. \end{aligned} \quad (236)$$

Indeed $-K_1 \Theta_1 = A_1$ in (220) gives the first equation, and $\zeta_{1,1} = -a/\chi$ in (112) gives the second. The held identities $\Omega_2 = \zeta_{2,1} = 0$ give the remaining two equations. For $S(q) = \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}$, the matrix in the exact geometry calculation is

$$M_3(t) = R_\vartheta S((h - h_b)/a) R_{-\vartheta_b} = M_2(t) M_2(t_b)^{-1}, \quad \dot{M}_3 = D_{<3} M_3, \quad M_3(t_b) = I. \quad (237)$$

Its determinant is one. Its inverse is $R_{\vartheta_b} S(-(h - h_b)/a) R_{-\vartheta}$. Indeed the rotation matrices have inverse rotations and $S(q)^{-1} = S(-q)$, while differentiating S gives $\dot{S} = \Omega_1 (J e_2 \otimes e_2) S$. Since $\zeta_1 = R_\vartheta e_2$, multiplication proves the ODE in (237) directly, including its signs. Integrating $\dot{h} = a\Omega_1$ over $16/\sigma_1$ gives $0 \leq (h - h_b)/a \leq 144$. The elementary inequality $\|S(q)\| \leq 1 + |q|$, obtained from $S(q) = I - q e_1 \otimes e_2$, proves

$$\|M_3\|, \|M_3^{-1}\| \leq N_3 := 145. \quad (238)$$

Use the original unit insertion vector $p_3 = e(s_3)$ and put $\zeta_3 = M_3^{-T} p_3$. Differentiating this exact inverse proves $\dot{\zeta}_3 = -D_{<3}^T \zeta_3$. Thus the moving phase and moving material envelope below are transported by the same actual older affine velocity, without changing the insertion coordinates.

The released functions F, P, f are retained without alteration: $F(s) = \sin s + \chi_F(s)(s - \sin s)$ on $[-\pi, \pi]$ and periodically extended, with χ_F smooth and even, $F(s) = s$ on $|s| \leq s_0$, $0 < s_0 < 1$; $P' = F$, P is even and periodic, and $\int_{-\pi}^{\pi} P = 0$. The radial $f \in C_c^\infty(B_1)$ satisfies $0 \leq f \leq 1$ and $f = 1$ on $B_{1/2}$. In particular the constant $P(0)$ on the affine cell is preserved. Keep the source radii

$$\ell_1 = \frac{s_0}{C_\ell \mu}, \quad \ell_2 = \frac{\hat{\lambda}_1^{-3}}{\mu}, \quad \ell_3 = \frac{\hat{\lambda}_2^{-3}}{\mu}, \quad C_\ell \geq 4. \quad (239)$$

Let $N = 1 + (3 - h_0)/a$, so $\|M_2^{\pm 1}\| \leq N$. The actual source frequency selection in (214) proves the strict inequalities established in the [exact third support nesting calculation](#):

$$\begin{aligned} N\hat{\lambda}_1^{-3} &< s_0/(2C_\ell), & N\hat{\lambda}_1^{-2} &< s_0, \\ 2N_3N\hat{\lambda}_2^{-3} &< \hat{\lambda}_1^{-3}, & N_3\sqrt{10}\hat{\lambda}_2^{-2} &< s_0. \end{aligned} \quad (240)$$

Here they have an exact spatial meaning. Put

$$y_3 = M_3^{-1}x, \quad g_3 = f(y_3/\ell_3), \quad k_3^{\text{ph}} = K_3\zeta_3, \quad s_3^{\text{ph}} = k_3^{\text{ph}} \cdot x.$$

The superscript distinguishes a phase vector and a phase variable from the source seed exponent k_3 and insertion angle s_3 . On the support of any third increment or derivative, $|x| \leq N_3\ell_3$. Consequently

$$|M_2^{-1}x| \leq NN_3\ell_3 < \ell_2/2, \quad |K_2\zeta_2 \cdot x| \leq K_2\sqrt{10}N_3\ell_3 < s_0.$$

Thus the second envelope is one and its profile is exactly affine. Moreover $x = M_2y_2$ with $|y_2| < \ell_2/2$, whence $|x| < N\ell_2/2$. The first two inequalities in (240) imply $|x| < \ell_1/2$ and $|K_1\zeta_1 \cdot x| < s_0$. This proves inclusion in the first affine cell as well. The base is affine there because $\mu\ell_1 = s_0/C_\ell < s_0$. All inequalities are strict, so the identities hold on an open neighborhood of the supports; differentiating a force introduces no missing boundary term. The actual older fields are therefore exactly $G_{<3} \cdot x$ and $D_{<3}x$ at every point needed below.

15.2 Third fields and the entire scalar and vector forces

Write $k = k_3^{\text{ph}}$, $\rho = |k|^2 = K_3^2 r_3^2$, $g = g_3$, $s = s_3^{\text{ph}}$, $D = D_{<3}$, and $G = G_{<3}$ in this subsection only. The actual third pair satisfies

$$\dot{\Theta}_3 = -\frac{Jk \cdot G}{\rho} \Omega_3 - d\rho \Theta_3, \quad \dot{\Omega}_3 = k_1 \Theta_3 - d\rho \Omega_3. \quad (241)$$

Define, with all one-variable profiles evaluated at s ,

$$T = w_3 \Theta_3, \quad Q = \frac{w_3 \Omega_3}{\rho}, \quad V_3 = TFg, \quad \Psi_3 = QPg, \quad v_3 = J\nabla \Psi_3 = QJ(kFg + P\nabla g). \quad (242)$$

For $t \leq t_b$ these three increments are defined to be zero. The flat activation and the smooth finite ODE solution make this zero extension smooth in space and time. On $[0, t_{a3}]$ set

$$\theta = \theta_b + V_1 + V_2 + V_3, \quad u = u_b + v_1 + v_2 + v_3, \quad p = 0.$$

The base and first two increments are precisely those in (116)–(117), continued by their actual holding equations through the third growth. The initial data remain exactly $\theta(x, 0) = -(A_0/\mu) \sin(\mu x_2) \chi_0(x)$ and $u(x, 0) = 0$. Define

$$Q_* = \frac{\dot{w}_3 \Omega_3 + w_3 k_1 \Theta_3}{\rho} - Q \frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D^T k. \quad (243)$$

Then $\dot{Q} = Q_* - d\rho Q$, including the varying phase length. The complete third scalar and full vector force increments are

$$E_{\theta,3} = \dot{w}_3 \Theta_3 F g + QP(J\nabla g \cdot G) + QT(F^2 - PF')g(Jk \cdot \nabla g) - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (244)$$

$$E_{u,3} = 2Dv_3 + Q_*J(kFg + P\nabla g) + (\nabla v_3)v_3 - TFge_2 - d(\rho v_3 + \Delta v_3). \quad (245)$$

In components, every derivative required in these formulas is

$$\nabla v_3 = QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \quad (246)$$

$$\Delta V_3 = T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (247)$$

$$\Delta v_3 = QJ\{\rho kF''g + \rho F'\nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + Fk\Delta g + P\nabla\Delta g\}. \quad (248)$$

The total forces are

$$f_\theta = f_{\theta,b} + \sum_{j=1}^3 E_{\theta,j}, \quad f_u = f_{u,b} + \sum_{j=1}^3 E_{u,j}, \quad (249)$$

with the exact base and old increments of (119)–(124) evaluated at the actual continued coefficients. There is no change of pressure hidden in the vector formula.

Proof of both equations and every interaction direction. For $\mathcal{D} = \partial_t + Dx \cdot \nabla$, (237) and its inverse give $\mathcal{D}(M_3^{-1}x) = 0$, and the phase equation gives $\mathcal{D}(k \cdot x) = 0$. Hence $\mathcal{D}g = 0$, $\mathcal{D}V_3 = \dot{T}Fg$, and $\mathcal{D}\Psi_3 = \dot{Q}Pg$. Subtracting the old scalar equation from the new one gives $\dot{T}Fg + v_3 \cdot G + v_3 \cdot \nabla V_3 - d\Delta V_3$. The two exact contractions are

$$v_3 \cdot G = QFg(Jk \cdot G) + QP(J\nabla g \cdot G), \quad v_3 \cdot \nabla V_3 = QT(F^2 - PF')g(Jk \cdot \nabla g).$$

Substitution of (241) cancels the first contraction against the coupling in $\dot{T}$ and proves (244).

Since $\text{tr } D = 0$, direct multiplication gives $DJ + JD^T = 0$. Therefore $\mathcal{D}(J\nabla\Psi_3) = Dv_3 + J\nabla(\mathcal{D}\Psi_3)$. The reverse interaction is $(v_3 \cdot \nabla)(Dx) = Dv_3$. Together with self-advection, buoyancy, and diffusion, this proves (245) after substituting (243). The ordinary product rule proves (246)–(248); in particular the last display contains all phase, envelope, and mixed terms.

The operator $\mathcal{D}$ uses the full old velocity $u_b + v_1 + v_2$, which equals Dx on the third support. It accounts for $(u_b + v_1 + v_2) \cdot \nabla V_3$ and $(u_b + v_1 + v_2) \cdot \nabla v_3$. The contractions with G and D account for $v_3 \cdot \nabla(\theta_b + V_1 + V_2)$ and $v_3 \cdot \nabla(u_b + v_1 + v_2)$. Although $v_2 = 0$, $v_3 \cdot \nabla V_2$ generally is not zero; it is exactly the $K_2\Theta_2\zeta_2$ part of $v_3 \cdot G$. The earlier increments already account for interactions among the base and first two layers. Outside a third support all its derivatives vanish. Adding these identities therefore proves (249) globally, with every ordered pair counted once. Finally each velocity is a rotated gradient, so mixed partial symmetry proves $\text{div } u = 0$. $\square$

For the exact coordinate correspondence put $C_3 = M_3^{-1}M_3^{-T}$ and $\tilde{g}(y) = f(y/\ell_3)$. The spatial gradient and Laplacian pulled back to (s, y) are

$$\begin{aligned} \mathcal{P}_3 &= K_3\zeta_3\partial_s + M_3^{-T}\nabla_y, \\ \mathcal{L}_3 &= K_3^2r_3^2\partial_s^2 + 2K_3(C_3p_3 \cdot \nabla_y)\partial_s + \sum_{l,m} (C_3)_{lm}\partial_{y_l}\partial_{y_m}. \end{aligned} \quad (250)$$

The chain rule gives the first identity, and multiplying its commuting spatial derivatives gives the second. In particular $k \cdot \nabla_x g = K_3(C_3p_3) \cdot \nabla_y \tilde{g}$ and $\Delta_x g = \text{tr}(C_3\nabla_y^2 \tilde{g})$. This proves the original-coordinate meaning of all mixed terms, without imposing a scalar metric or a fixed cutoff.

15.3 All spatial derivative norms and diffusion costs

For any scalar, vector, or tensor U , set $|U|_m = \sup_x \|D_x^m U(x)\|$ in the Euclidean multilinear operator norm, $m \geq 0$. For one-variable W set $[W]_m = \sup_s |W^{(m)}(s)|$, and set $f_m = \sup_y \|D^m f(y)\|$. Define only explicit, finite constants

$$\begin{aligned} \kappa_3^* &= \sqrt{2}K_3, & e_{3,m} &= (N_3/\ell_3)^m f_m, \\ \mathcal{B}_{3,m}(W) &= \sum_{q=0}^m \binom{m}{q} (\kappa_3^*)^q [W]_q e_{3,m-q}, & \mathcal{C}_{3,m}(P) &= \sum_{q=0}^m \binom{m}{q} (\kappa_3^*)^q [P]_q e_{3,m-q+1}. \end{aligned} \quad (251)$$

The symbol κ_3^* is a frequency bound, not a diffusivity. Repeated product and chain rules prove

$$|g_3|_m \leq e_{3,m}, \quad |W(s_3^{\text{ph}})g_3|_m \leq \mathcal{B}_{3,m}(W), \quad |P(s_3^{\text{ph}})\nabla g_3|_m \leq \mathcal{C}_{3,m}(P).$$

Indeed choose the q of m directional derivatives assigned to the profile, producing $\binom{m}{q}$ terms, each bounded by the corresponding summand. The derivative of a linear phase is k , and that of $M_3^{-1}x/\ell_3$ is M_3^{-1}/ℓ_3 ; their higher spatial derivatives vanish. This proves the stated operator norm bounds for arbitrary directions.

The actual coefficient bounds give

$$\begin{aligned} U_\Theta &= P_3^*, & U_\Omega &= 3K_3P_3^*/\sigma_2, & W_1 &= \Gamma_3[\eta']_0, \\ D^* &= 9\sigma_1(1+a^2), & G^* &= A_0 + 2\sigma_1^2 + \sqrt{10}K_2e^{18}P_2^*, \\ Q^* &= \frac{U_\Omega}{K_3^2}, & Q_*^* &= \frac{W_1U_\Omega + \kappa_3^*U_\Theta + 2D^*U_\Omega}{K_3^2}. \end{aligned} \quad (252)$$

In fact $\|D\| \leq |\dot{\alpha}| + \Omega_1$ and $|\dot{\alpha}| = a^2\Omega_1/\chi^2 \leq 9a^2\sigma_1$, which prove the bound for D . The three summands in G give its bound. The denominator $\rho \geq K_3^2$ and $|\dot{\rho}/\rho| \leq 2D^*$ prove the bounds for Q, Q_* . In particular activation contributes its exact physical factor $\Gamma_3[\eta']_0$ to both forces.

The full damping-plus-diffusion costs are bounded by

$$\begin{aligned} \mathcal{D}_{\theta,3,m} &= dU_\Theta \{(\kappa_3^*)^2 \mathcal{B}_{3,m}(F) + 2\mathcal{B}_{3,m+2}(F)\}, \\ \mathcal{D}_{u,3,m} &= dQ^* \{(\kappa_3^*)^2 \mathcal{B}_{3,m+1}(P) + 2\mathcal{B}_{3,m+3}(P)\}. \end{aligned} \quad (253)$$

These include $-d\rho V_3$ and $-d\rho v_3$ from modal damping as well as the Laplacians of the changed fields. The factor two bounds the two coordinate traces defining Δ in dimension two. Keeping this factor explicitly avoids identifying a Laplacian with a single second derivative. The complete third force estimates are

$$\begin{aligned} |E_{\theta,3}|_m &\leq W_1U_\Theta \mathcal{B}_{3,m}(F) + Q^*G^*\mathcal{C}_{3,m}(P) + \mathcal{D}_{\theta,3,m} \\ &\quad + Q^*U_\Theta \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{3,q+1}(P) \mathcal{B}_{3,m-q+1}(F), \end{aligned} \quad (254)$$

$$\begin{aligned} |E_{u,3}|_m &\leq (2D^*Q^* + Q_*^*)\mathcal{B}_{3,m+1}(P) + U_\Theta \mathcal{B}_{3,m}(F) + \mathcal{D}_{u,3,m} \\ &\quad + (Q^*)^2 \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{3,q+1}(P) \mathcal{B}_{3,m-q+2}(P). \end{aligned} \quad (255)$$

The nonlinear scalar term is bounded as $v_3 \cdot \nabla V_3$, and the nonlinear vector term as $(\nabla v_3)v_3$. The product rule with the displayed allocation of derivatives proves the two sums. All other terms follow individually from (244)–(248) and (251); hence the estimates include every activation, parent-interaction, self-interaction, buoyancy and diffusion term in the full vector equation. For time integrals one has, in particular,

$$\int_{t_b}^{t_{a3}} |\text{third damping-plus-diffusion force}|_m dt \leq \frac{16}{\sigma_1} \mathcal{D}_{\bullet,3,m}, \quad (256)$$

with the matching scalar or vector bound. The identical physical length factor integrates either full-force estimate.

All frequency and localization dependences are retained:

$$\frac{N_3}{\ell_3} = 145\mu\hat{\lambda}_2^3, \quad \kappa_3^* = \sqrt{2}\mu\hat{\lambda}_3, \quad dU_\Theta K_3^2 = d\nu^2\mu\hat{\lambda}_3^{9/8}, \quad dQ^* K_3^3 = \frac{3d\nu^2\mu}{\sigma_2}\hat{\lambda}_3^{9/8}. \quad (257)$$

The last two equalities identify coefficients occurring in the costs, not lower bounds for the entire forces. No cancellation of unrelated vector terms has been assumed. The finite positive- d range from the actual coefficient theorem is not replaced here by a bound uniform over an infinite frequency sequence.

For completeness, the bounds for the old and base forces remain fully quantitative on the slab ending at the third threshold. The formulas (128)–(133) hold with the original $N_1 = 1, N_2 = N$, radii, and global amplitude bounds (129); on the added interval the sharper $|\Theta_1| \leq 2\sigma_1^2/K_1$, $|\Omega_1| \leq 9\sigma_1$, $|\Theta_2| \leq e^{18}P_2^*$, $\Omega_2 = 0$ prove their continued validity. If $A_1^{\text{old}}, A_2^{\text{old}}$ are the explicit earlier control derivative bounds, one may use the following bounds on this slab:

$$\begin{aligned} A_1^{\text{ext}} &= \max\{A_1^{\text{old}}, 9a^2\sigma_1\}, \\ A_2^{\text{ext}} &= \max\{A_2^{\text{old}}, a^2(4\sigma_1^2 + 9dK_1^2\sigma_1) + 486a^3\sigma_1^2\}. \end{aligned} \quad (258)$$

To verify the second number, the holding equations give $\dot{\Omega}_1 = \sigma_1^2 a B / \chi - dK_1^2 \Omega_1$, so $|\dot{\Omega}_1| \leq 4\sigma_1^2 + 9dK_1^2\sigma_1$. Differentiating $\dot{\alpha} = -a^2\Omega_1/\chi^2$ and using $\dot{\chi}^2 = 2ha\Omega_1$, $h < 3$, and $\chi^2 \geq 1$ gives the stated additional term $486a^3\sigma_1^2$. Substitution into the explicit base norms (134) and old force norms completes the estimates for both total forces in (249), at every spatial order. Their integrals use this slab's actual interval length t_{a3} .

15.4 Mixed derivatives, affine core, and the threshold state

No stationary-envelope convention is used to estimate time derivatives. Set $A(t) = M_3(t)^{-1}/\ell_3$, introduce independent variables (t, x, s, y) , and define

$$\mathfrak{T} = \partial_t + (\dot{k} \cdot x)\partial_s + (\dot{A}x) \cdot \nabla_y, \quad \mathfrak{X}_i = \partial_{x_i} + k_i\partial_s + \sum_l A_{li}\partial_{y_l}. \quad (259)$$

For $\iota^*R = R(t, x, k(t) \cdot x, A(t)x)$, the chain rule gives

$$\partial_t^b \partial_x^\gamma \iota^* R = \iota^* (\mathfrak{T}^b \mathfrak{X}_1^{\gamma_1} \mathfrak{X}_2^{\gamma_2} R). \quad (260)$$

Induction on $b + |\gamma|$ proves this equality. The commutators vanish: $\dot{k}_i$ cancels the x_i derivative of $\dot{k} \cdot x$, and $\dot{A}_{li}$ cancels the corresponding derivative of $\dot{A}x$. This also proves consistency of the order of mixed derivatives. Apply these operators to the exact third fields and forces, starting for example with $Q(t)P(s)f(y)$ for the streamfunction. Every resulting term is a finite product of coefficient jets, explicit powers x^β , profile derivatives, and envelope derivatives. Its absolute value is bounded by replacing those factors by their coefficient suprema on $[t_b, t_{a3}]$, $(N_3\ell_3)^{|\beta|}$, $[P]_q$, $[F]_q$, and f_m , then summing all terms produced by (260). The coefficient jets themselves are evaluated recursively by (235), (236), (237), (241), and the explicit activation. Product and quotient rules, in increasing time order, use only the positive denominators a, K_3, χ^2, ρ and the fixed source scales. Their lower bounds in (234) therefore yield a finite explicit bound at every prescribed mixed order. The same recursion from (136)–(137) handles the base and old layers with the extended control bounds. This procedure includes every time derivative of the diffusion terms and of the changing material metric.

On a third affine cell, where $g_3 = 1$ and $|s_3^{\text{ph}}| < s_0$, the retained primitive is exactly $P(s) = P(0) + s^2/2$. Thus $\Psi_3 = Q(P(0) + s^2/2)$ there; taking its gradient, with the original constant still present in the full cutoff formula, gives

$$V_3 = TK_3\zeta_3 \cdot x, \quad v_3 = \frac{w_3\Omega_3}{r_3^2} J\zeta_3 \otimes \zeta_3 x, \quad \Delta V_3 = \Delta v_3 = 0. \quad (261)$$

The damping parts of the forces are still $-dK_3^2 r_3^2 V_3$ and $-dK_3^2 r_3^2 v_3$ there. On an uncut sine-profile region instead, $F'' = -F$ makes these damping-plus-diffusion contributions zero exactly. The remaining vector terms in (245) are retained at $p = 0$. This proves precisely how the actual damped pair maps into the released profile; it is not an unforced eigenfunction assertion.

At the actual third threshold, $\Theta_3(t_{a3}) = -P_3^*$ and $\Omega_3(t_{a3}) < 0$, with the exact phase $M_3(t_{a3})^{-T} e(s_3)$, the exact moving cutoff, and $w_3(t_{a3}) = 1$. The last equality follows from $\Gamma_3(t_{a3} - t_b) > L_3/3 > 1$ proved in Theorem 3. In the common affine cell the complete fields are

$$\theta = (G_{<3} + w_3 K_3 \Theta_3 \zeta_3) \cdot x, \quad u = \left(D_{<3} + \frac{w_3 \Omega_3}{r_3^2} J \zeta_3 \otimes \zeta_3 \right) x. \quad (262)$$

These are actual states of the full PDE just proved. The third vorticity has not returned to zero at this threshold; no third steering or return statement follows from the growth calculation.

15.5 Smooth finite-slab realization and honest force continuation

Let R_{base} denote the original spatial support radius called R_H in (116), as distinct from the temporal horizon $16/\sigma_1$ of the third-growth construction. The fields and forces are smooth and spatially supported in the same $B_{R_{\text{base}}}$ as the base. All third derivatives are supported within $|x| \leq N_3 \ell_3$; the proved inclusions put this ball inside the old base support. The velocity has finite energy since it is smooth and compact. For $N_0(x) = (2\pi)^{-1} \log |x|$, the compact smooth streamfunction $\psi = \psi_b + \Psi_1 + \Psi_2 + \Psi_3$ satisfies $N_0 * \Delta \psi = (\Delta N_0) * \psi = \psi$ by distributional integration by parts. Applying $J\nabla$ gives exactly $u = J\nabla N_0 * \omega$, $\omega = \Delta \psi$. Odd F , even P , and even material envelopes give odd θ, u, f_θ, f_u and even ω .

There is a smooth right collar at the threshold. The actual coefficient theorem continues the same F_2 growth equations through $t_b + 16/\sigma_1$, with t_{a3} strictly before that horizon. By (231) its remaining interval exceeds $2/\sigma_1$. Choose the explicit positive collar length $\epsilon_R = 1/\sigma_1$. Continue the exact coefficient solution and fields to $t_{a3} + \epsilon_R$; do not freeze their nonstationary endpoint. The upper rate in (230) holds on that entire horizon, as does the quotient bound (229). Consequently on the collar one has the explicit bounds

$$-\Theta_3 \leq e^{3\Gamma_3/\sigma_1} P_3^*, \quad |\Omega_3| \leq \frac{3K_3}{\sigma_2} e^{3\Gamma_3/\sigma_1} P_3^*. \quad (263)$$

Indeed integrate $(\log(-\Theta_3))' < 3\Gamma_3$ from the threshold to each collar time and then use the quotient bound. All spatial collar costs therefore follow from (252)–(255) by replacing U_Θ, U_Ω by the right sides of (263); the same coefficient-jet recursion proves the mixed costs. This does not impose the pre-threshold bound $-\Theta_3 \leq P_3^*$ after the crossing. The strict support inequalities continue on the proved horizon. Extend the original initial state constantly to the left of zero, with zero angle and inactive increments as before, and put $\epsilon_L = 1/\nu$. Multiplying the extended total residual forces by

$$\beta(t) = \eta(1 + t/\epsilon_L) \eta(1 + (t_{a3} - t)/\epsilon_R) \quad (264)$$

leaves them exactly unchanged on $[0, t_{a3}]$ and makes them smooth and compact in time as well as space. Flatness at the two outer endpoints proves smoothness of their zero extensions. Every additional mixed derivative follows from the ordinary product rule, with the explicit factors $\epsilon_L^{-b} [\eta^{(b)}]_0$ and $\epsilon_R^{-b} [\eta^{(b)}]_0$. The spatial fields solve the prescribed equations on the required slab $[0, t_{a3}]$; no equation outside that slab is asserted after the force cutoff. This proves the complete compact realization through the actual third threshold and retains the precise finite scope of the result.